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CAPSTONE PROJECT PROPOSAL

An Online Ordering System for Brand-new and Pre-loved Clothes with Product Listing, Checkout and Buyer - Seller Management

In partial fulfillment for the requirements in Capstone 1 and Research Subject

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

The growth of internet technology significantly transformed the way products were bought and sold in the global market. In the local context of the Philippines, this digital shift was particularly visible among small-scale enterprises that utilized social media and web-based platforms to promote their brands and interact with a wider customer base. Clothing businesses, in particular, benefited from this trend as consumers increasingly preferred the convenience of browsing fashion items online before committing to a purchase. Consequently, an online ordering system became more than just a luxury; it was a critical tool for businesses that aimed to improve operational efficiency and remain competitive in an increasingly digital age.

The clothing industry stood as one of the most active sectors in the e-commerce landscape because fashion products appealed to a diverse range of consumers with varying needs. Some buyers prioritized brand-new apparel for guaranteed quality and the latest styles, while others sought preloved clothing for its affordability and sustainability. Within the Philippine market, the rise of "ukay-ukay" or secondhand fashion reflected a broader cultural shift. What was once viewed merely as a budget option had gradually evolved into a conscious choice for environmentally-aware consumers. This evolution allowed small resellers to thrive, primarily operating through informal channels like Facebook Marketplace and Messenger.

However, the researchers observed that managing a dual inventory of both brand-new and preloved clothes presented unique operational hurdles. When a business relied solely on social media posts and private messages, the risk of confusion was high. Manual tracking often led to delayed responses, incomplete transaction records, and significant difficulty in monitoring real-time inventory. For a single seller juggling stock levels, payment verification via GCash, and delivery scheduling, the weight of manual labor was often overwhelming. A single mismanaged message could result in a lost order or a frustrated customer, highlighting a clear gap between the business's potential and its current operational capacity.

This was precisely the situation faced by **Margaux Collections**, a local boutique in Iloilo. While the business catered to a growing market for bikinis, coordinated sets, and preloved skirts, its ability to scale was hindered by its reliance on manual chatting. Each preloved item was unique, meaning that once it was sold through a private message, it had to be manually deleted or marked on social media to prevent other customers from inquiring. This process was not only time-consuming but also prone to human error. The researchers recognized that for Margaux Collections to maintain its brand identity—characterized by its elegant, "dark noir" aesthetic—it needed a professional and centralized digital platform.

The proposed **Online Ordering System for Brand-New and Preloved Clothes** was developed to bridge this gap. By providing a structured interface for product listings, the system allowed customers to view complete details, such as price, size, and item condition, without needing to send repeated inquiries. For the business owner, the system automated the checkout process and provided a reliable way to verify GCash transactions through reference number tracking. By integrating these features into one web-based platform, the researchers sought to modernize the operations of Margaux Collections, ensuring that the transition from browsing to checkout was as seamless and professional as the fashion items being sold. This study focused on demonstrating how a purpose-built system could reduce manual workload while simultaneously enhancing the shopping experience for the modern Filipino consumer.

1.2 Statement of the Problem

This study aimed to develop an Online Ordering System for Brand-New and Preloved Clothes with Product Listing, Checkout, and Buyer- Seller Management. The growth of online platforms in business had made it easier for sellers to reach customers, but it had also created significant challenges in handling orders, organizing product information, and maintaining clear communication. In the specific case of Margaux Collections, these challenges were exacerbated by the business's dual nature of selling both brand-new and preloved apparel. Each item required precise description, categorization, and monitoring, which became increasingly difficult under a manual framework. While the manual operation of the boutique allowed for a personal touch in customer service, it ultimately created a "glass ceiling" for the business's growth. The researchers identified that the primary issue was the lack of a synchronized system to manage two fundamentally different product types: brand-new items with multiple stocks and unique preloved pieces that were strictly one-off items.

This general problem was further analyzed through the following specific operational gaps that occurred during the business's manual phase:

First, the inefficiency of manual inquiry handling caused significant operational delays. Because product details such as exact measurements, fabric types, and condition reports were not readily available in a structured format, the owner was forced to spend a disproportionate amount of time answering repetitive questions via Messenger. This diverted her focus from critical activities like inventory sourcing and strategic marketing, leading to missed opportunities.

Second, the business faced a persistent risk of inventory discrepancy. In a manual, social-media-based setup, there was no real-time "lock" on items. A customer could inquire about a specific preloved skirt, but because the reservation process was not automated, the same item might have been promised to another buyer in a different chat thread. This lack of a centralized inventory status led to customer dissatisfaction and logistical confusion.

Third, the process of payment verification was labor-intensive and vulnerable to human error. Without an integrated checkout system, the owner had to manually cross-reference every GCash reference number against her transaction logs. During peak hours or "flash sales," managing dozens of simultaneous screenshots became overwhelming, increasing the likelihood of overlooking fraudulent claims or failed payments.

Lastly, the business lacked centralized data for informed decision-making. The owner had no immediate way to track total revenue or identify which categories—such as coordinates or accessories—were performing well. All vital sales information remained buried in scattered chat logs, making professional auditing and restocking plans nearly impossible to execute.

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study was to develop a web-based ordering system for brand-new and preloved clothes that could efficiently manage product listing, order processing, checkout, and buyer-seller communication in one centralized platform. The system aimed to provide sellers with a convenient way to upload, organize, update, and promote their clothing items while allowing buyers to easily browse products, compare options, and place orders online. It also sought to streamline the entire ordering process by providing secure checkout, order tracking, and automated notifications for confirmed purchases and updates. Furthermore, the study intended to improve buyer-seller interaction through built-in messaging features that would enable users to inquire about product details, negotiate prices, and clarify delivery arrangements. Ultimately, the system was designed to reduce manual

transactions, minimize errors, enhance customer convenience, and create a reliable online marketplace for both new and preloved clothing.

1.3.2 Specific Objectives

Specifically, This study aimed to:

1.Design a web-based system that allows the seller to upload, organize, and manage listings for brand-new and preloved clothing items.
2.Develop a checkout feature that enables customers to place orders easily and review their selected items before confirming their purchase.
3.Create a buyer-seller management module that supports communication between customers and the seller regarding product inquiries, order updates, and transaction concerns.
4.Implement an inventory management function that helps the seller monitor available stock and updated product status in real time.
5.Improve the accuracy and organization of order processing by reducing manual encoding and using a centralized digital platform.
6.Provide a user-friendly interface that makes it easier for customers to browse products, place orders, and complete transactions.
7.Ensure that the system supported both brand-new and preloved clothing items in a single online platform.
8.Develop an order tracking feature that allows customers to monitor the status of their purchases from confirmation to delivery.
9.Implement automated notifications to inform buyers and sellers about confirmed orders, payment updates, and changes in order status.
10.Provide a product search and filtering feature that enables customers to easily find clothing items based on category, size, price, and condition.
11.Develop a customer feedback and rating feature that allows buyers to share their experience and help improve seller credibility.
12.Ensure the security of customer information and transaction data through proper user authentication and data management.
13.Ensure that the system provided a simple, organized, and user-friendly interface for both buyers and sellers.
14.Evaluate the system's performance in terms of functionality, usability, reliability, and security through user feedback and testing.

1.4 Scope and Delimitation

This study focused on the development of an Online Ordering System for Brand-New and Preloved Clothes with Product Listing, Checkout, and Buyer-Seller Management for Margaux Collections. the system would cover the main operations of the business, particularly product listing, order placement, inventory tracking, payment handling, queuing, and
analytics. It was designed to help the owner managed clothing items more efficiently and to provide customers with a more organized and convenient way of browsing and ordering products. The study would specifically cater to the needs of Margaux Collections, which sold both brand-new items such as bikinis, coordinated sets, and accessories, as well as secondhand items such as pants, skirts, and shorts.

The system would be developed for use by the owner and her customers in the Philippines. It would use web-based technologies such as PHP, MySQL, HTML, CSS, and JavaScript to support the system's functionality. The platform would distinguish between brand-new and preloved clothing items so that products could be categorized properly and displayed in a clear manner. This separation would help both the seller and the buyers easily identify the type and condition of the items available for purchase.

The scope of the study included the implementation of features that supported product browsing, checkout, buyer seller communication, order management, and inventory monitoring. It also included queue management for pick-up orders and analytics to help the owner make better decisions based on sales and order data. These features were intended to improve the overall flow of business operations and reduced the problems caused by manual and social media based transactions.

On the other hand, this study had several delimitations. The system would not include integration with external logistics providers or third-party delivery tracking services. It would also not have advanced features such as automated size recommendations, as the focus of the study was on basic ordering and management functions. In addition, the project would prioritize web accessibility rather than the development of a mobile application. Since the business was operated solely by the owner, the system would be designed for a single-user business setup without staff handling transactions, inventory, or order processing.

The existing payment methods of Margaux Collections would remain unchanged. GCash would continue to be used for dropoff transactions, cash on delivery (COD) would be used for delivery orders, and in-person payment would be accepted for store pickup. These payment methods would stayed the same because they were already familiar to the owner and convenient for the customers. By maintaining the current payment setup, the system could be easier to adopt while still improving the organization and efficiency of the business. The core ordering and inventory functions needed by a single-owner clothing business rather than the broader feature set typically found in large-scale e-commerce platforms. The system was also limited to the payment and fulfillment methods already practiced by Margaux Collections — GCash reference verification, cash on delivery, in-person pickup, and rider-assisted pickup — instead of integrating third-party payment gateways such as PayMaya or credit card processors, since these were not part of the business's existing operations. Likewise, the study did not extend to multibranch or multi-seller functionality, as Margaux Collections operated as a single storefront managed by one owner.

1.5 Significance of the Study

This study was significant because it developed an Online Ordering System for Brand-New and Preloved Clothes with Product Listing, Checkout, and BuyerSeller Management that could improve the way Margaux Collections operated and served its customers.The proposed system was designed to benefit several stakeholders by making the ordering process more organized, efficient, and convenient. Since the business was currently managed entirely by the owner without anystaff, the system was expected to reduce the amount of manual work involved in handling orders, confirming payments,updating inventory, and responding to customer inquiries.

Through automation and centralized management, the business could become more efficient and easier to handle on a daily basis. For the Margaux Collections owner, the system provided a practical solution to the challenges of managing a small business alone. At present, the owner manually handled customer orders through Facebook Messenger, checked payments through GCash, and kept track of product availability on her own. This process could be time-consuming and might lead to errors, delays, or confusion, especially when multiple customers were inquiring or ordering at the same time.

With the proposed system, order processing and inventory management could be automated, allowing the owner to focus more on business operations and customer satisfaction. The system also provided sales insights and analytics that could help the owner make informed decisions regarding product restocking, especially in determining which items, such as brand-new bikinis and coordinated sets or preloved pants, skirts, and shorts, were in higher demand.

In this way, the system supported not only daily operations but also the long term growth of the business. For the customers, the system offered a more convenient and organized way of shopping. Instead of depending only on Facebook Messenger conversations, customers could use a dedicated web platform to browse products, view item details, and place orders more easily. This made the shopping experience more efficient because customers no longer needed to wait for repeated back-and-forth messages just to ask about product availability or order status. The system also provided a clearer payment workflow, since customers could follow the existing payment methods used by the business, such as GCash for drop-off orders, cash on delivery (COD) for delivery orders, and in-person payment for store pickup. In addition, the system could help customers tracked their orders and received queue numbers for pickup, which improved transparency and reduced confusion.

Overall, the proposed system gave customers a smoother and more reliable online shopping experience. For future researchers and students, this study served as a useful reference for understanding how web-based systems could support small business operations, especially in businesses that sold both brand-new and preloved fashion items. It could be used as a case study in web development, database management, and systems analysis because it combines practical business functions with technical implementation. The study might also helped future researchers who were interested in developing systems for micro enterprises, online clothing stores, or businesses that relied heavily on manual transaction handling. In addition, it provided ideas on how a single-owner business could transition from social media-based selling to a more structured digital platform. As a result, the study contributed not only to the improvement of Margaux Collections but also to the broader body of knowledge in information technology and business system development. The significance of this study lies in its ability to improve business efficiency, reduce manual workload, enhance customer convenience, and provide academic value for future researchers. By creating a centralized system for managing brand-new and preloved clothes, the study addressed the actual needs of a small business while also showing how technology could support growth and better decision-making. It demonstrated that even a small, ownermanaged business could benefit from a well-designed online ordering system that was tailored to its specific operations and payment practices.

1.6 Definition of Terms

Margaux Collections Ordering System – A web-based platform designed for Margaux Collections that allowed product browsing, ordering, queue management, and sales analytics. The system was intended to support the business by organizing both brandnew and preloved clothing items in one centralized platform.

Brand-new – Items that were new, unused, and sold in their original condition. In this study, brand-new items included bikinis, coordinated sets, and accessories. Preloved – Apparel that had been previously owned but was still in good wearable condition. In this study, preloved items included pants, skirts, and shorts.

Queue Number – An automated number assigned to each order to help organize processing and pickup. It was used to keep transactions in proper order, especially for drop-off locations and store pickups.

GCash Reference Number – A unique code generated after a GCash transaction, used by the system to verify that a customer's payment had actually been made before an order was confirmed.

Admin Panel – The restricted section of the system used by the owner to manage products, orders, users, and sales analytics.

Pick-up Via Rider – An order fulfillment method that allowed a customer to have their order picked up by a third-party rider on their behalf, in addition to store pickup, dropoff, and delivery.

Middleware – PHP-based components that managed communication between the frontend and backend of the system. These components handled authentication, sessions, authorization, and data validation to ensure that the system operated properly and securely.

API – PHP interfaces that supported system functions such as cart operations, order updates, payment status processing, and real-time dashboard information. These interfaces allowed different parts of the system to exchange data efficiently.

Database Gateway – The MySQL layer responsible for storing and retrieving data related to users, products, orders, payments, and transactions. It served as the connection between the system's interface and its database records.

GCash – A digital wallet used for advance payment in dropoff orders. Customers paid through GCash before the owner delivers the order to the designated drop-off store.

Cash on Delivery (COD) – A payment method in which customers pay in cash when their order was delivered. This method was used for delivery transactions in the business.

Drop-off Location – A specific store or address where the owner delivers orders for customers to pick up after payment through GCash.

Facebook Posts – Product listings published by the owner on her Facebook page, which were used as the main way of displaying available items to customers.

Messenger – The Facebook Messenger platform used by customers to communicate with the owner, ask about products, and place orders.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

The researchers conducted a comprehensive review of existing literature and studies to establish a solid theoretical and practical foundation for the development of the Margaux Collections Ordering System. This chapter explored various global and local perspectives on e-commerce, inventory management, and the evolving culture of secondhand fashion consumption. By analyzing these works, the researchers identified the technical gaps and operational requirements necessary to build a robust system tailored for a niche boutique.

2.1 Foreign Studies

Market Prices of Remanufactured, Used, and New Items

Neto, Bloemhof, and Corbett examined the market prices of remanufactured, used, and new items using data obtained from eBay. Their study focused on understanding how consumers valued products according to their condition and how these differences were reflected in market prices [1]. The researchers investigated the relationship between product condition, product descriptions, and selling prices to determine how these factors influenced the perceived value of products in the second-hand market. Their study provided insights into the pricing behavior of consumers and sellers in an online marketplace, where buyers rely heavily on available product information when evaluating items before making a purchase. The findings highlighted that used products possessed their own market value and were not necessarily perceived as inferior to new products. Instead, consumers could evaluate used products based on their condition, usefulness, quality, and overall suitability for their needs. This demonstrated that the value of a product was not determined solely by whether it was new or previously owned but could also depend on its characteristics and the information available to potential buyers [1].

The study was relevant to Margaux Collections because it emphasized the importance of determining appropriate prices for pre-loved products based on their actual condition and perceived value. In the second-hand market, customers may have different expectations regarding the price of an item depending on its appearance, durability, quality, and remaining usefulness. Therefore, sellers need to consider these characteristics when establishing prices for pre-loved clothing. For Margaux Collections, this implies that clothing items should be priced fairly according to their physical condition, fabric quality, appearance, style, brand, and overall appeal. Items that are well-maintained and show minimal signs of use may justify higher prices than those with noticeable wear or minor defects. Similarly, clothing with unique designs or desirable styles may attract buyers who perceive the items as valuable despite their pre-owned condition. By considering these factors, Margaux Collections can establish a pricing approach that reflects the characteristics of each item while remaining reasonable and understandable to customers.

Furthermore, the study demonstrated the importance of distinguishing between products with different conditions when evaluating their market prices [1]. This concept is applicable to the pre-loved clothing industry because second-hand garments are not always in the same condition. Some items may be nearly new, while others may show signs of previous use, such as fading, loose threads, minor stains, or slight fabric damage. If these differences are not properly considered, customers may perceive the prices as unreasonable or question whether the products are worth purchasing. For this reason, the Margaux Collections Ordering System can incorporate condition classifications such as *like new*, *good condition*, and *fair condition* to help customers understand the quality of the items being offered. These classifications should be supported by accurate descriptions of visible imperfections and other relevant characteristics. By providing consistent condition labels, the system can help customers compare products more easily and assess whether the listed prices correspond to their expectations.

In addition to pricing, product information plays an important role in helping consumers evaluate the value of second-hand goods. When purchasing pre-loved clothing online, customers cannot physically inspect the garments before placing an order. They must therefore depend on the descriptions, photographs, prices, and condition details provided by the seller. Inaccurate or incomplete information may lead to uncertainty and misunderstandings regarding the actual appearance or quality of an item. The findings of Neto, Bloemhof, and Corbett [1] provide a basis for considering the role of product descriptions in the online evaluation of used goods. For Margaux Collections, this insight can be applied by providing detailed product listings that include the item name, category, size, color, material, price, condition, and other relevant specifications. High-quality photographs from appropriate angles may also help customers examine the appearance of each garment before making a purchase. These features can provide buyers with a clearer understanding of the products and support more informed purchasing decisions.

Moreover, accurate product descriptions and transparent pricing can contribute to a more organized and professional online ordering experience. When customers can easily identify the price and condition of each item, they may find it easier to compare available clothing and select products that fit their preferences and budget. For instance, a

customer looking for an affordable jacket may compare several listings based on their prices, sizes, styles, and condition labels. If the system presents this information consistently, the customer can evaluate the available options without needing to request basic details individually from the seller. In contrast, incomplete listings may require additional communication and may delay the purchasing process. Therefore, the Margaux Collections Ordering System can improve product accessibility by organizing listings into appropriate categories and presenting essential product information in a clear and consistent format. Search and filtering features may further help customers locate items according to their preferred category, size, price range, or condition.

The study is also relevant to the issue of customer confidence in online second-hand transactions. Customers may hesitate to purchase pre-loved clothing when they are uncertain about the item's condition or the fairness of its price. Providing accurate information about the product can help address these concerns by allowing customers to evaluate the item before placing an order. Although Neto, Bloemhof, and Corbett [1] examined market prices and product descriptions rather than directly establishing that transparent pricing increases customer trust, their research provides a useful basis for designing product listings that support informed evaluation. In the context of Margaux Collections, displaying clear prices, realistic photographs, and honest condition descriptions may help reduce uncertainty and establish more consistent communication between the seller and customers. These practices can also help manage customer expectations by ensuring that the information presented in the system corresponds as closely as possible to the actual products being sold.

Another important application of the study is its relevance to the affordability and perceived value of pre-loved clothing. One of the potential advantages of purchasing second-hand garments is the opportunity to obtain usable clothing at prices that may be lower than those of comparable new products. However, affordability alone may not be sufficient to convince customers to purchase an item. Buyers may also consider whether the garment is fashionable, comfortable, durable, and suitable for their intended use. Consequently, the pricing strategy of Margaux Collections should balance affordability with the actual characteristics and perceived value of each item. A well-maintained garment offered at a reasonable price may appeal to customers seeking both quality and savings, while a garment with visible defects may require a lower price and a more detailed description. By presenting these differences clearly, the ordering system can help customers determine which products best meet their needs and expectations.

In terms of system development, the findings of Neto, Bloemhof, and Corbett [1] support the inclusion of product information management and transparent pricing as important features of the Margaux Collections Ordering System. The system can provide a structured way for the seller to record and update product details, assign condition labels, set prices, and organize available clothing items. This approach can reduce the likelihood of inconsistent product information across listings and make it easier to maintain an updated product catalog. Furthermore, an organized ordering system can allow customers to review available items before submitting their orders, reducing their dependence on informal inquiries for basic product details. The system may also provide a more convenient means of managing product availability and displaying current prices, thereby supporting a more systematic and accessible purchasing process.

The findings may also serve as a basis for future improvements in the pricing practices of Margaux Collections. By maintaining organized product records, the business may be able to review the prices of similar items and identify patterns in customer preferences. For example, the seller may compare the listed prices of garments with different condition classifications to determine whether the pricing structure is consistent with the quality of the products. Customer inquiries and purchasing patterns may also provide useful feedback about whether the prices and descriptions meet buyers' expectations. Although these activities were not necessarily tested in the original study, they represent practical applications of the broader principle that product characteristics and market information are relevant to the valuation of used goods. Incorporating these practices into the ordering process may help Margaux Collections improve its product presentation and make more informed pricing decisions over time.

Overall, the study of Neto, Bloemhof, and Corbett [1] provides relevant insights into the market valuation of used, remanufactured, and new products. Its examination of online marketplace prices and product descriptions highlights the importance of considering product condition and other characteristics when evaluating the value of second-hand

goods. These insights are applicable to Margaux Collections because pre-loved clothing items may differ considerably in terms of quality, appearance, style, and remaining usefulness. By incorporating transparent pricing, accurate condition labels, detailed product descriptions, appropriate photographs, and organized product listings, the Margaux Collections Ordering System can help customers evaluate available items more effectively. These features may contribute to a more convenient ordering process, clearer communication, and better-informed purchasing decisions. Thus, the study serves as a useful reference for developing an ordering system that presents pre-loved clothing in a structured, informative, and customer-oriented manner.

Kim, Jung, and Lee (2021)

Kim, Jung, and Lee [2] conducted a quantitative study examining consumers' value and risk perceptions of circular fashion and how these perceptions influenced their attitudes and behavioral intentions toward circular fashion products. The study focused on secondhand, upcycled, and recycled clothing, considering different dimensions of perceived value, including emotional value, social value, epistemic value, and environmental value, as well as perceived risks. The researchers investigated how consumers evaluated circular fashion products beyond their monetary cost and how different perceptions of value influenced their responses to these products. The study was relevant to the development of the Margaux Collections Ordering System because it demonstrated that circular fashion products were not evaluated solely on the basis of affordability. Consumers also considered the emotional significance, social meaning, and environmental benefits associated with these products. These findings supported the idea that preloved fashion could be presented as valuable, desirable, and fashionable rather than simply as previously owned clothing [2].

The study found that emotional, social, epistemic, and environmental value positively influenced consumers' attitudes toward circular fashion products. Among the examined dimensions, emotional value had the strongest influence on product attitudes. This suggested that consumers could develop favorable perceptions of circular fashion when the products provided personal satisfaction, social significance, useful information, and perceived environmental benefits. Emotional value could be associated with personal preferences, uniqueness, self-expression, and the satisfaction of discovering desirable fashion items. Similarly, social and environmental value could encourage consumers to view circular fashion as a meaningful and responsible form of consumption. These findings were relevant to Margaux Collections because they suggested that the way preloved products were presented could influence customers' perceptions of their value and desirability [2].

The findings also provided a basis for presenting preloved products through a professional and organized ordering system. Since consumers' perceptions of value influenced their attitudes toward circular fashion, the Margaux Collections Ordering System could provide sufficient product information to help customers evaluate items before making purchasing decisions. An organized product listing could include clear and high-quality photographs, accurate product descriptions, available sizes, prices, brands, colors, and detailed condition information. Providing these details could help customers understand the characteristics of each item and determine whether it suited their preferences. Although the original study did not directly examine an online ordering system, its findings offered a theoretical basis for improving product presentation and communicating the value of preloved fashion [2].

Kim, Jung, and Lee [2] also examined perceived risks associated with circular fashion products. Their findings indicated that sanitary risk and aesthetic risk negatively influenced consumers' attitudes toward circular fashion, whereas economic risk and performance risk did not significantly influence product attitudes in the overall analysis. These results highlighted the importance of addressing concerns about the cleanliness and appearance of secondhand clothing. Such concerns could cause hesitation among customers, particularly when purchasing products online because they could not physically inspect the items before placing an order. Therefore, providing sufficient and accurate product information could help customers better understand the condition of the products being offered [2].

For the Margaux Collections Ordering System, these findings supported the inclusion of detailed product descriptions, multiple product photographs, condition classifications, and other relevant product information. For example, descriptions could disclose visible stains, scratches, fading, missing components, or other signs of wear. Information regarding cleaning or product preparation could also be included when verified. By communicating these details clearly, the system could help customers establish realistic expectations before purchasing. These features could improve transparency and support informed purchasing decisions, although their actual effectiveness would need to be evaluated after the system was implemented.

The study further suggested the importance of presenting circular fashion products in a transparent and professional manner. When customers receive accurate information about an item's actual condition and characteristics, they can make decisions based on clearer expectations. For Margaux Collections, transparency in product presentation could help minimize misunderstandings between the seller and customers. Detailed product information could also support the ordering process by enabling customers to compare available items and assess their suitability before confirming an order. In this context, the ordering system could serve not only as a platform for processing purchases but also as a source of information that helps customers evaluate preloved products.

Furthermore, Kim, Jung, and Lee [2] provided insights into the environmental and social value associated with circular fashion. Circular fashion encourages the continued use and circulation of existing clothing, which may extend garment lifespans and reduce the need for newly manufactured products. For this reason, Margaux Collections could communicate the potential environmental benefits of purchasing preloved clothing through appropriate product descriptions or system content. Presenting these products as an alternative to buying newly manufactured clothing could help customers understand the role of reuse in more sustainable consumption. In addition, the distinctive characteristics of preloved products could contribute to their emotional and social value, especially when customers viewed them as unique fashion pieces that reflected their individual styles [2].

Overall, the study of Kim, Jung, and Lee [2] provided a relevant foundation for the development of the Margaux Collections Ordering System because it demonstrated that consumers' attitudes and behavioral intentions toward circular fashion were associated with multiple dimensions of perceived value and risk. The findings highlighted the importance of emotional, social, epistemic, and environmental value in shaping favorable product attitudes, while sanitary and aesthetic concerns could negatively affect consumer evaluations. These insights supported the development of an ordering system that presented products clearly, professionally, and transparently. By incorporating high-quality product images, organized product listings, accurate descriptions, condition information, and other relevant details, the Margaux Collections Ordering System could help customers evaluate preloved products and make informed purchasing decisions. Thus, the study supported the view that an effective ordering system should function not only as a platform for purchasing preloved fashion but also as a means of communicating product characteristics, perceived value, and the potential benefits of circular fashion [2].

Consumer Perceptions and Purchase Intentions Toward Second-Hand Luxury Goods

Lou, Chi, Janke, and Desch [3] conducted a quantitative study examining how consumers in the United States perceived second-hand luxury goods and how these perceptions influenced their purchase intentions. Their research investigated six factors associated with consumer decision-making: perceived economic value, emotional value, social value, quality value, green value, and perceived risk. The researchers collected survey data from 340 eligible respondents and used statistical analysis to examine the relationships between these factors and consumers' intentions to purchase second-hand luxury products. The study aimed to understand why consumers considered purchasing previously owned luxury goods and which perceived benefits encouraged or discouraged their buying decisions. The findings revealed that emotional value, social value, quality value, and green value positively influenced consumers' purchase intentions, while economic value and perceived risk did not have statistically significant effects in the study's model [3]. These results indicated that consumers' decisions to purchase second-

hand luxury products involved more than financial considerations and could also be influenced by personal satisfaction, social perceptions, product quality, and environmental concerns.

The study was relevant to the Margaux Collections Ordering System because it demonstrated the importance of understanding consumers' perceptions when presenting and selling pre-loved fashion products. Although the researchers focused on second-hand luxury goods, their findings offered useful insights into the broader second-hand fashion market. Consumers may evaluate pre-loved clothing based not only on its affordability but also on its appearance, condition, uniqueness, usefulness, and ability to satisfy their personal preferences. Therefore, the proposed Margaux Collections Ordering System should provide sufficient product information to help customers understand the characteristics and potential value of each item. Instead of presenting pre-loved clothing merely as discounted or previously owned products, the system can emphasize relevant product attributes through accurate descriptions, appropriate photographs, organized categories, and clear condition classifications. These features may help customers evaluate products according to their individual needs and preferences while providing a more informative and professional online shopping experience.

One of the important findings of Lou *et al.* [3] was that perceived quality positively influenced consumers' purchase intentions toward second-hand luxury goods. Perceived quality refers to consumers' assessment of a product's overall excellence, condition, appearance, and suitability for use. In the second-hand market, customers may be concerned about whether a previously owned item remains durable, attractive, and worth its listed price. These concerns can be particularly relevant when purchasing clothing online because buyers cannot physically inspect the fabric, stitching, color, or possible imperfections before completing an order. When product information is incomplete, customers may find it difficult to determine whether an item meets their expectations. For Margaux Collections, this finding supports the inclusion of detailed product descriptions, accurate photographs, size information, material details, and condition labels. Providing these details can help customers assess the apparent quality of each garment and make more informed purchasing decisions. However, because the original study examined second-hand luxury products, the same relationship should be considered a relevant basis for the proposed system rather than a directly established result for all categories of pre-loved clothing.

Emotional value was another factor that positively influenced purchase intentions in the study [3]. Emotional value refers to the feelings of satisfaction, enjoyment, excitement, or personal fulfillment that consumers associate with purchasing and owning a product. In second-hand fashion, customers may enjoy discovering distinctive clothing items, finding products that match their personal style, or purchasing garments that are not commonly available in mainstream stores. These experiences can make pre-loved shopping appealing beyond its potential financial benefits. For Margaux Collections, this suggests that product presentation should help customers recognize the style and distinctive characteristics of the available clothing. The ordering system can support this objective by displaying clear product images, descriptive item names, relevant style categories, and organized collections. For example, clothing may be grouped into categories such as casual wear, formal wear, vintage-inspired pieces, or seasonal selections when these categories accurately reflect the available inventory. Such organization may help customers discover items that match their preferences and make browsing more convenient. While the system cannot guarantee an emotional response, an appealing and organized product catalog can support a more engaging shopping experience.

The researchers also identified social value as a significant positive predictor of consumers' purchase intentions [3]. Social value concerns the extent to which a product contributes to a consumer's self-image, identity, social recognition, or desired impression on others. In the context of second-hand luxury goods, consumers may value particular products because of their brand reputation, recognizable design, exclusivity, or association with a preferred lifestyle. These symbolic characteristics can influence how consumers perceive the desirability of a product. This finding is relevant to Margaux Collections because pre-loved clothing may appeal to customers who want to express their individual style through distinctive and carefully selected garments. The ordering system can help communicate these characteristics by providing appropriate product categories, accurate brand information when available, and clear descriptions of each item's style and features. A consistent presentation can also make the

product catalog easier to navigate and allow customers to compare items according to their preferences. Nevertheless, product claims should remain factual, and the system should not imply that every pre-loved garment carries luxury status or social prestige.

Another important finding was the positive relationship between perceived green value and consumers' purchase intentions toward second-hand luxury goods [3]. Green value refers to the environmental benefits that consumers associate with a product or purchasing decision. Buying second-hand clothing can extend the useful life of existing garments and provide an alternative to purchasing newly manufactured products. This practice may contribute to more sustainable consumption by encouraging reuse and helping keep usable clothing in circulation. For Margaux Collections, this finding supports communicating the reuse and extended-use aspects of pre-loved fashion through appropriate product descriptions or general information about the business. The ordering system may explain that purchasing pre-loved garments gives existing clothing another opportunity to be used rather than discarded prematurely. This information can help customers understand the potential environmental value associated with second-hand shopping. However, environmental statements should be accurate and proportionate; the system should not claim that every purchase automatically reduces environmental impact by a specific amount unless that claim is supported by reliable evidence.

The study also examined perceived economic value, which relates to the financial benefits consumers expect when purchasing second-hand luxury products. Although affordability is often considered an important advantage of buying pre-owned goods, the researchers found that perceived economic value did not have a statistically significant effect on purchase intention in their U.S. sample [3]. This result suggests that lower prices alone may not be sufficient to explain why consumers intend to purchase second-hand luxury products. Instead, other factors, particularly emotional, social, quality, and environmental value, may be more influential in the context examined by the researchers. For Margaux Collections, this finding highlights the importance of presenting products through a combination of clear pricing and useful product information. The ordering system should display prices accurately and consistently while also allowing customers to assess product condition, appearance, size, and other relevant characteristics. By providing this information together, the system can help customers evaluate the overall suitability of an item rather than focusing exclusively on its price. It is important to note that the study's nonsignificant result does not mean that price is unimportant to all second-hand clothing customers; its influence may differ across consumer groups and product categories.

Perceived risk was another factor examined in the study. Perceived risk refers to the uncertainty or potential negative consequences consumers associate with a purchase, including concerns about product condition, reliability, authenticity, and the possibility that an item may not meet their expectations. For online second-hand purchases, this uncertainty may be increased by the inability to inspect the product physically before ordering. Customers may hesitate if they cannot determine whether a garment has stains, fading, loose stitching, missing components, or other imperfections. Lou *et al.* [3] found that perceived risk did not have a statistically significant effect on purchase intention in their model. This result should not be interpreted as evidence that perceived risk is irrelevant to every second-hand purchase. Instead, it indicates that perceived risk was not a significant predictor under the conditions and measurements used in their study. For Margaux Collections, providing accurate product information remains a practical consideration. The system can display visible imperfections, specify the condition of each item, provide relevant measurements, and present photographs that accurately represent the products. These features can help customers understand what they are considering purchasing and establish realistic expectations before placing an order.

The findings of Lou *et al.* [3] also provide a basis for considering the relationship between product information and the usability of an online ordering system. When customers browse an online catalog, they need to locate relevant products, compare alternatives, and review essential details before making a purchasing decision. If information is inconsistent or difficult to find, customers may need to contact the seller repeatedly to clarify prices, sizes, product condition, or availability. An organized ordering system can address these practical concerns by presenting product information in a standardized format. For Margaux Collections, each listing may include the product name,

category, size, color, material, price, condition, photographs, and availability status, where applicable. Search and filtering functions may also allow customers to narrow the product catalog according to their preferences. These features are proposed system applications rather than functions directly tested by the original researchers, but they are consistent with the broader objective of helping consumers evaluate second-hand products more effectively.

Furthermore, the study supports the importance of presenting pre-loved fashion products in a way that communicates their actual characteristics and potential value. A well-organized catalog can help customers distinguish among available products and understand the features that make each item suitable for their needs. Accurate descriptions can communicate the garment's condition, while photographs can help customers assess its visible appearance. Consistent pricing can make comparisons easier, and appropriate product categories can improve navigation throughout the ordering process. Together, these features may support customer confidence by reducing avoidable uncertainty about the items being offered. For Margaux Collections, this means that the ordering system should not be designed solely to record orders but should also provide a clear and informative interface through which customers can evaluate products before purchasing them.

The findings may also be useful when evaluating the effectiveness of the proposed Margaux Collections Ordering System. For example, future system evaluation may examine whether customers find the product descriptions understandable, whether condition labels provide sufficient information, and whether the product catalog makes it easy to compare available clothing. Feedback may also help determine whether customers consider the photographs, pricing information, and category organization useful during the purchasing process. These measures could help the business identify opportunities to improve product presentation and the overall ordering experience. However, the original study did not directly evaluate the Margaux Collections Ordering System, so any conclusions about the system's effectiveness would need to be supported by its own user testing and evaluation.

Overall, the study by Lou, Chi, Janke, and Desch [3] provides relevant theoretical and empirical support for understanding consumers' purchase intentions toward second-hand fashion products. The researchers found that emotional value, social value, perceived quality, and green value positively influenced purchase intentions among the U.S. consumers they studied, while perceived economic value and perceived risk were not statistically significant predictors in their model. These findings highlight the multidimensional nature of second-hand luxury purchasing and suggest that consumers may consider personal, social, product-related, and environmental benefits when deciding whether to purchase pre-owned goods. For Margaux Collections, the study supports the proposed inclusion of accurate product descriptions, clear condition labels, transparent pricing, appropriate photographs, and organized product categories in the ordering system. These features may help customers evaluate available clothing, understand product characteristics, and make better-informed purchasing decisions. Thus, the study provides a useful foundation for developing a more organized, informative, and customer-oriented ordering system for pre-loved fashion products.

Selling Second-Hand Luxury: Empowerment and Social Roles Turunen, Cervellon, and Carey (2020)

Turunen, Cervellon, and Carey [4] conducted a qualitative study entitled “Selling Second-Hand Luxury: Empowerment and Enactment of Social Roles,” which explored the meanings and values associated with selling second-hand luxury products. The researchers interviewed 18 women who had previously sold luxury-branded items to understand how selling influenced their perceptions of luxury, personal empowerment, and social roles. The

findings showed that selling luxury products could change the symbolic meaning of possessions by transforming items with personal significance into goods with financial value. The study also found that reselling could influence sellers' social identities. Some participants associated selling second-hand luxury products with a higher perceived social status, while others viewed it as a way to strengthen their identity as sustainable consumers. These findings demonstrated that second-hand selling involved more than financial transactions because it could also have personal, social, and environmental significance [4].

The study further revealed that selling previously owned luxury products could empower consumers by allowing them to recover financial value from their possessions. When individuals decided to sell items they had previously purchased for personal use, they reassessed the meaning and value of those possessions. This process transformed the items from personally meaningful belongings into products available for resale. The finding was relevant to the development of the Margaux Collections Ordering System because it highlighted the importance of recognizing the value of preloved products. Rather than presenting previously owned clothing merely as used goods, Margaux Collections could emphasize the products' brands, styles, distinctive characteristics, and actual condition. This approach could help customers recognize the potential value and appeal of each item [4].

Another important finding concerned the relationship between second-hand selling and sellers' social roles. Turunen et al. [4] found that reselling could influence how individuals perceived themselves and how they understood their participation in consumer culture. For some sellers, the activity supported a sense of higher social status, while for others, it reinforced their identity as sustainable consumers. This suggested that second-hand selling could have meanings beyond earning money from unwanted possessions. For Margaux Collections, this finding supported the potential value of presenting preloved fashion as an alternative way to obtain desirable clothing while encouraging the continued use of existing garments. By communicating relevant product information and presenting items attractively, the business could help customers appreciate the appeal of preloved fashion.

The study also provided a basis for considering the importance of product presentation in the resale market. Although the researchers primarily investigated the meanings of selling luxury products and the social roles of sellers, their findings highlighted how the value associated with possessions could change through resale. Applied to Margaux Collections, this insight suggested that product listings should provide information that helps customers understand the characteristics and appeal of each item. Relevant details could include the product name, brand, size, color, price, condition, and distinctive features. Clear photographs could further help customers evaluate the item's appearance before making a purchasing decision. These recommendations represent practical applications of the study rather than specific system features tested by the researchers.

The findings were also relevant to the organization and management of products within the Margaux Collections Ordering System. In an online resale environment, customers depend on the information provided by sellers to evaluate available products. A centralized ordering system could help the business owner record product details, categorize items, update availability, and display information in a consistent format. Instead of relying on scattered records or manually managed product information, the owner could maintain organized product listings and access relevant details more efficiently. These capabilities could support more systematic business operations and provide customers with a clearer way to browse available products.

Furthermore, the study suggested that second-hand selling could contribute to a seller's identity through the meanings associated with resale and sustainable consumption. Turunen et al. [4] found that some participants connected selling luxury products with a stronger identity as sustainable consumers. This finding was relevant to Margaux Collections because the business could communicate the potential benefits of extending the useful life of clothing through reuse. Appropriate product descriptions or system content could explain the value of purchasing preloved items as an alternative to buying newly manufactured clothing. However, environmental claims should be accurate and should not imply benefits that have not been established.

The findings also offered insights into the potential role of an ordering system in supporting the business owner. Although Turunen et al. [4] did not directly evaluate inventory-management software or online ordering systems,

their research demonstrated that selling second-hand luxury products could influence how sellers understood their possessions and their roles in the resale market. In the context of Margaux Collections, an organized system could help the owner manage product records, monitor item availability, and process customer orders more systematically. These capabilities could reduce difficulties associated with manual management and provide a more consistent process for presenting and selling preloved products. Any improvements in efficiency or customer satisfaction would need to be assessed after implementation.

Moreover, the study provided a conceptual basis for understanding how a second-hand fashion business could establish a recognizable identity. Since resale could carry financial, personal, and social meanings, Margaux Collections could present its products in a way that emphasizes their individual characteristics and value. Consistent product descriptions, organized categories, attractive photographs, and accurate condition information could contribute to a more professional presentation. These features could also help distinguish the business from informal resale practices by providing customers with a structured and accessible ordering experience.

Overall, Turunen, Cervellon, and Carey [4] provided relevant insights for the development of the Margaux Collections Ordering System by demonstrating that second-hand luxury selling involved financial, personal, social, and environmental dimensions. Their findings showed that reselling could transform the symbolic value of possessions, empower sellers through the recovery of financial value, and influence their social identities. These insights supported the importance of presenting preloved products clearly and communicating their characteristics accurately. By incorporating organized product listings, detailed descriptions, clear photographs, inventory-management functions, and order-processing features, the Margaux Collections Ordering System could support more systematic resale operations and a professional business identity. Thus, the study offered a useful conceptual foundation for understanding second-hand selling and applying its implications to the development of an organized ordering platform [4].

ONLINE SHOPPING VERSUS OFFLINE SHOPPING: A COMPARATIVE STUDY

Sarkar and Das [5] conducted a comparative study of online and offline shopping behavior to examine the differences between the two shopping methods and identify the factors influencing consumers' preferences and purchasing decisions. Their study discussed the advantages and disadvantages of online shopping and traditional shopping, focusing on factors such as convenience, accessibility, product selection, and the ability to evaluate products before purchasing. The study was relevant to the development of the Margaux Collections Ordering System because it provided a basis for understanding how customers might respond to an online platform for purchasing pre-loved clothing. Online shopping allows customers to browse available products and obtain information without visiting a physical store, while traditional shopping allows them to inspect products directly before making a purchase. These differences highlight the importance of designing an online ordering system that provides convenience while addressing customers' concerns about product information and evaluation.

One of the main advantages of online shopping is convenience. Customers can browse products, compare available options, and place orders without needing to travel to a physical store. This flexibility can help customers save time and access product information more easily. Online shopping platforms may also allow businesses to present their products to customers beyond their immediate physical location. These advantages are relevant to Margaux Collections because the proposed Ordering System could provide a more accessible way for customers to view available pre-loved clothing and submit their orders. Instead of depending entirely on physical visits or manual communication with the seller, customers could access the product catalog, review item details, and select products through the system. This could make the ordering process more organized and convenient for both customers and the business owner.

Another important consideration in the comparison between online and offline shopping is product selection and comparison. Online platforms can display multiple products in an organized catalog, allowing customers to review different items before deciding which ones to purchase. For Margaux Collections, this functionality would be useful because pre-loved clothing items may differ in terms of style, size, color, brand, price, and physical condition. An organized product catalog could help customers identify available items that match their preferences. The proposed system could also group products into appropriate categories and provide search or filtering functions, depending on the project's requirements. These features could reduce the time required to locate specific items and help customers compare the available options more efficiently.

Despite the convenience of online shopping, customers cannot physically examine products before completing their purchases. In traditional shopping, buyers can inspect an item's appearance, feel its material, assess its quality, and determine whether it meets their expectations. Online shoppers, in contrast, must rely on the photographs and descriptions provided by the seller. This limitation is particularly relevant to clothing because differences in size, color, fabric, fit, and appearance may influence purchasing decisions. For pre-loved clothing, customers may also need to consider visible signs of previous use, such as fading, stains, loose stitching, or minor damage. These concerns emphasize the importance of presenting accurate and complete product information in the Margaux Collections Ordering System.

To address these concerns, the proposed system could provide clear product photographs, detailed descriptions, accurate prices, size information, and condition classifications. Multiple photographs could help customers examine the visible appearance of a garment, while descriptions could explain its material, measurements, and relevant characteristics. Condition labels, such as *like new*, *good condition*, and *fair condition*, could help customers distinguish between items with different levels of wear. Any noticeable imperfections should also be disclosed to establish realistic expectations. Although these features cannot completely replace physical inspection, they could help customers evaluate available products more effectively and make better-informed purchasing decisions.

Product information is particularly important in an online ordering system because customers depend on the details presented through the platform when evaluating products. Incomplete descriptions or inaccurate photographs may cause uncertainty and lead to misunderstandings between the seller and customers. For Margaux Collections, each product listing could include the item name, category, price, size, color, condition, and availability status. Presenting these details consistently could help customers understand the characteristics of each garment before placing an order. Accurate product information could also reduce the need for customers to contact the seller repeatedly to clarify basic details. Consequently, a structured product catalog could improve the clarity of the purchasing process and help the business maintain consistent information across its listings.

The comparison between online and offline shopping also highlights the importance of usability. Traditional shopping allows customers to interact directly with products and sellers, whereas online shopping requires customers to navigate a digital interface to obtain information and complete transactions. Therefore, the Margaux Collections Ordering System should be designed so that its functions are easy to understand and use. Customers should be able to browse product categories, view item details, review prices, and submit orders without unnecessary difficulty. A clear interface, readable product information, consistent page layouts, and straightforward navigation could help reduce the effort required to complete a purchase. These design considerations are important because an online platform can provide convenience only when customers can use its features effectively.

In addition, the ordering process should provide clear information about the status of customer transactions. Features such as order summaries, order confirmations, and order status information could help customers understand whether their orders have been recorded and what stage of processing they have reached. An order summary could allow customers to review their selected items and prices before submitting an order, while confirmation information could help prevent uncertainty about whether the transaction was successfully recorded. These functions represent proposed applications for the Margaux Collections Ordering System rather than features directly evaluated in the comparative study. Nevertheless, they could contribute to a more structured and understandable ordering process.

Accessibility is another relevant consideration when comparing online and offline shopping. Customers who shop at a physical store generally need to visit the business location to examine available products. An online ordering system, however, could allow customers to review product listings remotely, subject to internet access and the business's service arrangements. For Margaux Collections, this could provide customers with a convenient way to explore pre-loved clothing without requiring an immediate physical visit. It could also make product information more readily available to customers who have limited time to communicate directly with the seller. By making the catalog accessible through a digital platform, the business could provide an additional channel for product discovery and order placement.

However, convenience alone may not be sufficient to create a satisfactory online shopping experience. Customers may still hesitate to purchase clothing online if they cannot determine whether an item meets their expectations. This is especially relevant to pre-loved fashion because each garment may have a different history of use and condition. Therefore, the Margaux Collections Ordering System should combine convenient access with transparent product presentation. Clear photographs, accurate descriptions, condition labels, consistent pricing, and organized product listings could help customers assess the available items before making a decision. These features could address some of the limitations associated with online product evaluation while preserving the convenience of digital ordering.

The study also provides a basis for considering how the proposed system should be evaluated. Although an online ordering platform may offer convenient access to products, its actual usefulness depends on whether customers can navigate it successfully and obtain the information they need. The Margaux Collections Ordering System could therefore be evaluated according to the clarity of its product listings, ease of navigation, accessibility of product information, and simplicity of the ordering process. User feedback could help identify difficulties that customers experience when browsing items or submitting orders. This evaluation would allow the developers to determine whether the system meets its intended objectives and identify possible improvements based on actual user experiences.

Overall, the comparative study by Sarkar and Das [5] provides a useful basis for understanding the differences between online and offline shopping and their implications for the Margaux Collections Ordering System. Online shopping offers potential advantages in convenience, accessibility, and product comparison, while traditional shopping allows customers to inspect products physically before purchasing them. These differences are particularly important in the pre-loved clothing market, where customers need sufficient information to assess the appearance and condition of individual garments. The proposed Margaux Collections Ordering System could address these needs through organized product categories, clear photographs, accurate descriptions, transparent pricing, condition classifications, and a straightforward ordering process. By combining convenient digital access with complete product information, the system could help customers make informed purchasing decisions and provide a more organized and accessible shopping experience.

SUSTAINABLE SUPPLY CHAIN MANAGEMENT

Svensson [6] examined sustainable supply chain management (SSCM) by introducing a conceptual framework and an empirical example related to the clothing industry. The study explored how supply chains could be understood in relation to the use of nonrenewable, nonrecycled, renewable, and recycled resources. Svensson explained the concepts of first-order, second-order, and higher-order supply chains to emphasize the importance of considering product life cycles beyond the initial point of consumption [6]. The study highlighted that sustainable supply chain

management required businesses to consider how products and resources moved through different stages of their life cycles. This perspective was relevant to Margaux Collections because the business focused on pre-loved clothing, allowing previously owned garments to remain in circulation and provide value to subsequent customers. The proposed Margaux Collections Ordering System could support this process by organizing product listings, maintaining inventory information, and making available garments easier for customers to discover and purchase.

One important concept presented by Svensson [6] was that the life cycle of clothing products could extend beyond their initial purchase and use. In a traditional product cycle, clothing may be viewed primarily in terms of its production, distribution, and purchase by the first consumer. However, the study emphasized the importance of considering subsequent supply chains and the continued use of products beyond their original consumption stage. This concept was particularly relevant to the pre-loved clothing industry because garments could retain practical and economic value after their first owner no longer needed them. Through resale, these products could be transferred to new customers and continue to serve their intended purpose. For Margaux Collections, this provided a conceptual basis for supporting clothing reuse through an organized ordering platform. By presenting available garments clearly and facilitating customer orders, the system could help connect existing clothing items with potential buyers and support their continued use.

Svensson [6] also discussed the distinction between supply chains that depend substantially on nonrenewable and nonrecycled resources and those that incorporate a greater share of renewable or recycled resources. This distinction illustrated the importance of examining resource use throughout a product's life cycle rather than focusing exclusively on its initial production and sale. In the context of pre-loved fashion, resale provides an opportunity to extend the useful life of existing garments instead of treating them as products that no longer have value after their first ownership. For Margaux Collections, this perspective supported the business's role in facilitating the continued use of previously owned clothing. The proposed ordering system could contribute to this objective by displaying product information, photographs, prices, and condition descriptions that help customers evaluate available items. Although the original study did not directly evaluate an online ordering system, its conceptual framework was relevant to understanding the relationship between clothing resale and extended product use.

Another important implication of the study concerned the management of product information throughout the supply chain. Effective product management requires accurate records and accessible information to support operational decisions. This consideration was relevant to Margaux Collections because pre-loved clothing inventories may contain unique garments with different sizes, colors, brands, prices, and physical conditions. Unlike mass-produced products that may be replenished in identical quantities, individual pre-loved items may have limited availability. Consequently, inaccurate inventory records could result in duplicate listings, outdated product information, or orders for items that have already been sold. The Margaux Collections Ordering System could address these concerns by maintaining organized product records and updating availability when items are added, reserved, or sold. Such functions could help the business owner monitor inventory more accurately and maintain consistency between the products displayed in the system and the actual items available.

The study's emphasis on sustainable supply chain management also provided a basis for considering operational efficiency in the proposed system. Businesses need appropriate methods for coordinating activities and maintaining information that supports their daily operations. For Margaux Collections, a centralized ordering and inventory system could reduce dependence on scattered records and repetitive manual updates. By storing product information in an organized database, the system could allow the owner to retrieve details more easily, monitor available items, and review order records systematically. These capabilities could reduce administrative difficulties and help the business manage its inventory more consistently. Although these particular system functions were not directly tested by Svensson [6], they represent practical applications of the broader principle that effective information management contributes to the coordination of business activities.

Furthermore, accurate inventory information could help the business make more informed operational decisions. When product records are maintained consistently, the owner can identify which garments remain available, which have already been sold, and which items may require additional promotion. This information could assist in

monitoring stock movement and identifying discrepancies between inventory records and actual availability. For a pre-loved clothing business, where many products may be unique or available in only one quantity, maintaining accurate records is especially important. The proposed Margaux Collections Ordering System could therefore support inventory monitoring and order management by keeping product details organized and accessible. These functions could improve the reliability of business records and help prevent avoidable errors during the ordering process.

Svensson's [6] discussion of sustainable supply chains was also relevant to the continued circulation of clothing products. Reselling usable garments can provide another opportunity for existing clothing to be worn rather than discarded prematurely. This practice is consistent with the broader idea of considering product use beyond the initial consumer. For Margaux Collections, the resale process could be supported by an ordering system that presents garments clearly and allows customers to identify products suited to their preferences. Accurate product descriptions and condition labels could help customers understand what each garment offers, while organized categories could make the catalog easier to browse. By helping customers discover available items, the system could facilitate the movement of pre-loved garments from listing to purchase and support the business's clothing reuse activities.

In addition to supporting product reuse, an organized digital system could contribute to accountability in business operations. Systematic records of products, availability, and orders could provide the business owner with a clearer overview of inventory changes and customer transactions. For example, recording when an item is listed or sold could help identify discrepancies and maintain accurate stock information. These records could also support the review of previous transactions and help the owner understand the movement of products through the business. For Margaux Collections, such capabilities could make daily operations more manageable and improve the consistency of information used in business decisions. However, the system's actual contribution to efficiency and accountability would need to be assessed through implementation and evaluation.

The study also suggested the importance of considering sustainability as part of regular business practices rather than treating it as an isolated objective. For Margaux Collections, this means that the proposed ordering system could support the business's reuse-oriented activities through its everyday functions. Organized product listings, updated inventory records, and systematic order processing could help the business manage pre-loved clothing more efficiently. These features could support the continued availability of garments for resale while reducing avoidable administrative errors. Nevertheless, the implementation of a digital ordering system alone would not guarantee environmental sustainability. Its contribution would depend on how effectively the business manages its inventory, facilitates reuse, and incorporates responsible practices into its operations.

Overall, Svensson [6] provided a conceptual foundation for understanding sustainable supply chain management and its relevance to the clothing industry. The study emphasized that product life cycles extend beyond initial consumption and that subsequent supply chains should be considered when examining sustainability [6]. These concepts were applicable to Margaux Collections because the resale of pre-loved clothing allows existing garments to continue providing value to new owners. The proposed Margaux Collections Ordering System could support this process through organized product listings, accurate inventory records, updated product availability, and systematic order management. Although the system could not independently guarantee sustainable outcomes, it could help the business coordinate its resale activities and manage clothing products more effectively. Therefore, the study supported the development of an ordering system that combines operational organization with the broader objective of extending the useful life of pre-loved clothing.

Clothing Reuse and Environmental Impact — Cuc and Vidovic (2014)

Cuc and Vidovic [7] examined the environmental sustainability of textile waste recycling and its potential to address environmental and economic concerns associated with discarded textiles. Their study highlighted the importance of recycling textile waste as a way to conserve natural resources, reduce waste disposal costs, and support sustainable development. This research was relevant to the Margaux Collections Ordering System because the business focuses on selling preloved clothing. By facilitating the sale of previously owned garments, the proposed system could support clothing reuse and encourage more responsible fashion consumption.

The study explained that textile waste recycling could help reduce the demand for virgin materials and minimize the environmental problems associated with textile disposal [7]. The production of new clothing requires raw materials and other resources, while discarded garments contribute to the growing volume of textile waste. Allowing wearable clothing to be resold and used by other consumers could help maximize the value of existing products. For Margaux Collections, this finding supported the resale of preloved garments as a business approach that could contribute to reducing unnecessary disposal and promoting more efficient resource use.

Cuc and Vidovic [7] also discussed the potential economic and environmental benefits of recovering and reselling wearable textiles. Reusing garments allows products that remain suitable for wear to continue serving their original purpose instead of being immediately discarded. This approach is relevant to second-hand fashion businesses because it provides consumers with access to existing clothing while creating opportunities for the continued circulation of garments. Through the Margaux Collections Ordering System, customers could browse available preloved products and place orders, helping make reusable clothing more accessible to potential buyers.

The environmental significance of preloved clothing could be communicated through the Margaux Collections Ordering System. Product pages could include short sustainability statements explaining how purchasing second-hand garments supports the continued use of existing clothing. These messages could help customers recognize the environmental considerations associated with their purchasing decisions. Although Cuc and Vidovic [7] focused on textile recycling and waste management rather than online ordering systems, their findings provided a basis for incorporating sustainability information into the proposed platform.

The study further established the importance of considering the economic, social, and environmental dimensions of textile waste management [7]. In relation to the proposed system, customers could be provided with detailed product descriptions, photographs, garment condition information, and relevant care recommendations. These features could help buyers assess the suitability of preloved garments before placing an order. Proper garment care could also help preserve clothing quality and usability, although this specific system feature represents a proposed application rather than a directly evaluated outcome of the study.

Including garment care information could strengthen the informational function of the Margaux Collections Ordering System. In addition to displaying products and processing orders, the system could provide general recommendations for washing, drying, storage, and handling, depending on the garment's material and care requirements. Such information could help customers maintain their purchases and potentially extend the garments' useful lives. These recommendations would complement the broader sustainability objectives of clothing reuse and responsible ownership.

Another implication of the study was the importance of presenting textile recycling and reuse as part of a broader approach to environmental sustainability. Consumers may better understand the value of preloved clothing when they are informed about the potential benefits of keeping wearable products in circulation. Therefore, Margaux Collections could communicate its commitment to clothing reuse through product descriptions and other relevant sections of the ordering system. Such communication could help customers recognize the environmental purpose of the business alongside the affordability and variety of its products.

The integration of sustainability-related information could also strengthen the identity of Margaux Collections as a preloved clothing business. By emphasizing garment reuse and responsible consumption, the business could appeal to customers who value the continued use of existing clothing resources. The ordering system could support this objective by combining product information, convenient ordering functions, and sustainability messages within one

digital platform. These proposed features would connect the commercial activities of the business with the environmental principles discussed by Cuc and Vidovic [7].

Furthermore, the proposed system could use digital technology to make information about preloved products more accessible to customers. By providing organized product listings, garment details, and information about clothing reuse, the platform could help customers make informed purchasing decisions. The system could therefore serve both an operational purpose, through the management of product orders, and an informational purpose, through the communication of sustainability-related content. However, the extent of its environmental contribution would depend on factors such as whether second-hand purchases replace purchases of new clothing and whether the garments remain in use.

Overall, Cuc and Vidovic [7] provided a relevant foundation for the sustainability aspect of the Margaux Collections Ordering System by discussing the potential environmental and economic benefits of textile waste recycling. Their study highlighted the importance of conserving resources, managing textile waste, and considering the broader effects of textile recovery. These findings supported the proposed inclusion of preloved product listings, detailed garment information, care recommendations, and sustainability messages in the ordering system. By facilitating the sale of wearable second-hand clothing and providing customers with relevant product information, the Margaux Collections Ordering System could support the continued use of garments and encourage more responsible fashion consumption.

CONSUMERS' INTENTION TO BUY SECOND-HAND CLOTHING

Koay, Cheah, and Lom [8] examined the factors influencing consumers' intention to purchase second-hand clothing in an online shopping environment. Their study integrated the theory of planned behavior with consumption value concepts to understand how consumers developed intentions to purchase pre-loved garments. The researchers collected survey responses from 290 consumers in Malaysia and analyzed the data using partial least squares structural equation modeling. The study examined factors such as attitudes toward second-hand clothing, social norms, moral norms, perceived behavioral control, and consumption values. The findings indicated that attitudes, injunctive norms, descriptive norms, moral norms, and perceived behavioral control positively influenced consumers' purchase intentions. The researchers also found that emotional value and environmental value positively influenced consumers' attitudes toward second-hand clothing [8]. These findings were relevant to the Margaux Collections Ordering System because they demonstrated that consumers' decisions to purchase pre-loved clothing could be influenced by their personal perceptions, social environment, values, and confidence in the purchasing process.

One of the major findings of Koay *et al.* [8] was that consumers' attitudes toward second-hand clothing positively influenced their intention to purchase such products. Attitude refers to an individual's favorable or unfavorable evaluation of a particular behavior. Consumers who viewed second-hand clothing positively were more likely to express an intention to purchase it. These positive attitudes could be associated with the perceived usefulness, attractiveness, affordability, and suitability of pre-loved garments. This finding was relevant to Margaux Collections because the way products were presented through the ordering system could influence customers' impressions of the available items. Clear photographs, accurate product descriptions, organized categories, and appropriate condition labels could help customers understand the characteristics of each garment. By presenting products professionally and consistently, the system could help customers evaluate available clothing and determine whether particular items matched their preferences.

The study also identified emotional value as a factor that positively influenced consumers' attitudes toward second-hand clothing [8]. Emotional value refers to the feelings and personal experiences consumers associate with a

product or purchasing decision. In the context of pre-loved fashion, customers may enjoy discovering unique garments, finding clothing that reflects their personal style, or obtaining items that they consider meaningful or distinctive. These experiences may contribute to the appeal of second-hand shopping beyond its practical benefits. For Margaux Collections, this finding suggests that product presentation should allow customers to explore available garments and recognize their individual characteristics. The proposed ordering system could support this objective through appealing product photographs, informative descriptions, and organized product categories. Although the system cannot guarantee that customers will develop positive emotional responses, these features could help create a more engaging and convenient browsing experience.

Another important finding concerned environmental value. Koay *et al.* [8] found that environmental value positively influenced consumers' attitudes toward second-hand clothing. Environmental value refers to the perceived environmental benefits associated with a product or purchasing decision. Purchasing pre-loved clothing allows existing garments to remain in use and provides an alternative to purchasing newly manufactured clothing. Consumers who value responsible consumption may therefore view second-hand fashion favorably because it provides an opportunity to participate in clothing reuse. This finding was relevant to Margaux Collections because the business specializes in pre-loved garments. The ordering system could support this objective by making available clothing items easier to discover and purchase. It could also provide factual information about clothing reuse and the continued use of existing garments. However, specific claims about environmental savings should only be made when supported by reliable evidence.

Koay *et al.* [8] further found that injunctive norms and descriptive norms positively influenced consumers' intention to purchase second-hand clothing. Injunctive norms refer to consumers' perceptions of whether important people approve of a particular behavior, while descriptive norms refer to their perceptions of whether other people engage in that behavior. These findings indicate that purchasing decisions may be influenced by perceived social approval and observations of other people's actions. In the context of pre-loved fashion, consumers may become more willing to consider second-hand clothing when they perceive it as an acceptable and familiar shopping option. For Margaux Collections, this supports the importance of maintaining a professional and consistent presentation of products. An organized catalog, accurate descriptions, and clear photographs could help communicate the value and characteristics of available garments. Although these system features cannot directly control social influence, they could help present pre-loved clothing in a way that supports a positive customer experience.

The researchers also identified moral norms as a positive influence on consumers' intention to purchase second-hand clothing [8]. Moral norms refer to an individual's personal sense of responsibility regarding a particular behavior. Some consumers may associate purchasing pre-loved clothing with responsible consumption, reuse, and avoiding unnecessary disposal of usable garments. This finding was relevant to Margaux Collections because reselling previously owned clothing provides an opportunity for garments to be used by additional customers. The proposed ordering system could support this process by presenting available items clearly and allowing customers to review and order products more conveniently. Providing accurate information about the nature of pre-loved clothing could also help customers understand the business's resale activities. These features may be useful to consumers whose purchasing decisions are influenced by personal values and attitudes toward clothing reuse.

Perceived behavioral control was another important factor examined in the study. Koay *et al.* [8] found that perceived behavioral control positively influenced consumers' intention to purchase second-hand clothing. This concept refers to the extent to which individuals believe that they have the ability and resources to perform a particular behavior. In online shopping, customers may feel more capable of completing a purchase when the ordering process is understandable, product information is accessible, and the required steps are manageable. Conversely, a confusing interface, incomplete descriptions, or unclear ordering instructions may create difficulties that discourage customers from proceeding with a transaction. This finding was particularly relevant to the Margaux Collections Ordering System because the platform needed to provide a straightforward and user-friendly purchasing process.

To support perceived behavioral control, the proposed Margaux Collections Ordering System could allow customers to browse product categories, examine item details, review prices, select products, provide the required order information, and submit orders through clearly defined steps. Consistent navigation and understandable instructions could reduce the effort required to use the system. Features such as order summaries and confirmation messages could also help customers understand whether their orders had been recorded successfully. If order tracking were included in the project's scope, status information could provide further clarity regarding order progress. These features represent proposed applications of the study's findings rather than functions directly tested by the researchers. Nevertheless, they are consistent with the importance of making the purchasing process accessible and manageable for potential customers.

The study also provided insights into the importance of understanding consumers' purchasing decisions beyond product availability and price. Although customers may consider affordability when purchasing pre-loved clothing, the findings indicated that attitudes, social norms, moral norms, and perceived behavioral control were also relevant to purchase intention [8]. This suggests that businesses should consider both the products they offer and the experience customers encounter when evaluating and purchasing them. For Margaux Collections, the ordering system should therefore provide more than a basic product catalog. It should also make product information easy to understand, present garments consistently, and allow customers to complete orders without unnecessary complications. Clear product descriptions, transparent pricing, condition classifications, and organized listings could help customers evaluate the suitability of individual items before making a decision.

Furthermore, the findings of Koay *et al.* [8] provided a basis for considering the relationship between system usability and customers' willingness to purchase pre-loved clothing. If customers experience difficulties locating products or understanding the ordering process, they may be less willing to proceed with a transaction. Therefore, the proposed system should be evaluated according to how effectively users can navigate the interface, access product details, and submit orders. User testing could help determine whether product descriptions are understandable, whether categories are appropriately organized, and whether customers can complete the required ordering steps without difficulty. Such evaluation would provide evidence of the system's actual usability rather than assuming that its digital format automatically improves customer experience.

The study also suggested that online second-hand clothing businesses should consider both the practical and psychological aspects of purchasing. From a practical perspective, customers need accessible product information, convenient navigation, and a manageable ordering process. From a psychological perspective, customers may be influenced by their attitudes toward second-hand clothing, their personal values, and the social environment in which purchasing decisions occur. For Margaux Collections, the proposed ordering system could address the practical aspects by providing organized product listings, accurate item descriptions, clear photographs, and straightforward order submission. These functions could support the purchasing process, while the presentation of pre-loved garments could help customers assess their attractiveness and suitability. However, the actual influence of these features on purchase intention would need to be evaluated among the business's intended users.

Overall, Koay, Cheah, and Lom [8] provided relevant empirical evidence concerning the factors that influence consumers' intention to purchase second-hand clothing. Their findings showed that attitudes, injunctive norms, descriptive norms, moral norms, and perceived behavioral control positively influenced purchase intention, while emotional value and environmental value positively influenced attitudes toward second-hand clothing [8]. These findings were applicable to the Margaux Collections Ordering System because they highlighted the importance of product appeal, personal values, social considerations, and ease of purchasing. The proposed system could address these considerations through organized product listings, accurate descriptions, clear photographs, condition labels, transparent pricing, and user-friendly navigation. By making pre-loved garments easier to evaluate and order, the system could provide a more accessible and informative shopping experience. Therefore, the study offered a useful foundation for developing an ordering system that considers both the functional requirements of online shopping and the behavioral factors associated with consumers' intention to purchase pre-loved clothing.

Mindful Consumption of Second-Hand Clothing

Mohammad, Quoquab, and Mohamed Sadom [9] examined mindful consumption of second-hand clothing by investigating the roles of electronic word-of-mouth (eWOM), consumer attitudes, and consumer engagement in purchasing behavior. Their study highlighted the importance of online information and consumer interactions in shaping perceptions of preloved fashion products. These findings were relevant to the development of the Margaux Collections Ordering System because customers purchasing clothing online may rely on information and experiences shared by other consumers when evaluating products and deciding whether to place an order.

The study emphasized the role of electronic word-of-mouth in influencing consumer perceptions and purchasing decisions [9]. Since customers cannot always physically inspect second-hand clothing before purchasing, they may depend on product descriptions, photographs, reviews, and feedback to evaluate an item's condition and suitability. For Margaux Collections, providing detailed product information and opportunities for customer feedback could help reduce uncertainty and enable customers to make more informed purchasing decisions. This would be particularly useful for preloved garments, where information about condition, appearance, and previous use may influence customer expectations.

Another important aspect of the study was the relationship between consumer engagement and purchasing behavior [9]. Engagement allows customers to interact with a business and participate more actively in the shopping process. In the proposed Margaux Collections Ordering System, this could be supported through customer feedback forms, product ratings, reviews, and communication channels. These features could allow customers to share their experiences, ask questions, and obtain additional information about products before completing their orders.

The study also provided insights into the importance of customer feedback in online shopping environments. Reviews and ratings may help prospective buyers understand other customers' experiences and evaluate the products offered by a seller. For Margaux Collections, a customer feedback feature could provide useful information about product quality, order processing, and overall service. However, reviews and ratings should be presented as customer opinions rather than guarantees of product quality. Combining customer feedback with accurate product descriptions and photographs could help make the ordering system more informative and transparent.

Furthermore, the findings were relevant to the development of interactive features for a preloved clothing marketplace. Customers may want to confirm that the garments they intend to purchase match the descriptions provided by the seller. Features such as customer inquiries, seller responses, and detailed garment condition information could help address these concerns. By providing accessible communication tools, the Margaux Collections Ordering System could help customers obtain relevant information before placing an order and potentially improve their shopping experience.

Consumer engagement could also contribute to the continuous improvement of the proposed system. Feedback collected from customers could help Margaux Collections identify common concerns, understand customer preferences, and determine which aspects of the ordering process need improvement. For example, customer comments could provide insights into the clarity of product descriptions, the usefulness of photographs, and the convenience of placing orders. These proposed functions could help the business improve its services and develop a more customer-oriented online platform.

The study was also relevant to an ordering system that offers both brand-new and preloved clothing. Customers may require different types of information depending on the products they intend to purchase. For preloved clothing, they may pay particular attention to the garment's condition, visible defects, and signs of wear. The system could therefore provide detailed product descriptions, clear photographs, and appropriate condition labels. When combined with customer feedback and communication features, these functions could help buyers compare available products and make purchasing decisions based on the information provided.

Moreover, the findings of Mohammad, Quoquab, and Mohamed Sadom [9] supported the importance of considering consumer attitudes and engagement when designing an online clothing platform. An effective ordering system should not only display available products and manage customer orders but should also provide accessible information that supports purchasing decisions. For Margaux Collections, integrating product listings, customer inquiries, feedback mechanisms, and order management into one platform could improve the organization of business operations while providing customers with a more convenient shopping experience.

Overall, Mohammad, Quoquab, and Mohamed Sadom [9] provided valuable insights into mindful consumption of second-hand clothing by examining electronic word-of-mouth, consumer attitudes, and consumer engagement. Their findings were relevant to the Margaux Collections Ordering System because they highlighted the importance of consumer information and participation in online shopping. These insights supported the proposed inclusion of detailed product descriptions, customer reviews, feedback mechanisms, and communication features. By incorporating these functions, the system could help customers make more informed purchasing decisions, encourage interaction between customers and the business, and provide a more transparent and user-friendly platform for ordering brand-new and preloved clothing.

Secondhand Clothing Purchase Intentions: Generation Z's Perspective

Herjanto, Amin, Purinton, and Lindle II [10] examined the factors influencing Generation Z consumers' intentions to purchase second-hand clothing. Their study investigated how consumers' attitudes toward clothing hygiene and fashion style, previous purchasing experiences, fashion innovativeness, materialism, and financial pressure influenced their intentions to purchase pre-loved garments. The researchers collected 180 usable survey responses from Generation Z participants and analyzed the data using partial least squares structural equation modeling (PLS-SEM). The findings indicated that attitudes toward second-hand clothing and previous purchasing experiences directly influenced purchase intention. The study also found that materialism, financial pressure, and fashion innovativeness influenced attitudes toward second-hand clothing as a fashion choice [10]. These findings were relevant to the Margaux Collections Ordering System because they demonstrated that younger consumers' purchasing decisions could be influenced by product perceptions, personal experiences, fashion preferences, and financial considerations.

One of the important findings of Herjanto *et al.* [10] was that consumers' attitudes toward second-hand clothing influenced their intention to purchase these products. The researchers considered attitudes related to both hygiene and fashion style, recognizing that consumers could evaluate pre-loved clothing according to different considerations. Hygiene concerns could affect whether consumers felt comfortable purchasing garments that had previously been owned by others, while fashion-related attitudes could influence whether they considered second-hand clothing attractive and suitable for their personal style. This finding was significant to Margaux Collections because customers might evaluate both the physical condition and visual appeal of the garments offered through the ordering system. Providing accurate condition descriptions, clear product photographs, and relevant information about each item could help customers assess the products before making purchasing decisions.

The study also highlighted the importance of previous purchasing experiences. Herjanto *et al.* [10] found that past experiences directly influenced second-hand clothing purchase intention, considering experiences associated with daily use and particular occasions. This suggests that consumers' familiarity with second-hand clothing may contribute to their willingness to purchase it again. Customers who have previously purchased and used pre-loved garments may have developed expectations about product quality, comfort, appearance, and value. For Margaux Collections, this finding supports the importance of providing a consistent and informative ordering experience. Accurate product listings, reliable order handling, and clear information about garment condition could help

customers understand what they are purchasing and establish realistic expectations. Although a well-designed system cannot guarantee repeat purchases, it could support a more dependable shopping experience.

Another important aspect of the study was the relationship between fashion style and consumers' attitudes toward second-hand clothing. Herjanto *et al.* [10] examined how consumers perceived second-hand clothing as a fashion choice rather than solely as previously owned merchandise. Their findings indicated that factors such as fashion innovativeness, materialism, and financial pressure influenced attitudes toward second-hand clothing style. This was relevant to Margaux Collections because customers might be interested in pre-loved garments for their appearance, uniqueness, and suitability for their personal preferences. The ordering system could support product evaluation by displaying clothing items through clear photographs and informative descriptions that highlight their style, color, brand, size, and other relevant characteristics. These features could help customers identify garments that match their fashion preferences.

Financial pressure was another factor examined in the study. Herjanto *et al.* [10] found that financial pressure influenced consumers' attitudes toward second-hand clothing as a fashion choice. Pre-loved garments may provide consumers with opportunities to explore different styles while managing their spending. However, consumers' decisions may also depend on whether the products meet their expectations regarding appearance, condition, and usefulness. For Margaux Collections, this finding emphasizes the importance of displaying prices clearly alongside product information. Customers should be able to assess the cost of an item in relation to its condition and characteristics. Transparent pricing and organized product listings could help customers compare available options and make purchasing decisions based on their preferences and budgets.

The study further examined fashion innovativeness, which relates to consumers' openness to new fashion ideas, styles, and trends. Herjanto *et al.* [10] identified fashion innovativeness as a factor influencing attitudes toward second-hand clothing style. This finding suggests that consumers who are interested in exploring fashion may consider pre-loved clothing as another way to discover distinctive garments and express their personal style. For Margaux Collections, the implication is that product presentation should make the available selection easy to explore. Organized categories, appealing photographs, and informative descriptions could help customers discover items that suit their interests. By making the product catalog accessible and understandable, the ordering system could support customers in exploring the range of available pre-loved clothing.

The findings were also relevant to product presentation in an online shopping environment. Customers using the Margaux Collections Ordering System would depend on the information displayed on the platform to evaluate products before placing orders. Unlike customers in physical stores, online shoppers may not be able to touch garments, examine fabric details directly, or inspect signs of wear in person. Consequently, product photographs and descriptions would be important sources of information. The system could provide images from appropriate angles and descriptions that identify relevant characteristics, including size, color, brand, material when known, price, and physical condition. For pre-loved garments, visible imperfections should also be disclosed where applicable. These features could help customers form more accurate expectations and make better-informed purchasing decisions.

Hygiene perceptions identified in the study were particularly relevant to the resale of clothing. Some customers may have concerns about cleanliness when considering garments that have previously been owned by another person. Such concerns could affect their attitudes toward second-hand clothing and their willingness to purchase it. For Margaux Collections, this highlights the importance of communicating product condition and any relevant garment-care information accurately. Where cleaning or garment preparation procedures are part of the business's actual practices, the business could provide truthful information about those procedures. The ordering system should avoid unsupported assurances and should ensure that any hygiene-related information reflects the actual condition and handling of the products. Clear and accurate information could help customers evaluate the items with greater confidence.

The study also supported the importance of developing an accessible and organized digital platform for the intended customers. Generation Z consumers who are familiar with online services may expect product information to be easy to access and navigation to be straightforward. If customers encounter confusing categories, incomplete listings, or complicated ordering steps, they may find it difficult to evaluate products and complete purchases. Therefore, the Margaux Collections Ordering System should allow users to browse products, review item details, check prices, and submit orders through a clear sequence of steps. Features such as understandable navigation, organized product categories, and order confirmation could help users understand the process and reduce avoidable difficulties during ordering.

Furthermore, the findings of Herjanto *et al.* [10] provided a basis for considering the relationship between product value and customer expectations. Consumers may evaluate second-hand clothing according to its fashion appeal, hygiene, affordability, and relevance to their previous experiences. Therefore, the ordering system should present these characteristics through accurate and consistent product information rather than relying on price alone. Customers could use the information to determine whether an item meets their expectations and fits their intended use. For Margaux Collections, this approach could improve the clarity of product listings and support a more informed purchasing process. The actual effect of these features on purchase intention, however, would need to be assessed through user feedback or system evaluation.

Overall, Herjanto, Amin, Purinton, and Lindle II [10] provided relevant insights into Generation Z's intentions to purchase second-hand clothing. Their study demonstrated that attitudes toward hygiene and fashion style, previous purchasing experiences, fashion innovativeness, materialism, and financial pressure were important considerations in understanding consumers' responses to pre-loved fashion [10]. These findings were applicable to the Margaux Collections Ordering System because they highlighted the importance of presenting products accurately, communicating their characteristics, displaying transparent prices, and providing an accessible ordering process. By incorporating clear photographs, detailed descriptions, condition information, organized categories, and straightforward ordering functions, the proposed system could help customers evaluate available garments according to their individual preferences and expectations. Therefore, this study provided a useful foundation for developing an ordering system that considers the fashion-related, experiential, and financial factors associated with Generation Z's interest in second-hand clothing.

Cross-Cultural Examination of Apparel Online Purchase Intention: SOR Paradigm Nam, Cho, and Kim (2021)

Nam, Cho, and Kim [11] examined consumers' online apparel purchase intentions using the Stimulus-Organism-Response (S-O-R) paradigm. Their study investigated how e-service quality, particularly website design quality and responsiveness, influenced consumer trust and purchasing intentions. The findings indicated that website design quality and responsiveness were associated with consumer trust, which served as an important predictor of purchase intention. These findings were relevant to the development of the Margaux Collections Ordering System because they emphasized that customers' purchasing decisions could be influenced not only by the clothing products offered but also by the quality, responsiveness, and usability of the online shopping platform.

The study highlighted the importance of website design quality in shaping customers' perceptions of an online apparel store [11]. A well-organized website could help customers locate products, understand product information,

and navigate the platform more conveniently. For Margaux Collections, this finding supported the need for a consistent and organized presentation of brand-new and preloved clothing. Product listings could include clear photographs, readable descriptions, prices, sizes, and relevant condition information. These features could help customers evaluate the available products and develop greater confidence in the ordering process.

Product presentation was particularly important for preloved clothing because customers could not physically examine garments before purchasing them online. Clear photographs could help customers assess the appearance, color, and design of an item, while detailed descriptions could provide information about its size, material, and condition. For previously owned garments, descriptions could also identify visible signs of wear or defects. Although Nam, Cho, and Kim [11] focused on e-service quality, trust, and purchase intention rather than specifically examining preloved clothing, their findings provided a useful basis for improving the quality of product information presented through the proposed system.

The study also emphasized the importance of responsiveness in online apparel shopping [11]. Responsiveness refers to the ability of an online retailer to respond appropriately to customer needs and inquiries. In the Margaux Collections Ordering System, this principle could be applied through accessible contact information, customer inquiry features, timely responses to questions, and clear communication regarding orders. Such functions could help customers obtain the information they need before making a purchase and could contribute to a more reliable shopping experience.

Another important finding was the role of consumer trust in influencing online purchase intention. Nam, Cho, and Kim [11] found that trust was a key factor connecting website design quality and responsiveness with consumers' intentions to purchase apparel online. This was relevant to Margaux Collections because customers needed confidence in the accuracy of product information and the reliability of the ordering process. A professional interface, transparent product details, clear ordering instructions, and appropriate communication features could help establish a more trustworthy online shopping environment.

The study further demonstrated the value of a convenient and user-friendly interface. When customers could navigate a website easily, they could spend less effort locating products and accessing important information. For the Margaux Collections Ordering System, clearly labeled menus, organized product categories, readable text, and straightforward navigation could make browsing more convenient. These design elements could also help customers with different levels of online shopping experience understand how to use the platform and complete their orders.

In addition, product categories and search functions could improve the organization of the ordering system. Customers could browse clothing according to product type or other available attributes, making it easier to find items that match their preferences. Search and filtering options could allow users to narrow product listings based on factors such as size, price, brand, or clothing category, provided that the relevant information was available in the system. These proposed features could reduce the effort required to locate suitable products and support a more efficient browsing experience.

The ordering process was another important consideration in the development of the system. Customers who had selected their preferred clothing items needed a clear and understandable method for submitting their orders. A complicated process could create confusion and discourage customers from completing their transactions. Therefore, the Margaux Collections Ordering System could provide straightforward order forms, clear instructions,

order summaries, and confirmation messages. These features could help customers understand the steps involved in ordering and provide greater clarity regarding the status of their transactions.

The findings also supported the importance of maintaining a consistent and professional online shopping environment. Website design quality and responsiveness were associated with consumer trust in the study [11]. For Margaux Collections, consistent product layouts, accurate descriptions, functional navigation, and accessible customer support could help communicate professionalism and reliability. These considerations could be especially valuable for preloved products, for which customers may have concerns about garment condition and whether an item matches its online description. Providing transparent information could help address these concerns and support more informed purchasing decisions.

Furthermore, the S-O-R paradigm provided a theoretical foundation for understanding how the proposed ordering system could influence customer behavior. In this framework, external stimuli, such as website design and service responsiveness, could influence an individual's internal state, represented in the study by consumer trust. Trust could then influence the individual's behavioral response, including the intention to purchase apparel online [11]. Applied to Margaux Collections, this framework suggested that improving the interface and responsiveness of the system could help establish customer trust and potentially encourage purchasing intentions. However, the actual effects of these features would depend on how customers perceived and used the completed system.

Overall, Nam, Cho, and Kim [11] provided relevant evidence concerning the relationship between website design quality, responsiveness, consumer trust, and online apparel purchase intention. Their findings supported the development of a well-organized, responsive, and user-friendly Margaux Collections Ordering System. The proposed system could incorporate clear product listings, detailed descriptions, organized categories, convenient navigation, accessible communication features, and a straightforward ordering process. By integrating these features, the platform could provide a more convenient and trustworthy shopping experience for customers purchasing brand-new and preloved clothing. The study therefore offered a useful theoretical and empirical basis for considering interface quality and customer trust in the design of the proposed ordering system.

Thrift Shopping for Clothes: To Treat Self or Others? Park, Kwon, Zaman, and Song (2020)

Park, Kwon, Zaman, and Song [12] examined the motivations influencing consumers' thrift shopping behavior, particularly their reasons for purchasing second-hand clothing. Their study explored the personal and social factors that encouraged consumers to engage in thrift shopping and identified motivations associated with the value consumers obtained from the experience. The researchers highlighted treasure hunting, seeking name-brand products, and responsible citizenship as important motivations related to thrift shopping [12]. These findings were relevant to the development of the Margaux Collections Ordering System because they demonstrated that consumers could be attracted to pre-loved clothing for reasons beyond affordability. The opportunity to discover distinctive garments, obtain desirable brands, and participate in more responsible consumption could also influence consumers' interest in second-hand fashion. Therefore, the proposed ordering system needed to present products in a way that allowed customers to evaluate their characteristics and discover items that matched their preferences.

One important finding of Park *et al.* [12] was the role of treasure hunting in thrift shopping. Treasure hunting refers to the enjoyment consumers may experience when searching for and discovering desirable or distinctive products. Unlike conventional retail shopping, where products are often displayed in standardized collections, thrift shopping can involve finding individual garments with unique styles, brands, and characteristics. This experience may

encourage consumers to explore a wider range of products in search of items that suit their interests. The finding was relevant to Margaux Collections because its pre-loved clothing inventory could include garments with different styles, appearances, sizes, and conditions. The ordering system could support product discovery by providing organized listings, clear photographs, and informative descriptions. These features could help customers browse available garments and identify products that appeal to their individual preferences.

The concept of treasure hunting also emphasized the importance of an organized and engaging browsing experience. Customers who enjoy discovering unique clothing may appreciate a platform that allows them to explore products conveniently. If product information is incomplete or listings are difficult to navigate, customers may experience difficulty identifying items that interest them. For Margaux Collections, a centralized product catalog could make browsing more systematic while allowing customers to explore the available collection. Organizing products according to categories such as clothing type, size, brand, or other relevant characteristics could help users narrow their choices. Regularly updating the catalog could also make newly available products easier to discover. Although an online ordering system cannot fully reproduce the experience of physically searching through a thrift store, it can provide an accessible way for customers to explore second-hand clothing.

Another important motivation identified by Park *et al.* [12] was consumers' interest in seeking name-brand products. Recognizable brands may attract thrift shoppers because of their designs, reputation, or perceived value. Consumers may view second-hand shopping as an opportunity to obtain clothing from preferred brands at prices that are potentially more affordable than those of brand-new products. This finding was relevant to Margaux Collections because brand information could help customers evaluate available garments and identify products that match their preferences. The ordering system could include brand names in product listings when the information is available and verified. Displaying brand information alongside photographs, prices, sizes, and condition descriptions could help customers compare products and make more informed decisions. However, the system should not claim that an item is authentic unless there is an appropriate basis for verifying its authenticity.

The study also identified responsible citizenship as an important motivation associated with thrift shopping [12]. Responsible citizenship reflects a concern for broader responsibilities and the consequences of individual consumption choices. In the context of second-hand clothing, consumers may associate thrift shopping with the reuse of existing garments and more responsible consumption practices. Purchasing pre-loved clothing provides an opportunity to extend the useful life of garments by allowing them to be used by additional owners. This finding was relevant to Margaux Collections because the business focuses on selling previously owned clothing. The ordering system could support this activity by making available garments easier to discover, evaluate, and purchase. It could also provide factual information about clothing reuse, helping customers understand the role of second-hand fashion in keeping existing garments in circulation.

The distinction between personal and other-oriented motivations was significant to the proposed Margaux Collections Ordering System. Personal motivations may include the satisfaction of finding a desirable item, discovering distinctive styles, or obtaining a preferred brand. Other-oriented motivations may involve concerns about responsible consumption and the continued use of existing products. These motivations suggest that customers may approach thrift shopping with different priorities. Some customers may focus on price and personal style, while others may also consider the broader implications of purchasing pre-loved clothing. Therefore, the ordering system should provide comprehensive product information that accommodates different customer needs. Clear prices and condition descriptions could support practical evaluations, while photographs and brand details could help customers assess style and product appeal. Information about clothing reuse could also communicate the purpose of the business without assuming that all customers share the same motivations.

Product presentation was another important consideration derived from the study. In physical thrift stores, customers can directly inspect garments, compare items, and discover products while browsing through clothing racks. In an online environment, customers depend more heavily on photographs, descriptions, and organized product listings. The Margaux Collections Ordering System could provide a digital catalog in which each garment has its own product page containing relevant information. Photographs taken from appropriate angles could help customers

examine an item's appearance, while descriptions could provide details about its size, color, brand, price, and physical condition. Information about visible imperfections should also be included when applicable. These features could help customers evaluate products more effectively and develop realistic expectations before placing an order.

Furthermore, the study supported the importance of making product discovery convenient. Since thrift shoppers may search for specific brands or enjoy discovering unexpected items, an online ordering system should allow users to explore the available inventory efficiently. Product categories and search or filtering functions, where included in the system's scope, could help customers narrow their choices according to their preferences. For example, customers could search for particular brands or browse products according to clothing type and size. These functions could reduce the effort required to locate relevant items and make the catalog easier to navigate. By improving access to product information, the system could support customers who prefer to search for specific items as well as those who wish to explore the collection more broadly.

The findings also had implications for inventory management. Pre-loved clothing often consists of individual garments available in limited quantities, and an item may not have an identical replacement once it has been sold. Maintaining accurate inventory records is therefore important for ensuring that customers see reliable product availability information. The Margaux Collections Ordering System could help the business maintain organized records of each garment and update availability when an item is added, reserved, or sold, depending on the implemented system functions. Accurate records could reduce the likelihood of accepting orders for unavailable products and help the business owner monitor the current inventory. These proposed system features represent practical applications of the study's insights rather than functions directly evaluated by Park *et al.* [12].

Another implication of the study concerned the relationship between product discovery and the overall customer experience. Treasure hunting suggests that discovering a desirable item can be an important part of thrift shopping [12]. An online ordering system could support this motivation by allowing customers to explore available products through organized categories, clear photographs, and consistent descriptions. Customers could compare garments and identify items that suit their preferences without needing to visit a physical location. A clear and accessible interface could make browsing more convenient, especially when the inventory includes products with different styles, brands, sizes, and conditions. By combining product organization with informative listings, Margaux Collections could provide a shopping experience that supports both purposeful searching and the discovery of distinctive garments.

The study also indicated that thrift shopping should not be presented exclusively as a way to purchase inexpensive clothing. Although affordability may be important to many consumers, the motivations identified by Park *et al.* [12] suggest that shoppers may also value treasure hunting, name-brand products, and responsible citizenship. For Margaux Collections, this means that product listings should communicate the overall characteristics of each garment instead of focusing only on its price. Customers should be able to assess the item's appearance, condition, brand information, and cost before deciding whether it meets their needs. Providing this information clearly could help customers understand the value of individual products and make more informed purchasing decisions.

Overall, Park, Kwon, Zaman, and Song [12] provided useful insights into the motivations associated with thrift shopping behavior. Their study identified treasure hunting, seeking name-brand products, and responsible citizenship as important motivations for consumers engaging in thrift shopping [12]. These findings were relevant to the Margaux Collections Ordering System because they demonstrated that consumers may be interested in pre-loved clothing for personal, product-related, and responsibility-oriented reasons. The proposed system could address these considerations through organized product categories, clear photographs, accurate descriptions, brand information where available, transparent pricing, and updated product availability. By making the collection easier to browse and individual garments easier to evaluate, the system could provide a more convenient and informative shopping experience. Therefore, the study supported the development of an ordering system that presents pre-loved clothing as products that may offer personal value, distinctive characteristics, and opportunities for continued garment use.

Trust Management in Electronic Marketplaces — Kim and Ahn (2007)

Kim and Ahn [13] examined trust management in electronic marketplaces and investigated how buyers developed trust in online marketplaces and individual sellers. Their study emphasized that trust was an important factor in online transactions because buyers often interacted with sellers with whom they had little or no previous experience. The researchers examined how marketplace reputation, website usability, website security, seller expertise, and buyers' previous transaction experiences influenced trust and transaction intentions. These findings were relevant to the development of the Margaux Collections Ordering System because customers needed to feel confident in the platform and the business before placing orders for brand-new and preloved clothing.

The study explained that buyers' trust in an electronic marketplace could be influenced by the reputation and characteristics of the website, as well as their experiences with the seller [13]. A website that was easy to use and provided a secure and organized environment could help customers develop greater confidence in the platform. For Margaux Collections, this finding supported the development of a professional and user-friendly ordering system with organized product listings, accurate product descriptions, clear ordering instructions, and accessible business information. These features could help customers understand the purchasing process and assess the reliability of the business.

Another important finding was that buyers evaluated trust in the marketplace and trust in individual sellers separately [13]. Customers might trust the online platform's ability to organize transactions while still needing confidence in the seller's reliability. In the context of Margaux Collections, the ordering system could support this distinction by presenting accurate business information, complete product details, and clear procedures for submitting and processing orders. Providing consistent information about available clothing, prices, and garment conditions could help customers evaluate products and understand what to expect from their purchases.

The study also highlighted the importance of seller characteristics in building consumer trust. Kim and Ahn [13] identified seller expertise as a factor associated with trust in an electronic marketplace. For Margaux Collections, this principle could be applied through the accurate presentation of product information, appropriate handling of customer inquiries, and consistent management of orders. Since the business offers both brand-new and preloved clothing, product listings could include relevant details about size, material, price, and condition. For preloved garments, photographs and descriptions could also disclose visible signs of wear or defects. These practices could help customers make informed decisions and establish more realistic expectations before purchasing.

Website usability and security were also important considerations in the study [13]. A usable ordering system could allow customers to browse products, view item details, and submit orders without unnecessary difficulty. For Margaux Collections, clearly labeled menus, organized categories, readable product information, and a straightforward checkout process could improve usability. Security-related measures could include appropriate protection of customer information, controlled access to administrative functions, and careful handling of transaction records. These proposed measures could help support the reliability of the ordering system and protect information associated with customer transactions.

The researchers further considered the role of buyers' previous experiences in the development of trust and their intention to transact [13]. Positive experiences with an electronic marketplace could influence how customers evaluate subsequent transactions. For Margaux Collections, accurate order processing, clear communication, and reliable updates regarding order status could help provide a more consistent customer experience. Maintaining organized transaction records could also help the business review previous orders, monitor outstanding transactions, and address customer concerns. Over time, consistent service could contribute to customers' confidence in the business, although repeat purchases would also depend on product availability, prices, and individual preferences.

The study was particularly relevant to payment verification in the Margaux Collections Ordering System. Since the proposed business process involved advance payments through GCash, the system needed a reliable way to record and review payment information before an order was processed. Requiring customers to submit their GCash reference numbers could help associate payment details with the corresponding orders. The business could then use the reference number, together with the available transaction information, to check whether the payment had been received. This proposed feature could improve the organization of payment records and help distinguish between orders awaiting verification and those whose payments had been confirmed.

The use of GCash reference numbers could also help reduce reliance on informal payment confirmations, such as screenshots sent through Messenger. Although screenshots may provide supporting information, they should not be treated as conclusive proof that a payment has been received. A reference number could serve as an identifier for checking the corresponding transaction, but it would still need to be verified against the available payment records. For Margaux Collections, this process could help reduce recording errors, improve accountability, and support more systematic order processing. However, these payment procedures are applications proposed for the ordering system and were not specific features tested by Kim and Ahn [13].

Furthermore, the study supported the importance of transparency in online transactions. Customers need sufficient information to evaluate products and understand the procedures involved in placing an order. This is especially relevant to preloved clothing because garments may differ in condition, appearance, and signs of previous use. Providing accurate photographs, detailed descriptions, prices, and condition information could help reduce uncertainty before purchase. Clear payment instructions, order confirmations, and understandable order status updates could further help customers follow the progress of their transactions and understand when their orders had been accepted or processed.

Overall, Kim and Ahn [13] provided a theoretical and empirical basis for understanding the role of trust in electronic marketplaces. Their study showed that marketplace reputation, website usability, website security, seller expertise, and buyers' previous experiences were important considerations in trust formation and transaction intentions. These findings were relevant to the Margaux Collections Ordering System because customers needed confidence in both the online platform and the business offering the clothing products. The proposed inclusion of organized product listings, accurate garment information, accessible communication, secure handling of customer data, transaction records, and GCash payment verification could help support transparency and accountability. By applying these principles, the Margaux Collections Ordering System could provide a more reliable and user-friendly environment for ordering brand-new and preloved clothing.

Online Feedback Mechanism in Electronic Marketplaces Dellarocas (2003)

Dellarocas [14] examined the role of online feedback mechanisms in electronic marketplaces in the study titled “The Digitization of Word-of-Mouth: Promise and Challenges of Online Feedback Mechanisms,” published in 2003. The study discussed how online feedback systems could help establish trust, reduce uncertainty, and encourage cooperation between buyers and sellers in digital transactions. Online feedback mechanisms allowed consumers to share their experiences through ratings, reviews, and evaluations, providing information that other customers could consider before making purchasing decisions. The study was relevant to the development of the Margaux Collections Ordering System because customers purchasing pre-loved clothing online might have limited opportunities to physically inspect products or directly evaluate a seller's reliability before completing a transaction. Therefore, feedback mechanisms could serve as an additional source of information to help customers assess the credibility of the business and make more informed purchasing decisions.

Dellarocas [14] explained that online feedback could provide valuable information in situations where buyers and sellers interacted without direct physical contact. In traditional shopping environments, customers could examine clothing, assess its condition, and communicate directly with the seller before purchasing an item. In contrast, online customers depended largely on product photographs, descriptions, and other information provided through the

platform. Feedback from previous customers could offer additional insights into their actual purchasing experiences and help potential buyers evaluate the reliability of the seller. This was particularly relevant to Margaux Collections because customers might have concerns regarding the quality, appearance, condition, and accuracy of information provided about pre-loved clothing. By providing a structured way for customers to share their experiences, the ordering system could help address some of these concerns and support greater transparency.

One important implication of the study was the relationship between online feedback and seller reputation. Feedback mechanisms could help potential buyers evaluate a seller's previous performance and develop expectations regarding future transactions [14]. Consistently positive feedback could contribute to a favorable perception of a seller, while negative feedback could alert customers to possible problems. For Margaux Collections, this principle supported the possible inclusion of a customer rating and review feature in the proposed ordering system. Customers who had completed purchases could provide feedback regarding product accuracy, clothing condition, seller communication, and their overall ordering experience. Future customers could then use these reviews as an additional reference when deciding whether to purchase an item.

Online feedback could also encourage sellers to maintain consistent service quality. When customers could evaluate their purchasing experiences, sellers had an incentive to consider how their actions affected their reputation in the marketplace [14]. This could encourage more accurate product descriptions, clearer communication, and more careful order processing. For Margaux Collections, customer feedback could help the business owner determine whether the products received matched the photographs and descriptions displayed in the system. Reviews could also identify recurring customer concerns and provide information that could be used to improve product listings and customer service. However, feedback would be most useful when it reflected genuine purchasing experiences and was managed through clear and fair procedures.

The importance of feedback was particularly significant for pre-loved clothing because individual garments could differ in their physical condition and show varying signs of previous use. Customers might want to know whether the received item matched its stated condition and whether visible imperfections had been properly disclosed. Although photographs and detailed descriptions could help customers evaluate products before purchasing, feedback from previous buyers could provide an additional perspective on their transactions. The Margaux Collections Ordering System could therefore include a feedback mechanism that allowed eligible customers to evaluate their experiences after completing an order. This feature could help prospective buyers make more informed decisions while encouraging the business to maintain accurate and transparent product information.

Dellarocas [14] also discussed challenges associated with online feedback mechanisms. Although customer reviews could help establish trust, they were not automatically reliable in every situation. Feedback could be incomplete, biased, misleading, or insufficient to represent a seller's overall performance. Consequently, an online ordering system should not rely exclusively on ratings as proof of product quality or seller reliability. For Margaux Collections, customer reviews should complement other measures, including accurate product descriptions, clear photographs, transparent pricing, condition labels, and organized order records. Combining these features could give customers a more complete basis for evaluating the products and the purchasing process.

Another important implication of the study was the potential of feedback to reduce information asymmetry between buyers and sellers. In online transactions, sellers generally had direct access to the products they offered, while buyers had to evaluate those products through the information available on the platform. Feedback from previous transactions could provide additional perspectives based on actual purchasing experiences [14]. Within the Margaux Collections Ordering System, customers could use product descriptions to assess an item's stated characteristics and consult reviews to learn about other buyers' experiences. This combination could help customers evaluate whether the business provided accurate information and managed orders appropriately.

Customer feedback could also serve as a source of information for improving the operations of Margaux Collections. Reviews and comments might reveal recurring problems involving product descriptions, photographs, ordering procedures, communication, or order fulfillment. By examining these comments, the business owner could identify

aspects of the service that required improvement. For example, repeated concerns about incomplete product information could indicate the need to standardize product descriptions, while comments about unclear ordering procedures could indicate the need to improve system instructions. These applications represent potential uses of customer feedback in the proposed system based on the broader principles discussed by Dellarocas [14].

Furthermore, an organized feedback mechanism could support accountability by providing a record of customers' experiences. When feedback was associated with completed transactions, it could give the business owner useful information about service performance over time. The owner could review recurring concerns, identify strengths, and assess whether changes to the ordering process improved the customer experience. For Margaux Collections, this could complement order management by providing a more systematic approach to monitoring customer satisfaction. Appropriate safeguards would still be necessary to protect customer information, prevent abusive reviews, and ensure that feedback was handled fairly.

The study was also relevant to creating a professional and trustworthy online shopping experience. Customers might hesitate to purchase from an unfamiliar seller if they had little information about the business's previous transactions. A system that presented accurate product information, clear ordering instructions, and genuine customer feedback could help address some of this uncertainty. For Margaux Collections, combining organized product listings, transparent condition labels, reliable order records, and customer reviews could make the purchasing process easier to understand. These features could contribute to customer confidence, although they could not guarantee that every transaction would be free from problems.

Overall, Dellarocas [14] provided a relevant theoretical foundation for considering online feedback mechanisms in the Margaux Collections Ordering System. The study explained how digital feedback could support trust, cooperation, seller reputation, and information sharing in electronic marketplaces [14]. These insights supported the potential inclusion of customer ratings and reviews as complementary features of the proposed system. By allowing customers to share genuine purchasing experiences, the system could provide future buyers with additional information and help the business owner identify opportunities for improvement. Combined with accurate product descriptions, clear photographs, transparent pricing, and organized order management, customer feedback could contribute to a more informative, accountable, and trustworthy online shopping experience for pre-loved clothing.

Trust, Perceived Usefulness, and Perceived Ease of Use in Online Shopping — Gefen, Karahanna, and Straub (2003)

Gefen, Karahanna, and Straub [15] examined the relationship between trust and the Technology Acceptance Model (TAM) in online shopping. Their study integrated trust with perceived usefulness and perceived ease of use to explain consumers' acceptance of electronic-commerce websites. The researchers emphasized that consumers' decisions to use online shopping platforms were influenced not only by the usefulness and simplicity of the system but also by their trust in the online seller. These findings were relevant to the development of the Margaux Collections Ordering System because customers needed a convenient, understandable, and reliable platform for browsing products and placing orders for brand-new and preloved clothing.

The study highlighted the importance of perceived usefulness in determining whether consumers would accept and use an online shopping system [15]. Perceived usefulness refers to the extent to which users believe that a particular technology helps them accomplish their tasks more effectively. In an online shopping environment, a useful system allows customers to access product information, compare available items, and complete orders conveniently. For Margaux Collections, this principle supported the development of a system that could centralize product listings, customer ordering, and transaction information. By providing these functions in one platform, the proposed system could make the purchasing process more organized and accessible to customers.

Another important concept examined in the study was perceived ease of use. Gefen, Karahanna, and Straub [15] explained that users' acceptance of online shopping technology was influenced by how easy they perceived the system to be. A platform with understandable instructions, clear navigation, and straightforward functions could reduce the effort required to complete shopping activities. For the Margaux Collections Ordering System, this finding supported the use of a simple interface that would allow customers to browse products, view item details, and submit orders without unnecessary difficulty. A user-friendly design could also help customers who had limited experience with online ordering platforms.

The study further supported the importance of clear navigation and accessible system functions. When customers could easily identify product categories, locate relevant information, and understand the ordering process, they could interact with the platform more efficiently. In contrast, complicated navigation and unclear instructions could create confusion and discourage continued use. For Margaux Collections, clearly labeled menus, organized product listings, and straightforward ordering instructions could help simplify the shopping experience. These features could also reduce the need for customers to seek assistance through separate communication channels when completing basic ordering tasks.

Product browsing was another area in which perceived ease of use and usefulness could be applied. The Margaux Collections Ordering System could organize clothing products into categories that allow customers to find items according to their preferences. Separating brand-new clothing from preloved clothing could help customers identify the type of products they wanted to purchase. Search and filtering functions could further assist users in locating products based on available attributes such as size, price, brand, and clothing category. These proposed features could reduce the time and effort required to browse the available inventory and make the system more convenient to use.

The study also provided a basis for emphasizing the importance of complete and understandable product information. Customers need sufficient information to evaluate products before making purchasing decisions. Photographs, prices, sizes, descriptions, and garment condition details could help users understand the characteristics of the clothing being offered. This was particularly relevant to preloved garments because customers could not physically inspect the items before ordering. By providing clear and accessible product information, the Margaux Collections Ordering System could help customers compare available products and make more informed decisions.

Trust was another central factor in the study by Gefen, Karahanna, and Straub [15]. Trust is important in online shopping because customers may be uncertain about the reliability of sellers and the fulfillment of transactions conducted through the internet. When customers believe that an online seller is dependable, they may be more willing to share the information necessary for a transaction and proceed with a purchase. For Margaux Collections, trust could be supported by accurate product descriptions, transparent prices, clear ordering procedures, appropriate handling of customer information, and reliable payment verification. These measures could help customers understand how transactions were handled and what they could expect from the business.

The study's findings were also relevant to the checkout and ordering process. A complicated checkout procedure could make it difficult for customers to complete their purchases and could negatively affect their perceptions of the system. Therefore, the Margaux Collections Ordering System could provide a straightforward sequence of steps for reviewing selected products and submitting orders. An order summary could display relevant information, including product names, quantities, prices, and other necessary details before submission. This function could allow customers to review their selections, identify possible errors, and confirm their order information before proceeding.

The design of the Margaux Collections Ordering System could therefore reflect the combined principles of usefulness, ease of use, and trust. The system could provide practical shopping functions while maintaining a simple and understandable interface. Organized product categories, clear product photographs, complete descriptions, accessible prices, and straightforward ordering procedures could make the platform useful to customers. At the same time, transparent transaction procedures and appropriate payment verification could help support customer

confidence. Applying these principles could improve the overall usability of the proposed system while addressing the informational and operational needs of the business.

Furthermore, the study provided a theoretical basis for reducing unnecessary communication between customers and the seller. When customers could independently access product information and submit orders through a centralized platform, they would have less need to repeatedly ask for basic details through messaging applications. Product availability, prices, sizes, and descriptions could be presented directly within the system, allowing customers to review the information before making a purchase. This could make the ordering process more convenient for customers and help the business manage inquiries and orders more systematically.

Overall, Gefen, Karahanna, and Straub [15] provided important theoretical support for the development of the Margaux Collections Ordering System by demonstrating the relevance of trust, perceived usefulness, and perceived ease of use to online shopping acceptance. Their research supported the idea that customers' acceptance of an online shopping platform depended on both its practical value and their confidence in using it. These concepts were relevant to the proposed system because customers needed to browse clothing, evaluate product information, submit orders, and complete transactions through an accessible and reliable platform. By incorporating organized product categories, detailed product displays, straightforward navigation, clear ordering instructions, order summaries, and appropriate payment verification, the Margaux Collections Ordering System could provide a more convenient and trustworthy shopping experience for customers purchasing brand-new and preloved clothing.

Consumer Acceptance of Electronic Commerce — Pavlou (2003)

Pavlou [16] investigated consumer acceptance of electronic commerce by examining how trust, perceived risk, perceived usefulness, and perceived ease of use influenced consumers' willingness to engage in online transactions. The study explained that consumers could experience uncertainty when purchasing products electronically because they could not always inspect products physically or directly verify the reliability of the seller. Such uncertainty could involve concerns about product accuracy, payment security, personal information, and order fulfillment. Using two empirical studies, Pavlou examined the relationships among these factors and consumers' intentions to transact online. The findings supported the importance of trust and perceived risk in understanding electronic-commerce acceptance [16]. These findings were relevant to the development of the Margaux Collections Ordering System because customers needed to feel confident when purchasing brand-new and preloved clothing through an online platform.

One of the major contributions of the study was its explanation of how trust could influence consumers' willingness to conduct online transactions. Customers could be more willing to purchase products when they believed that the seller was reliable, the information provided was accurate, and the transaction would be handled appropriately. Conversely, uncertainty about the seller or the purchasing process could discourage customers from completing their orders. For Margaux Collections, this emphasized the importance of developing an ordering system that presented product information clearly and provided an organized and understandable purchasing process. Accurate product details, transparent pricing, and clear ordering instructions could help customers understand what they were purchasing and what to expect after submitting an order.

Pavlou [16] also examined perceived risk as an important factor in electronic commerce. Perceived risk refers to consumers' concerns about the possible negative consequences of conducting a transaction. In online shopping, these concerns may involve receiving a product that differs from its description, experiencing problems with

payment, or encountering difficulties with order fulfillment. Such concerns are particularly relevant to preloved clothing because customers cannot physically inspect an item before purchasing it. They may be uncertain about the garment's actual condition, appearance, size, or quality. Therefore, the Margaux Collections Ordering System should provide clear product photographs, accurate descriptions, condition labels, and complete pricing information. These features could help reduce uncertainty and allow customers to make more informed purchasing decisions.

Another important aspect of Pavlou's study was the role of perceived usefulness and perceived ease of use in online transaction acceptance [16]. Perceived usefulness refers to the extent to which consumers believe that a system helps them accomplish their goals, while perceived ease of use concerns how easily they can operate the system. These concepts were applicable to Margaux Collections because customers would need to browse products, review item details, select their preferred clothing, and submit orders without unnecessary difficulty. A well-organized interface, understandable navigation, and a straightforward ordering procedure could make the system more convenient for customers. By reducing the effort required to complete an order, the system could provide a more accessible and satisfactory shopping experience.

The study also supported the importance of presenting transaction information in a clear and consistent manner. Customers should be able to review the products they selected, the corresponding prices, and the details of their orders before submitting a transaction. An order summary could help customers identify possible mistakes and confirm that their selections were correct. Furthermore, an order confirmation could inform customers that their requests had been received by the system. For Margaux Collections, these functions could improve transparency throughout the ordering process and reduce confusion between customers and the business owner.

Payment verification was another relevant consideration in applying Pavlou's findings to the proposed system. Since Margaux Collections used GCash for advance payments, the ordering process needed a structured method of recording payment-related information. A field for the GCash reference number could help associate a customer's payment with the appropriate order. The system could also allow the business owner to record whether a payment had been verified. However, recording a reference number alone would not establish that a payment was successful; the business owner would still need to verify the transaction through the appropriate payment records. This approach could help organize payment information, reduce confusion between orders, and improve transaction monitoring.

In addition, the study provided a basis for considering the importance of order status information. Customers may become uncertain when they do not know whether their orders have been received, whether their payments have been verified, or whether their items are ready for pickup or delivery. The Margaux Collections Ordering System could address this concern by maintaining clear order statuses and updating them as the transaction progresses. These updates could help customers understand the current stage of their purchases while allowing the business owner to monitor outstanding orders more efficiently. By keeping order information organized and accessible, the system could support more consistent communication between the business and its customers.

The findings were also relevant to the management of customer and transaction records. Maintaining accurate records of orders, payment verification, and order fulfillment could help the business monitor its daily activities and identify discrepancies. These records could provide a clearer basis for checking whether an order had been processed correctly and whether the corresponding payment had been verified. For Margaux Collections, organized transaction records could reduce reliance on informal messages and scattered manual documentation. They could also support accountability by making it easier to review the details of individual transactions when questions or problems arose.

Furthermore, Pavlou's study highlighted the importance of considering both system functionality and customer confidence when developing an electronic-commerce platform [16]. A system might provide convenient ordering features, but customers could still hesitate to use it if product information was incomplete or the transaction process was unclear. Similarly, a trustworthy seller could encounter difficulties if the ordering platform was confusing or difficult to navigate. Therefore, the Margaux Collections Ordering System needed to combine usability with transparency and reliable transaction management. Clear product listings, accurate condition descriptions,

understandable order summaries, payment verification records, and order status updates could collectively contribute to a more dependable online purchasing experience.

Overall, Pavlou [16] provided a relevant theoretical and empirical foundation for the development of the Margaux Collections Ordering System by demonstrating the importance of trust, perceived risk, perceived usefulness, and perceived ease of use in electronic commerce. The study suggested that consumers' willingness to transact online was influenced not only by the availability of products but also by their perceptions of the platform and the risks associated with purchasing electronically [16]. These insights supported the proposed system's emphasis on transparent product information, clear pricing, user-friendly navigation, organized order records, payment verification, and order status monitoring. By integrating these features, the Margaux Collections Ordering System could help reduce customer uncertainty, improve the clarity of transactions, and provide a more convenient and trustworthy process for purchasing preloved and brand-new clothing.

Consumer Motivations for Purchasing Second-Hand Products Guiot and Roux (2010)

Guiot and Roux [17] investigated the motivations behind consumers' decisions to purchase second-hand products instead of new alternatives. Their study developed a framework for understanding second-hand shopping motivations and identified economic, recreational, and critical dimensions of consumer behavior. The findings suggested that consumers could be attracted to second-hand products for several reasons, including affordability, product uniqueness, personal enjoyment, and the opportunity to discover items that differed from those available in conventional retail stores. These findings were relevant to the Margaux Collections Ordering System because they demonstrated that preloved clothing could appeal to customers for reasons beyond lower prices.

The study highlighted the importance of economic motivations in second-hand shopping [17]. Consumers may consider used products as practical alternatives that allow them to obtain desired items at more affordable prices. For Margaux Collections, this finding supported the need to display product prices clearly and provide sufficient information about the condition and characteristics of each garment. By presenting brand-new and preloved clothing with transparent pricing and detailed descriptions, the ordering system could help customers compare available products and determine which items best suited their preferences and budgets.

Guiot and Roux [17] also examined recreational and product-related motivations associated with second-hand shopping. Some consumers may enjoy searching for distinctive products or discovering items that are not readily available in conventional retail stores. Preloved clothing can offer different styles, designs, brands, and fashion choices that appeal to individual preferences. This finding supported the presentation of preloved garments as distinctive fashion products rather than merely inexpensive alternatives to brand-new clothing. Through organized product listings and clear photographs, Margaux Collections could help customers discover clothing items that matched their personal styles.

Personal and emotional motivations could also influence second-hand consumption. Some shoppers may enjoy the experience of searching for products with distinctive characteristics or associations. For Margaux Collections, detailed product descriptions could highlight relevant information such as brand, style, color, material, and other notable features. These details could help customers evaluate the characteristics of each item and understand its potential appeal. Providing complete product information could therefore improve the shopping experience by allowing customers to assess products according to both practical needs and personal preferences.

The study also provided a basis for understanding the relationship between second-hand shopping and alternative consumption practices [17]. Purchasing used clothing allows garments to be transferred to new owners rather than being discarded immediately. For Margaux Collections, the resale of wearable clothing could support the continued use of existing garments. The ordering system could communicate this aspect of the business by including

sustainability-related information that explains the purpose of clothing reuse. Such information could help customers recognize the potential environmental value of purchasing preloved products, although the actual environmental benefits would depend on factors such as garment use and whether second-hand purchases replace purchases of new clothing.

Another important consideration was product availability. Second-hand products may include distinctive items that customers cannot easily find in conventional retail stores. However, their availability may be limited because the business may have only one unit of a particular preloved garment. For Margaux Collections, maintaining accurate and updated product listings could help customers identify which items were available for purchase. Organized categories, product photographs, and clear availability information could make it easier for customers to browse the inventory and select suitable products.

The study also supported the importance of providing accurate information about second-hand products. Because preloved garments may differ in condition, customers need sufficient details to assess their quality and suitability before ordering. Items may have visible signs of wear, stains, fading, loose threads, or other imperfections. The Margaux Collections Ordering System could address these concerns by including clear condition descriptions and photographs that accurately represent each item. Disclosing visible defects could help customers establish realistic expectations and reduce potential misunderstandings.

This principle was particularly relevant because Margaux Collections offered both brand-new and preloved clothing. Customers needed to distinguish between the two categories to compare their prices, characteristics, and conditions accurately. Consistent product labels could help prevent confusion and make the shopping experience more transparent. Product pages could include information about size, brand, material, price, and condition, allowing customers to evaluate available garments based on their individual needs.

Overall, Guiot and Roux [17] provided valuable insights into the different motivations behind second-hand shopping. Their research demonstrated that consumers' decisions could involve economic considerations, recreational experiences, product characteristics, and other motivations associated with alternative consumption. These findings were relevant to the Margaux Collections Ordering System because they supported the presentation of preloved clothing as an accessible and distinctive fashion option. By providing transparent prices, accurate product descriptions, clear photographs, condition labels, and organized product categories, the proposed system could help customers evaluate available garments and make informed purchasing decisions.

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Perceived Value and Trust in Online Resale Fashion, Zhang, Xu, and Wang (2023)

The study examined how perceived value and seller trust influenced consumers' purchasing decisions in the online second-hand fashion market. It focused on how buyers evaluated the quality, price, and emotional value of preloved clothing before deciding whether to make a purchase. The findings suggested that consumers were more willing to purchase second-hand fashion products when they believed that the items offered reasonable value for their price and condition. These findings were relevant to the development of the Margaux Collections Ordering System because customers needed sufficient information to evaluate the value of brand-new and preloved clothing before placing their orders.

The study emphasized the importance of perceived value in online second-hand shopping. Consumers considered whether the price of an item was reasonable in relation to its quality, condition, and other characteristics. For Margaux Collections, this finding supported the need to display product prices clearly and provide accurate descriptions of each garment. Information such as brand, size, material, and condition could help customers assess

whether a product was worth its listed price. By presenting these details consistently, the ordering system could help customers compare available products and make informed purchasing decisions.

Another important finding was the role of seller trust in influencing purchase decisions. Customers might hesitate to purchase preloved clothing when they were uncertain about the seller's reliability or the accuracy of the information provided. Transparent product descriptions, clear photographs, and honest disclosures of visible defects could help address these concerns. For the Margaux Collections Ordering System, presenting accurate information about each garment could help customers develop realistic expectations about the products they intended to purchase.

The study also highlighted the relationship between perceived value and consumer confidence. Although affordable prices could attract potential buyers, price alone might not be sufficient to encourage a purchase when customers lacked confidence in the product or seller. For Margaux Collections, this supported the importance of combining competitive pricing with complete product information and transparent ordering procedures. Customers could evaluate preloved clothing more confidently when they understood its condition and the characteristics of the item before completing an order.

Furthermore, the findings emphasized the importance of consistency in product presentation. When product listings followed a standard format, customers could compare items more easily and identify the information relevant to their purchasing decisions. The Margaux Collections Ordering System could therefore use a consistent structure for displaying photographs, prices, sizes, descriptions, and condition labels. This approach could improve the organization of product listings and help customers understand the differences between brand-new and preloved garments.

The study was also relevant to the development of a trustworthy online shopping experience. Customers needed to believe that the products they received would reasonably correspond to the descriptions and photographs displayed in the system. For preloved clothing, this required particular attention to visible signs of wear, stains, fading, or other imperfections. Providing honest condition descriptions and clear photographs could reduce uncertainty and help prevent misunderstandings between the business and its customers.

Overall, the study provided useful insights into the importance of perceived value and seller trust in online second-hand fashion shopping. Its findings supported the development of the Margaux Collections Ordering System by emphasizing transparent pricing, accurate product descriptions, clear condition information, and consistent product presentation. By incorporating these features, the proposed system could help customers evaluate the value of available garments, reduce uncertainty during online shopping, and make more informed purchasing decisions. These measures could also help establish a more transparent and reliable shopping experience for customers purchasing brand-new and preloved clothing.

Visual Presentation and Buyer Confidence in Online Clothing Shopping, Saha and Das (2019)

Boardman and McCormick [19] investigated how product presentation influenced consumer decision-making in apparel retailing. Their research examined how different presentation features affected consumers' visual attention and responses when evaluating clothing products. Using eye-tracking and qualitative interviews, the researchers found that consumers benefited from having more visual information available when assessing apparel products. Images of models attracted the most attention, followed by mannequin images and zoom functionality. These findings demonstrated the importance of visual presentation in helping consumers evaluate clothing products and make purchasing decisions [19]. The study was relevant to the development of the Margaux Collections Ordering System because customers purchasing clothing online needed sufficient visual information to assess the appearance and characteristics of available items.

The study highlighted the importance of providing multiple forms of visual information when presenting clothing products online. Unlike physical stores, online shopping platforms do not allow customers to touch fabrics, inspect garments directly, or try on clothing before making a purchase. Customers must therefore rely on photographs and written descriptions to understand a product's appearance and features. For Margaux Collections, this finding supported the inclusion of clear product photographs that showed clothing items from different angles. Providing detailed images could help customers examine the garment's color, design, fabric appearance, and visible condition before deciding whether to purchase it.

Product imagery was particularly important for preloved clothing because individual items could have different levels of wear and varying physical conditions. Customers might need to identify fading, stains, loose threads, or other visible imperfections before placing an order. The Margaux Collections Ordering System could address these concerns by allowing the business owner to upload multiple photographs for each product and provide an accurate description of its condition. This approach could help customers develop realistic expectations about the items being offered and reduce uncertainty during the purchasing process.

Furthermore, the findings of Boardman and McCormick [19] supported the importance of presenting products in an organized and visually appealing manner. When product photographs and descriptions were arranged clearly, customers could evaluate available clothing items more efficiently. For Margaux Collections, product listings could include the product name, brand, size, color, price, condition, and relevant photographs. Organizing these details consistently could help customers compare available items and identify products that matched their preferences. This could also improve the overall usability of the ordering system by making product information easier to locate and understand.

The study also provided a basis for considering the role of zoom functionality and detailed product views in online clothing presentation. Enlarged photographs could help customers examine specific garment features that might not be easily visible in standard images. In the context of preloved fashion, these details could include fabric texture, stitching, labels, and visible signs of previous use. Although photographs cannot completely replace physical inspection, providing detailed visual information could help customers make better-informed purchasing decisions. Therefore, the Margaux Collections Ordering System could benefit from image-display features that allow customers to view product photographs clearly.

In addition, effective visual presentation could contribute to customers' confidence in evaluating products online. When photographs accurately represented the actual items, customers could compare the displayed appearance with the accompanying descriptions and assess whether a product met their expectations. Conversely, unclear photographs or insufficient product information could make it more difficult for customers to evaluate clothing items. For Margaux Collections, maintaining accurate and consistent product presentations could support transparency and help prevent misunderstandings between the business owner and customers.

Overall, Boardman and McCormick [19] provided relevant support for the development of the Margaux Collections Ordering System by demonstrating the importance of visual product presentation in apparel-related purchasing decisions. Their findings supported the use of clear product photographs, detailed product views, and organized product information to help customers evaluate clothing before purchasing. These insights were particularly applicable to preloved fashion, where customers needed to assess each item's appearance and visible condition without physically inspecting it. By incorporating multiple product photographs, accurate descriptions, and organized listings, the Margaux Collections Ordering System could improve product visibility, reduce uncertainty, and provide customers with a more informative and convenient online shopping experience.

Listing Organization and Purchase Decision-Making, Chen, Liu, and Wu (2021)

Chen, Liu, and Wu [20] examined the relationship between product organization, inventory visibility, and consumer decision-making in online fashion retail. Their study focused on how the arrangement and presentation of product listings could influence customers' shopping experiences and purchasing decisions. The findings suggested that clearly organized product listings could help customers identify relevant product information and compare available items more efficiently. These findings were relevant to the development of the Margaux Collections Ordering System because the platform needed to present brand-new and preloved clothing in an organized manner that allowed customers to browse products and make informed purchasing decisions.

The study emphasized the importance of inventory visibility in online shopping [20]. Customers needed to know whether a product was available before deciding to purchase it. When availability information was unclear or outdated, customers could experience confusion and spend additional time determining whether an item could still be ordered. For Margaux Collections, displaying clear product availability statuses could help customers identify items that were available for purchase and avoid selecting products that were no longer in stock. Maintaining updated inventory information could also help the business manage orders more accurately.

Another important consideration was the organization of product categories. When different products were displayed without a clear structure, customers could find it difficult to identify items that matched their preferences. For the Margaux Collections Ordering System, separating brand-new clothing from preloved clothing could help customers distinguish between the two product types. Organized categories could also make browsing more convenient and allow customers to focus on the items most relevant to their needs.

The study also highlighted the importance of presenting essential product information clearly. Details such as size, price, product type, and availability could influence how customers evaluated their options. For Margaux Collections, these details could be displayed consistently on product listings and individual product pages. Providing accessible information could help customers compare clothing items without repeatedly contacting the seller to request basic details.

Furthermore, clear product condition labels could be particularly important when presenting preloved clothing. Since previously owned garments may differ in their condition, customers needed to understand whether an item was brand-new or preloved before making a purchase. The system could provide condition descriptions and photographs that accurately represented each garment. These features could help customers evaluate the products more fairly and reduce confusion about the differences between the available items.

The organization of product listings could also improve the overall efficiency of the shopping process. Customers could spend less time searching for products when categories, descriptions, and availability information were presented consistently. For Margaux Collections, a structured product catalog could make it easier for customers to locate clothing, review item details, and proceed with their orders. This could contribute to a more convenient shopping experience while helping the business present its inventory systematically.

Overall, Chen, Liu, and Wu [20] were cited as emphasizing the importance of product organization and inventory visibility in online shopping. These concepts were relevant to the Margaux Collections Ordering System because customers needed clear product categories, accurate availability statuses, and accessible information about clothing items. By organizing brand-new and preloved products separately and displaying essential details consistently, the proposed system could help customers browse efficiently, compare available options, and make informed purchasing decisions. These features could also support more accurate inventory management and improve the organization of the business's online ordering operations.

Sustainability Messaging and Appeal of Preloved Fashion, Mair and Gazzola (2020)

Gazzola et al. [21] examined sustainability and circular economy principles in the fashion industry, focusing on consumer perceptions and generational differences. Their study investigated how sustainability-related concerns influenced fashion consumption, particularly among younger consumers. The researchers highlighted the growing importance of sustainable practices and circular economy principles in response to changing consumer lifestyles and expectations [21]. This study was relevant to the development of the Margaux Collections Ordering System because it provided a basis for understanding how preloved clothing could be presented as a sustainable and valuable alternative to newly manufactured garments.

The study emphasized the importance of sustainability in shaping the fashion industry's development. Circular fashion encourages the reuse of existing garments and the extension of their useful lifespans. These practices can help reduce unnecessary clothing disposal and promote more responsible resource utilization. For Margaux Collections, this concept was relevant because selling preloved clothing allows existing garments to remain in use. The proposed ordering system could support this practice by making available products easier to browse, evaluate, and purchase.

Gazzola et al. [21] also highlighted the relevance of sustainability to younger consumers' fashion preferences. Their findings supported the importance of considering environmental awareness and changing consumption patterns when developing fashion-related business strategies. For Margaux Collections, this suggested that preloved clothing could be presented not only according to price and appearance but also in terms of its potential contribution to more responsible consumption. However, customers would still need accurate information about each item's condition, quality, and price before making a purchasing decision.

The study further supported the importance of presenting sustainability information alongside clear product details. The Margaux Collections Ordering System could include brief statements explaining that purchasing preloved clothing supports garment reuse and may help reduce textile waste. Such information could help customers understand the environmental relevance of their purchases while allowing them to evaluate the practical value of each product. Combining sustainability information with accurate photographs, condition labels, and transparent pricing could provide a more complete product presentation.

Overall, Gazzola et al. [21] provided a relevant foundation for the Margaux Collections Ordering System by discussing sustainability, circular economy principles, and changing consumer perceptions in the fashion industry. Their research supported the relevance of extending garment lifespans and promoting more responsible fashion consumption. These insights could be applied through organized product listings, accurate inventory records, clear descriptions, and optional sustainability information. By combining convenient online ordering with transparent product presentation, the system could support the continued circulation of preloved clothing while helping customers make informed purchasing decisions.

2.2 Local Studies

Local studies provided the empirical basis for the proposed system because they showed how Filipino businesses actually used digital tools in practice. These studies were especially valuable because they reveal the operational benefited, limitations, and adoption patterns of systems designed for small enterprises in the Philippines. In the case of Margaux Collections, the local studies reviewed here supported the need for a more structured ordering platform that could improve efficiency, accuracy, and customer trust. They also demonstrated that even small-scale

businesses could benefit significantly from digital solutions when these were designed according to their actual workflow, resources, and operational challenges.

Developing a Point-of-Sale Inventory Management System for Lords Grace General Merchandising

Antonio, Reyes, and Torres [1] developed a PHP- and MySQL-based point-of-sale and inventory management system for Lord's Grace General Merchandising, a small retail business in the Philippines. The project was completed as an undergraduate capstone at the Polytechnic University of the Philippines and demonstrated the application of information technology in addressing common operational challenges experienced by small enterprises. These challenges included managing product records, monitoring inventory levels, recording sales transactions, generating receipts, and preparing business reports. The study illustrated how accessible web technologies could provide a practical solution for improving business operations without requiring expensive commercial software.

Small businesses often operate with limited financial resources, personnel, equipment, and technical support. Despite these limitations, they need reliable methods for maintaining product information, monitoring available stock, documenting sales, and making informed business decisions. Web-based information systems can address these requirements by centralizing business records and making information easier to access, update, and retrieve. The system developed by Antonio et al. [1] demonstrated the potential of a database-driven application to support the daily activities of a small retail enterprise.

The system incorporated several essential modules, including product management, inventory tracking, sales recording, and reporting. The product management module allowed authorized users to add new products, modify product information, update existing records, and remove unnecessary entries. Through a centralized database, product information could be organized systematically, reducing dependence on handwritten records, separate notebooks, and scattered files. This arrangement made it easier to search for product information, correct errors, and maintain consistent records.

The inventory tracking module supported the monitoring of available stock and helped identify products that required replenishment. Maintaining updated inventory records was important because inaccurate stock information could lead to unexpected shortages, delayed purchases, and customer dissatisfaction. A system that identifies products approaching low-stock levels can help business owners determine when to replenish supplies before inventory is completely exhausted. Consequently, inventory monitoring supports more organized purchasing decisions and improves the availability of products.

The sales recording module provided a structured method for documenting customer transactions. It recorded information about purchased products, quantities, and transaction amounts while supporting receipt generation. Connecting sales records with inventory information also helped ensure that stock quantities reflected completed transactions. This integration reduced the possibility of maintaining outdated inventory records after products had been sold.

The reporting module supported the preparation of sales summaries over specific periods. Organized sales records could help business owners review business performance, identify frequently purchased products, and recognize changes in customer demand. This information could support decisions concerning purchasing, inventory replenishment, and product selection. By transforming transaction records into useful reports, the system provided a more systematic basis for business decision-making.

The study also demonstrated the advantages of digital recordkeeping compared with manual inventory and sales management. Manual methods can result in misplaced documents, duplicated entries, delayed updates, calculation errors, and inconsistencies between recorded and actual stock. These problems become more difficult to manage

when employees perform several responsibilities simultaneously or when business transactions occur frequently. A centralized digital system can reduce these difficulties by providing a consistent method for storing, updating, and retrieving business information.

Accurate inventory information is also important for customer satisfaction. When product records are not updated immediately after a sale, customers may be informed that an item is available even though it has already been purchased. This situation can lead to order cancellations, delays, and loss of customer confidence. Maintaining updated product and transaction records helps businesses provide more accurate information about product availability and respond to customer inquiries more efficiently.

Another contribution of the study was its demonstration of the practicality of PHP and MySQL for small-business applications. PHP can support the development of web-based interfaces and business functions, while MySQL can store structured information about products, inventory, sales, and users. These technologies provide a foundation for developing database-driven applications that can be accessed through a web browser. For small businesses, this approach offers an opportunity to implement essential digital functions without adopting an unnecessarily complex or expensive commercial platform.

The study further highlighted the importance of designing information systems according to the actual requirements of a business. A system should prioritize practical functions and ease of use rather than including features that do not contribute to daily operations. A complicated interface may discourage users from adopting the system, particularly when they have limited technical experience or must perform several business responsibilities. Therefore, usability and reliability are important considerations in the development of information systems for small enterprises.

The findings of Antonio et al. [1] are relevant to the proposed Margaux Collections Ordering System because both businesses require organized product records, inventory monitoring, sales recording, and efficient transaction management. Margaux Collections handles both brand-new and pre-loved fashion items, creating inventory requirements that differ from those of a conventional retail business. Brand-new products may be available in several quantities, sizes, and colors, whereas pre-loved garments may be unique or limited to a single piece. This distinction requires accurate product records and consistent updates whenever an item is reserved or sold.

The proposed Margaux Collections Ordering System can adopt similar functions by providing centralized product management, inventory monitoring, sales recording, order processing, and report generation. Product records may include the item name, product code, category, brand, size, color, condition, price, quantity, and product image. For pre-loved clothing, additional information concerning measurements, fabric condition, and visible flaws may be recorded to provide more complete product descriptions. These details can help the business owner organize listings and respond to customer inquiries more efficiently.

Inventory status management is particularly important for pre-loved products because an individual garment may not have an identical replacement. If an item is sold but remains listed as available, another customer may attempt to purchase the same product. To minimize this problem, the proposed system can distinguish among available, reserved, sold, and unavailable items. Updating product status when transactions occur can reduce duplicate orders and improve the accuracy of inventory records.

The proposed system can also integrate sales recording with order monitoring. Each transaction may include customer information, purchased products, quantities, prices, payment status, and delivery details. Order statuses such as pending, confirmed, paid, packed, shipped, completed, and cancelled can help the owner monitor the progress of customer transactions. These functions can reduce dependence on handwritten notes, chat histories, and memory while making transaction information easier to retrieve.

Furthermore, sales reports can help Margaux Collections identify fast-moving and slow-moving products. Frequently purchased products may indicate customer preferences and strong demand, while items that remain

available for extended periods may require additional promotion or changes in pricing. Organized reports can therefore support purchasing decisions, product selection, and business planning.

The study also supports the importance of developing a system that follows the actual workflow of Margaux Collections. The proposed application should allow the owner to add products, edit product information, update inventory, process orders, record sales, and review transaction statuses without unnecessary steps. Clear navigation, organized forms, readable product listings, and visible inventory indicators can improve usability. In addition, regular data backups, appropriate access controls, and system testing can help maintain record accuracy and reliability.

Overall, the study by Antonio et al. [1] provides a relevant local basis for developing the Margaux Collections Ordering System. It demonstrates the practical application of web-based technologies in supporting product organization, inventory control, sales recording, reporting, and operational efficiency in a small Philippine retail business. Its approach supports the development of a centralized and user-friendly system tailored to fashion retail, particularly in managing limited-stock and unique pre-loved products. By applying these principles, the proposed system can help reduce manual recording errors, improve inventory visibility, and support more efficient order processing.

Analysis of Inventory Management Systems of Small-Sized Restaurants in Quezon Province

Santos, Dela Cruz, and Lopez [2] examined inventory management practices among small-sized restaurants in Quezon Province, Philippines. Their study, published in the *Philippine Journal of Hospitality and Service Management*, focused on the methods used by restaurant owners to monitor supplies, identify inventory-related problems, and improve stock control and operational efficiency. The research considered issues involving purchasing, inventory recording, stock monitoring, spoilage, and the maintenance of accurate records. Its subject was relevant to small enterprises because these businesses often operate with limited personnel, financial resources, and time while still requiring dependable information for daily operations.

Inventory management is an important business function because it helps organizations determine the quantity of products or supplies available, identify items that require replenishment, and maintain records of inventory movements. Although restaurants and retail businesses handle different types of products, both depend on accurate stock information to support purchasing decisions, prevent shortages, and maintain customer satisfaction. The study by Santos et al. [2] therefore provides a useful basis for understanding inventory management challenges that may also occur in small fashion retail businesses.

Manual inventory methods, including handwritten logs, notebooks, paper-based stock lists, and spreadsheets, can be inexpensive and easy to implement. However, these methods may become difficult to maintain when inventory changes frequently. Purchases, sales, consumption, and disposal require consistent record updates. When these activities are not recorded immediately, the available information may no longer correspond to the actual inventory. Consequently, business owners may experience recording errors, incomplete information, delayed updates, and discrepancies between physical stock and recorded quantities.

The study discussed how inaccurate inventory information could contribute to overstocking, shortages, unnecessary purchasing expenses, and operational disruptions. In restaurant operations, excessive purchasing of perishable goods can result in spoilage, wasted resources, and financial losses. Conversely, insufficient supplies can interrupt food preparation and limit the availability of menu items. Purchasing decisions based mainly on personal estimates or memory may further increase these risks. Accurate and updated inventory records are therefore important for making timely purchasing decisions and maintaining operational efficiency.

The study also considered inventory rotation practices, including First In, First Out (FIFO) and First Expiry, First Out (FEFO). FIFO prioritizes the use of items received earlier, while FEFO prioritizes products with the earliest expiration dates. These methods are particularly important when managing perishable goods because they help

reduce waste and prevent older supplies from remaining unused. Although Margaux Collections does not primarily handle perishable products, the broader principle of systematically tracking stock movement remains relevant. Maintaining records of when products are received, reserved, sold, or removed from inventory can improve stock control and reduce inconsistencies.

Another important aspect of inventory management is the identification of fast-moving and slow-moving items. Products that are frequently used or sold may require closer monitoring to prevent shortages, while items that remain in stock for extended periods may indicate the need for changes in purchasing or promotional strategies. Digital records can help business owners identify these patterns by organizing inventory movements and transaction histories. This information supports decisions based on actual business records rather than estimates alone.

Compared with manual methods, digital inventory management systems can provide more organized records and faster access to inventory information. These systems allow users to record purchases, sales, and inventory adjustments systematically. When properly maintained, a centralized database can reduce duplicated entries, improve the consistency of stock records, and make it easier to identify products that require replenishment. It can also improve communication among employees by allowing authorized users to refer to the same inventory information instead of maintaining separate records.

The study emphasized the importance of simplicity, system reliability, and perceived usefulness in the adoption of digital inventory management systems. These factors are particularly relevant to small businesses because owners often manage several responsibilities, including purchasing, sales, customer service, financial management, and inventory monitoring. A system that requires extensive training or involves unnecessary steps may discourage consistent use. Therefore, users should be able to perform routine tasks, such as adding products, updating quantities, recording transactions, and checking inventory, through a straightforward interface.

System reliability is equally important because business owners must be able to depend on the accuracy and availability of their records. A system that loses information, produces incorrect calculations, or fails to update stock properly can create problems similar to those associated with manual recordkeeping. Reliable data storage, consistent inventory updates, regular backups, and appropriate access controls can help protect business information and maintain users' confidence in the system.

Perceived usefulness also influences whether small-business owners accept a digital solution. Owners are more likely to use a system when its benefits are directly related to their daily activities. These benefits may include reducing recording errors, saving time, improving inventory visibility, and making purchasing decisions easier. Consequently, digital systems should focus on functions that address existing operational problems rather than introducing unnecessary features.

The findings of Santos et al. [2] are relevant to the proposed Margaux Collections Ordering System because both restaurant and fashion retail businesses need accurate product information, updated inventory records, and reliable methods for monitoring stock movements. Although their products differ, both businesses can experience problems when recorded information does not match actual availability. In fashion retail, an outdated inventory record may lead to a customer attempting to purchase a product that has already been sold or reserved. Such discrepancies can result in order cancellations and customer dissatisfaction.

Inventory control is especially important for Margaux Collections because its products include both brand-new and pre-loved fashion items. Brand-new products may be stocked in multiple quantities, sizes, and colors, while pre-loved garments may have unique characteristics and limited availability. Each product therefore requires an organized record containing relevant information, such as the item name, product code, category, size, color, brand, condition, price, and quantity. For pre-loved items, measurements and notes about visible flaws may also be necessary.

The proposed system should provide mechanisms for updating inventory status whenever products are reserved, sold, or removed from the available listings. For example, once a customer successfully purchases a unique pre-loved garment, its status should be changed to sold so that it is no longer presented as available to other customers. Similarly, reserved items should be distinguishable from products that remain available for purchase. These functions can improve inventory accuracy and reduce the risk of duplicate orders.

The proposed system should also support sales tracking and order monitoring. Transaction records can contain customer details, purchased products, quantities, payment information, and delivery status. Order categories such as pending, confirmed, paid, packed, shipped, completed, and cancelled can help the owner determine which transactions require attention. Integrating these records with inventory information can improve the consistency of product availability and order processing.

Furthermore, sales and inventory reports can help the owner evaluate product performance. Reports on total sales, remaining stock, popular products, and slow-moving items can provide useful information for future purchasing, pricing, and promotional decisions. For example, frequently purchased products may indicate customer preferences, while products that remain unsold for extended periods may require improved presentation or additional promotion. Such information can help Margaux Collections respond more effectively to customer demand.

The study also supports the development of a system that is simple, reliable, and aligned with the actual workflow of the business. The proposed application should prioritize product management, inventory updating, sales recording, order monitoring, and stock-status management. Clear labels, straightforward navigation, organized forms, and visible status indicators can help the owner complete routine tasks efficiently. These functions should be designed around the specific requirements of fashion retail rather than adopting features that are unnecessary for the business.

In addition, the system should accommodate the differences between brand-new and pre-loved products. Brand-new products may require quantity tracking across multiple sizes or colors, while pre-loved products may require separate records for individual garments. Distinguishing between these product types can improve inventory organization while preserving a manageable user experience. Reliable data storage, regular backups, and consistent updating procedures can further help maintain accurate records.

Overall, the study by Santos et al. [2] provides a relevant basis for developing the Margaux Collections Ordering System. It highlights the importance of accurate inventory records, systematic stock monitoring, and practical digital solutions for small enterprises. Its discussion of manual recordkeeping problems and the importance of simplicity, reliability, and usefulness supports the development of an ordering and inventory system tailored to a small fashion business. Applying these principles could help Margaux Collections reduce inventory discrepancies, monitor product availability, improve order processing, and make more informed business decisions.

Impact of Inventory Management on Financial Performance of MSMEs

Gaoat et al. [3] examined the relationship between inventory management, operational efficiency, and financial performance among micro, small, and medium enterprises (MSMEs) in Santiago City, Philippines. The study employed a quantitative correlational research design and gathered data from MSME owners and managers to determine how inventory management practices were associated with business performance. The researchers considered financial indicators such as sales, operating margin, profit, return on assets, liquidity, and inventory turnover. The findings indicated a strong positive relationship between effective inventory management and operational efficiency, as well as a moderate positive relationship between inventory management and financial performance. These results highlighted the importance of inventory control as a business function associated with profitability, liquidity, productivity, and long-term sustainability.

The study was relevant to small enterprises because MSMEs generally operate with limited capital, personnel, storage space, and technical resources. Compared with larger corporations, small businesses may have less capacity to absorb losses resulting from unsold products, damaged goods, inaccurate records, or excessive purchasing. Slow-moving inventory can tie up working capital and limit the resources available for acquiring products that customers demand. Consequently, inventory management can influence both daily operations and the financial stability of a business.

Gaoat et al. [3] found that effective inventory management was associated with better operational and financial performance. Proper inventory control enabled business owners to monitor product movement, determine available stock, reduce unnecessary purchases, and identify products requiring replenishment. These practices helped businesses use their resources more efficiently by directing capital toward products with stronger demand rather than allowing excessive funds to remain tied up in unsold merchandise.

Overstocking is one of the problems that effective inventory management can help prevent. When businesses purchase more products than they can sell within a reasonable period, capital becomes tied up in inventory. Excessive stock may also require additional storage space and increase the time needed to manage products. In fashion retail, products may become less desirable as styles and consumer preferences change, potentially requiring discounts or promotional activities to encourage sales. A reliable inventory system can help business owners identify slow-moving items and develop appropriate strategies, such as promotions, discounts, improved product presentation, or adjustments to purchasing decisions.

Understocking presents a different challenge. When businesses fail to maintain sufficient quantities of popular products, customers may be unable to purchase the items they want. Repeated stockouts can result in missed sales, customer dissatisfaction, and the loss of potential repeat purchases. Accurate inventory records help business owners identify products requiring replenishment and reduce the likelihood of discovering shortages only after customers have placed orders.

The study also emphasized the importance of using inventory information to guide business decisions. Inventory records can provide more than a count of available products. When combined with sales data, they can help business owners identify sales trends, purchasing patterns, product performance, and customer preferences. For example, a business can determine which products sell quickly, which items remain in stock for extended periods, and which product categories generate greater sales revenue.

Such information is particularly important for MSMEs because they often operate with limited working capital. Business owners need to avoid repeatedly purchasing products with weak demand while ensuring that popular products remain available. Accurate inventory data allow purchasing, pricing, promotional, and product-selection decisions to be based on recorded transactions rather than memory or assumptions. This approach can improve resource allocation and support more informed business planning.

Effective inventory management can also strengthen a business's ability to respond to changing customer demand. Fast-moving products require closer monitoring and timely replenishment, while slow-moving items may need additional promotion or adjustments in pricing and presentation. This responsiveness is important for small businesses because customer preferences can change quickly and financial resources may be limited.

Inventory accuracy is also associated with customer satisfaction. Customers expect businesses to provide reliable information about product availability, prices, conditions, and delivery arrangements. If a product is listed as available after it has been sold, the business may be unable to fulfill the order. Such situations can result in cancellations, delays, and customer complaints. Updated inventory records can help businesses provide more dependable service and reduce the risk of accepting orders that cannot be fulfilled.

The financial implications of inventory management extend beyond preventing overstocking and stockouts. Effective control can help reduce losses associated with damaged products, misplaced items, unnecessary purchases,

and avoidable storage expenses. It can also support liquidity management by reducing the amount of capital tied up in unsold merchandise. Furthermore, accurate inventory and sales records provide useful information for evaluating product performance, determining appropriate prices, and assessing the effects of discounts and promotions.

These findings are particularly relevant to fashion businesses because clothing products vary according to brand, design, color, size, condition, and style. Items that appear similar may have different prices, measurements, or conditions. If these characteristics are not recorded correctly, business owners may encounter difficulties when processing orders or updating product availability. Maintaining detailed product records is therefore important for ensuring that customers receive the correct items and that sales transactions are recorded accurately.

For Margaux Collections, the study supports the development of an Ordering System with inventory tracking, sales summaries, and reporting capabilities. The business handles both brand-new and pre-loved fashion items, which require careful monitoring of product characteristics, prices, quantities, and availability. Brand-new items may be available in several units, while pre-loved garments may be unique or limited to a single piece. These differences increase the risk of duplicate orders and inaccurate availability when inventory is managed manually.

A digital ordering and inventory system can help ensure that product records are updated when items are added, reserved, sold, or removed. It can distinguish among available, reserved, sold, and unavailable products, reducing the possibility of accepting orders for items that are no longer available. This function is particularly important for unique pre-loved garments, which may not have identical replacements.

The system can also help the owner identify fast-moving and slow-moving products. Frequently purchased items may indicate strong customer demand, effective pricing, or successful product presentation. Products that remain unsold for extended periods may require additional promotion, improved photographs, more detailed descriptions, or price adjustments. By reviewing inventory reports and sales summaries, the owner can make better-informed decisions about which products to promote and which items to acquire in the future.

Sales reports can further support the evaluation of business performance. The proposed system may summarize transactions by date, product category, item condition, price range, or product type. These reports can help the owner identify products that generate frequent sales and determine which categories contribute substantially to revenue. Although an ordering system cannot guarantee higher profits, it can provide information that supports better purchasing, pricing, and promotional decisions.

The relationship between operational efficiency and financial stability is another important contribution of the study. Proper inventory management can reduce avoidable costs and improve the utilization of available resources. For Margaux Collections, each product represents a financial investment. Money spent acquiring merchandise remains committed to that inventory until the product is sold. If items remain unsold for extended periods, the business may experience slower cash flow and reduced capacity to acquire new products.

A digital inventory system can provide the owner with a clearer overview of product movement and sales activity. By identifying which products sell frequently and which remain unsold, the owner can evaluate how effectively business resources are being used. The information can also help identify products requiring additional attention and prevent repeated purchasing decisions that result in excessive slow-moving stock.

Inventory management can additionally improve customer service by making product information easier to retrieve. When records are centralized, the owner can respond to inquiries about product availability, prices, condition, and order status without searching through multiple notebooks, messages, or files. Accurate information can reduce delays and support a more organized purchasing experience.

The findings also suggest that inventory management should be treated as a continuous business process rather than a one-time activity. Product records must be updated whenever merchandise is received, listed, reserved, sold, returned, or removed. A system that is accurate initially may become unreliable if transactions are not recorded

consistently. Therefore, the effectiveness of the proposed Ordering System will depend partly on regular procedures for entering product information, recording sales, and reviewing inventory records.

The system should also accommodate changes in the product range and business operations of Margaux Collections. The business may introduce additional fashion categories, expand its pre-loved selection, or modify its order and delivery procedures. Allowing the owner to create new product categories and update existing records can help the system remain useful as the business develops.

Gaoat et al. [3] also provided a basis for understanding the relationship between inventory management and long-term business sustainability. Financial stability depends partly on controlling costs, maintaining sufficient liquidity, responding to customer demand, and using resources efficiently. A system that improves visibility into stock and sales can support these objectives by providing organized information for planning, evaluating performance, and monitoring business activities.

When considered alongside the studies by Antonio et al. [1] and Santos et al. [2], the research by Gaoat et al. [3] strengthens the justification for developing the Margaux Collections Ordering System. Antonio et al. [1] demonstrated the application of a PHP- and MySQL-based system to product management, sales recording, inventory monitoring, and reporting in a small Philippine retail business. Santos et al. [2] examined inventory management practices and the importance of organized stock control. Gaoat et al. [3] extended this discussion by examining the relationship between inventory management, operational efficiency, and financial performance among MSMEs.

Together, these studies support the development of a digital ordering and inventory system that addresses both operational and financial requirements. For Margaux Collections, such a system can help maintain accurate stock records, reduce manual recording errors, monitor sales, identify product demand, and support purchasing decisions. These capabilities can improve business organization and provide information that assists the owner in managing available resources.

Overall, Gaoat et al. [3] provided a research basis for treating inventory management as an important business function rather than merely an administrative responsibility. The proposed Margaux Collections Ordering System should therefore do more than record customer orders. It should also support inventory monitoring, sales analysis, and informed business decision-making. By maintaining updated product records, generating sales summaries, identifying fast-moving and slow-moving items, and monitoring product availability, the system can help the business improve operational efficiency and support its long-term sustainability.

Digital Ukaynomics: Analysis on the Market for Secondhand Clothing Sold through Online Channels

Tomoling [4], in the undergraduate thesis titled *Digital Ukaynomics: Analysis on the Market for Secondhand Clothing Sold through Online Channels*, examined the online secondhand clothing market in the Philippines. The study focused on consumers purchasing pre-loved garments through digital platforms, particularly Facebook groups, which have become important channels for buying and selling secondhand clothing. By examining consumer characteristics and purchasing behavior, the research provided insights into how digital platforms have influenced traditional ukay-ukay consumption and created opportunities for online sellers.

The study highlighted how digital platforms allowed ukay-ukay sellers to reach customers beyond the geographic boundaries of physical stores and public markets. Traditional ukay-ukay shopping generally required customers to visit a physical location and inspect garments personally. Online selling transformed this process by allowing buyers to browse photographs, descriptions, and product posts, communicate with sellers, compare available items, and arrange deliveries without visiting a store. This development made secondhand clothing more accessible and integrated it into online shopping practices.

Tomoling [4] employed a quantitative research design using a Google Forms questionnaire distributed through Facebook groups dedicated to buying and selling secondhand clothing. The study collected responses from 248 online secondhand clothing consumers, including 200 female and 48 male respondents. Most respondents were employed or studying. These characteristics provided a description of the study sample, although they should not be interpreted as representing every consumer in the Philippine online ukay-ukay market.

Facebook groups offered several advantages for online secondhand clothing transactions. Sellers could publish new listings, announce restocks, conduct live-selling sessions, and communicate directly with prospective buyers. Customers could ask questions, request measurements, check product availability, and clarify payment or delivery arrangements. These groups also enabled members to share recommendations and feedback about sellers. Although less formal than conventional e-commerce platforms, they provided an environment in which buyers and sellers could organize reservations, payments, and deliveries.

The study identified affordability as an important motivation for purchasing secondhand clothing online. Ukay-ukay products allowed consumers to acquire clothing at prices that were often lower than those of brand-new garments. This was particularly relevant to students and consumers with limited disposable income, as well as buyers who wanted to purchase multiple items at a relatively low cost. Consumers could also discover branded, fashionable, or distinctive garments at prices below those of comparable new products.

However, the study indicated that affordability was not the only factor influencing online secondhand clothing purchases. Consumer demand was also associated with demographic characteristics, purchasing preferences, online shopping experience, and logistical considerations. A low product price did not necessarily guarantee a completed purchase because customers also considered delivery expenses, seller reliability, product descriptions, and the convenience of the transaction.

Using an ordinary least squares regression model, Tomoling [4] identified several factors associated with consumer demand for secondhand clothing sold online. These included age, sex, income, willingness to pay for brand-new clothing, willingness to pay for secondhand clothing, delivery fees, previous buying experience, and browsing time. These variables demonstrated that online ukay-ukay consumption was influenced by economic, demographic, behavioral, and logistical factors rather than by product price alone.

Demographic characteristics such as age and sex were examined in relation to consumer demand. Consumers may differ in their shopping habits, fashion preferences, and interest in secondhand clothing. Income also influences purchasing capacity and the types of garments consumers can afford. Although some consumers are attracted to ukay-ukay because of its affordability, others may be interested in unique designs, branded items, vintage clothing, or the environmental benefits associated with reusing garments.

Willingness to pay for brand-new and secondhand clothing provided further insight into consumer preferences. Some consumers may prioritize new products because of their expectations regarding condition, brand, or availability. Others may be willing to purchase pre-loved garments when these offer desirable characteristics, such as good condition, uniqueness, durability, or style. Therefore, the online secondhand clothing market includes consumers with different budgets and purchasing priorities.

Delivery fees were another important factor influencing online demand. Even when a garment was inexpensive, a high delivery charge could increase the total purchasing cost and discourage customers from completing a transaction. This was particularly relevant to lower-priced products, for which delivery could represent a substantial portion of the final expense. Reasonable shipping fees, combined orders, pickup arrangements, and delivery promotions may therefore improve the attractiveness of online secondhand clothing purchases.

Previous buying experience also influenced consumer behavior. Customers who had purchased secondhand clothing online may have become more familiar with seller reliability, payment procedures, delivery arrangements, sizing concerns, and product quality. This experience could reduce uncertainty and increase confidence in future

transactions. Conversely, first-time buyers may be more cautious because they cannot physically inspect the garment before purchasing it.

Browsing time was another factor examined in the study. Consumers who spent more time browsing online listings had more opportunities to discover products, compare prices, evaluate sellers, and identify garments that matched their preferences. Browsing also allowed customers to observe product updates and discover newly listed items. Therefore, the presentation and organization of online product listings can influence how easily customers find products that meet their needs.

Tomoling [4] also used logarithmic regression to examine the elasticity of demand for online secondhand clothing. The study reported that secondhand clothing was a normal good and had inelastic demand. A normal good is one for which demand generally increases as consumer income increases. This finding suggested that secondhand clothing could remain desirable even when consumers had greater purchasing power. Consumers might continue purchasing pre-loved garments because of their affordability, uniqueness, familiarity, or other preferences.

Inelastic demand means that the percentage change in quantity demanded is smaller than the percentage change in price, within the conditions examined. This does not mean consumers are completely unaffected by price changes. Rather, it suggests that other considerations, including product characteristics, convenience, and shopping habits, may also influence purchasing behavior. For online ukay-ukay sellers, this highlights the importance of offering attractive products and dependable service rather than relying exclusively on low prices.

The findings suggested that sellers could strengthen customer interest through accurate product descriptions, clear photographs, measurements, transparent condition information, reasonable delivery fees, and reliable communication. These practices could help customers make informed decisions and reduce uncertainty when purchasing garments online. Consistent service and organized listings could also encourage repeat purchases and strengthen customer trust.

The study further highlighted the importance of logistical factors in online secondhand clothing transactions. Customers must rely on the information provided by sellers and wait for delivery rather than immediately inspecting and taking home their purchases. Difficulties involving product availability, payment confirmation, shipping schedules, or delivery costs may therefore discourage purchases. Efficient order processing and accurate product information are important for creating a dependable online shopping experience.

Tomoling [4] also discussed the legal context of the Philippine secondhand clothing market, including Republic Act No. 4653, which prohibits the commercial importation of used clothing and rags. The study considered the relationship between this legal restriction and the continued practice of buying and selling secondhand clothing through physical and digital marketplaces. The expansion of online selling demonstrated the importance of understanding the economic and consumer behavior associated with the secondhand clothing sector. Any legal interpretation or discussion of compliance, however, must be based on the relevant legislation and applicable regulations.

The research contributed to the understanding of secondhand clothing consumption in the Philippines by demonstrating how digital platforms had expanded access to ukay-ukay products. Through online channels, consumers could browse garments, communicate with sellers, compare prices, and arrange purchases without visiting physical markets. These platforms provided additional opportunities for sellers to reach potential customers and organize their product listings and transactions.

The study also demonstrated the value of economic analysis in understanding online ukay-ukay consumption. By examining consumer demand, income, delivery fees, browsing behavior, and purchasing experience, it provided insights into the factors associated with online buying decisions. These findings are useful for understanding the needs of sellers who want to improve product presentation, customer communication, and transaction management.

The findings of Tomoling [4] are directly relevant to Margaux Collections because the business sells both brand-new and pre-loved fashion items. Online customers may need to view product photographs, check garment measurements and condition, compare prices, confirm availability, and coordinate payment and delivery. Managing these activities through separate messages and informal records can become difficult as the number of products and customer inquiries increases.

The proposed Margaux Collections Ordering System can address these needs by providing organized product listings and centralized records. Each product can have a unique code and include its name, category, size, color, brand, condition, price, quantity, and photographs. For pre-loved garments, measurements and descriptions of visible flaws can help customers understand the product before ordering. This information can improve product presentation and reduce misunderstandings between the business and its customers.

The system can also help manage product availability and customer orders. When a customer reserves or purchases a unique pre-loved garment, the product status can be updated to prevent duplicate orders. Order records can include customer information, selected items, payment status, and delivery details. These functions can help the owner track transactions more consistently and reduce dependence on separate chat histories or handwritten notes.

Furthermore, the study's discussion of delivery fees and purchasing experience highlights the importance of organizing transaction details. A centralized system can help the owner record delivery arrangements, monitor order progress, and provide more accurate information to customers. It can also support the preparation of sales reports that identify frequently purchased products and categories that attract customer interest.

Overall, Tomoling [4] provided a relevant basis for understanding the online secondhand clothing market and the consumer factors associated with purchasing pre-loved garments through digital channels. The study highlighted the importance of affordability, product information, delivery costs, shopping experience, and reliable communication. These findings support the development of the Margaux Collections Ordering System as a practical tool for organizing product listings, monitoring availability, managing orders, and improving the online purchasing experience.

By integrating product management, inventory tracking, and order processing, the proposed system can help Margaux Collections respond more efficiently to customer inquiries and transactions. It can also provide the owner with organized sales and inventory information for future planning. Thus, the study supports the development of a user-friendly ordering system that addresses the particular requirements of a small fashion business selling both brand-new and pre-loved products.

Pre-Loved Clothing and Local Fashion Industry in Kalibo

Itubay [5] examined the coexistence of pre-loved clothing businesses and local fashion enterprises in Kalibo, Aklan, in the study *"Pre-Loved Clothing and Local Fashion Industry: Its Marketability, Sustainability and Challenges."* The research explored how these two sectors managed competition, sustainability concerns, and changing market conditions within the local fashion industry. Using focus group discussions with 15 business owners, consisting of seven pre-loved clothing sellers and eight local fashion business owners, the study provided insights into the marketing strategies, operational challenges, and sustainability practices of small fashion enterprises.

The findings showed that social media platforms, particularly Facebook, Instagram, and TikTok, played an important role in promoting products and engaging customers. Pre-loved clothing sellers used live-selling sessions, multiple Facebook pages, and e-commerce platforms such as Shopee to reach potential buyers. In comparison, local fashion businesses relied more heavily on word-of-mouth referrals and personal relationships, although they also

increasingly adopted digital marketing strategies. Both sectors recognized product quality, flexible pricing, and personalized customer service as important factors in attracting and retaining customers.

The study also highlighted the importance of sustainability and inventory management in small fashion businesses. Pre-loved clothing sellers needed to update their product selections according to fashion trends, customer preferences, and social media influences. Since secondhand clothing items often differed in size, style, brand, and condition, sellers had to inspect products carefully before offering them for sale. These activities demonstrated the importance of maintaining accurate product information and organizing inventory to ensure that available items could be presented and sold effectively.

Despite their different business models, both pre-loved clothing sellers and local fashion enterprises experienced challenges involving product sourcing, quality control, customer expectations, and delivery schedules. Pre-loved clothing sellers encountered difficulties obtaining consistent supplies and maintaining product quality, while local fashion businesses experienced challenges involving employee performance, motivation, and recruitment. These findings demonstrated that small fashion enterprises needed effective operational strategies to maintain product quality, satisfy customers, and remain competitive.

The study further revealed that business owners addressed these challenges through supplier partnerships, financial assistance from family members, personalized customer service, and community engagement. Rather than presenting pre-loved clothing businesses as necessarily harmful to local fashion enterprises, the research suggested that both sectors could coexist through product differentiation, complementary market positioning, and sustainable business practices [5].

The findings are relevant to the development of the Margaux Collections Ordering System because the business handles both brand-new and pre-loved fashion products. These products require organized records of their categories, sizes, prices, conditions, and availability. A digital ordering system could help the business maintain updated product information, monitor stock levels, and reduce errors when processing customer orders. It could also help manage unique pre-loved items by preventing products that have already been sold or reserved from being offered to other customers.

Furthermore, the study emphasized the importance of digital platforms in expanding the reach of small fashion businesses. An organized system could complement Margaux Collections' online selling activities by providing reliable information about product availability and order status. Sales records and inventory reports could also help the owner identify popular products, monitor slow-moving items, and make better purchasing and promotional decisions. Therefore, the study provides a local research basis for developing an ordering system that supports inventory accuracy, customer service, and the operational efficiency of a small fashion enterprise.

From Necessity to Identity: Entrepreneurial Perspectives on Key Motivations for Filipino Second-Hand Clothing Consumers

The study *"From Necessity to Identity: Entrepreneurial Perspectives on Key Motivations for Filipino Second-Hand Clothing Consumers,"* published in *Frontiers in Sustainability* in 2022, examined the motivations behind Filipino consumers' participation in the secondhand clothing market [6]. Through observational research and in-depth interviews, the study explored how consumers understood and experienced the purchase of pre-loved clothing. It demonstrated that secondhand clothing consumption was influenced not only by affordability but also by personal identity, self-expression, environmental awareness, and sustainable fashion values.

The research showed that ukay-ukay had gradually developed from a practice commonly associated with economic necessity into a more widely accepted form of fashion consumption. Although affordability remained an important motivation, consumers increasingly viewed secondhand clothing as an opportunity to express their individuality and discover distinctive garments. Younger consumers, in particular, could use pre-loved clothing to create personal

styles that differed from the standardized designs commonly found in mass-produced fashion. This indicated that consumers' purchasing decisions were influenced by both economic considerations and personal preferences.

The study identified economic necessity as one of the principal motivations for purchasing secondhand clothing. Pre-loved garments allowed consumers to obtain clothing at relatively affordable prices while maintaining their desired appearance. This was particularly important for individuals who needed to manage limited budgets and competing financial responsibilities. However, the research emphasized that affordability did not fully explain the popularity of ukay-ukay. Consumers also valued the uniqueness, quality, and styling possibilities of individual garments.

Another important finding concerned the relationship between secondhand clothing and environmental sustainability. Consumers recognized that purchasing pre-loved garments could extend the useful life of clothing and reduce the need for immediate disposal. By allowing garments to circulate among different owners, secondhand clothing markets provided an alternative to continuously purchasing newly manufactured apparel. The study therefore connected ukay-ukay with broader principles of clothing reuse, circular fashion, and responsible consumption [6].

The research also examined identity construction as a motivation for secondhand clothing consumption. Consumers could use distinctive garments to communicate their personalities, creativity, fashion preferences, and social values. Unlike conventional retail products that were often available in large quantities, pre-loved clothing collections could contain rare, vintage, branded, or one-of-a-kind items. Discovering these garments allowed consumers to create personalized outfits and experience a sense of achievement in finding products that matched their individual preferences.

Social media further strengthened these consumer motivations. Digital platforms allowed sellers to showcase curated collections, publish product photographs, conduct live-selling sessions, and communicate directly with customers. These platforms also enabled buyers to share their purchases, recommend sellers, and present their personal styles to online communities. As a result, digital selling became more than a means of completing transactions; it also supported the promotion of fashion identity, customer engagement, and sustainability-related messages.

The study also suggested that secondhand clothing entrepreneurs needed to adapt their marketing strategies to changing consumer expectations. Sellers could strengthen their market position by emphasizing product uniqueness, quality, condition, and styling potential rather than focusing exclusively on low prices. Accurate product descriptions, clear photographs, and transparent communication could help customers evaluate items and make informed purchasing decisions. These practices were particularly important in online transactions, where buyers could not physically inspect garments before ordering [6].

The findings are relevant to Margaux Collections because the business offers both brand-new and pre-loved fashion items to customers with different preferences and budgets. The proposed Ordering System could support these differences by allowing products to be classified according to category, condition, size, price, and availability. Clear and organized product information would help customers evaluate available items and allow the owner to manage product listings more effectively.

In addition, the study supports the inclusion of inventory monitoring and sales reporting features in the proposed system. By reviewing sales records, the owner could identify which product categories attracted more customers and which items remained unsold. This information could guide decisions regarding product selection, pricing, promotions, and future sourcing. Therefore, the study provides a local research basis for developing an ordering system that supports changing consumer preferences and the digital operations of a pre-loved clothing business.

The Philippine Ukay-Ukay Culture as Sustainable Fashion

Biana [7] examined ukay-ukay as a cultural practice and sustainable fashion activity in the Philippines in the study *"The Philippine Ukay-Ukay Culture as Sustainable Fashion,"* published in the *DLSU Business & Economics Review* in 2020. The research explored how secondhand clothing consumption was connected to affordability, personal expression, Filipino cultural values, and environmental awareness. It presented ukay-ukay as more than an alternative for consumers with limited purchasing power, emphasizing its development into a recognized form of fashion consumption.

The study explained that ukay-ukay involved searching through collections of pre-loved clothing to discover affordable, fashionable, and distinctive items. This process reflected the participatory nature of thrift shopping, in which consumers examined different garments and selected products according to their preferences. The practice also demonstrated resourcefulness and creativity because shoppers could find items that were difficult to obtain in conventional retail stores.

The findings indicated that secondhand clothing consumers included individuals who were educated, employed, and financially capable of purchasing brand-new garments. This challenged the perception that ukay-ukay was exclusively associated with poverty or financial difficulty. Instead, consumers also participated in secondhand clothing markets because of their interest in affordable fashion, unique products, and sustainable consumption. The study consequently presented ukay-ukay as a market influenced by overlapping economic, cultural, and environmental motivations [7].

The research further highlighted the role of ukay-ukay in extending the useful life of garments. Through resale and reuse, clothing items could continue to provide value to new owners rather than being discarded immediately. This practice was consistent with circular fashion principles, which emphasize keeping products in use for longer periods through reuse, repair, resale, and recycling. Although secondhand clothing alone could not eliminate all environmental problems associated with fashion, the study recognized its potential contribution to more responsible consumption.

Another important aspect of the study was the changing perception of secondhand clothing among Filipino consumers. Ukay-ukay had expanded from traditional market stalls into specialized stores, thrift boutiques, online marketplaces, and social media shops. This development increased the accessibility of pre-loved clothing and allowed sellers to reach customers beyond their immediate communities. Digital platforms also enabled sellers to present their products through photographs, descriptions, styling content, and online promotions.

The study emphasized that the popularity of ukay-ukay reflected a combination of affordability, individuality, and environmental consciousness. Consumers could obtain garments at lower prices while discovering products that matched their personal styles. The availability of different brands, designs, colors, and styles also encouraged consumers to create distinctive outfits. These factors contributed to the growing acceptance of secondhand clothing as a legitimate fashion choice rather than merely a substitute for new apparel [7].

The research also recognized the importance of responsible business practices. Sellers needed to consider product quality, garment condition, pricing, and customer expectations when offering pre-loved clothing. Transparent product information could help buyers understand what they were purchasing, while effective marketing could communicate the value and uniqueness of available items. The study further suggested that combining traditional selling practices with digital marketing could help businesses respond to changing consumer behavior.

The findings are relevant to the Margaux Collections Ordering System because the business sells products that differ in condition, quantity, style, and availability. Brand-new products may have multiple units, while pre-loved garments may consist of individual pieces with unique characteristics. These differences make accurate product classification and inventory monitoring essential. A digital system could maintain detailed product records and update availability when items were reserved or sold.

The study also supports the inclusion of organized product listings and sales reports in the proposed system. Product descriptions containing information about condition, size, price, and category could help customers make informed decisions. Meanwhile, sales reports could help the owner determine which products were popular and which items required additional promotion. By supporting accurate records and organized transactions, the system could help Margaux Collections improve its daily operations while accommodating the specific characteristics of pre-loved fashion products.

Overall, Biana's study provides a local research basis for understanding the cultural, economic, and environmental significance of ukay-ukay in the Philippines. Its findings reinforce the importance of organized product information, customer-oriented selling practices, and digital tools in managing a pre-loved clothing business. These considerations support the development of the Margaux Collections Ordering System as a practical tool for improving inventory control, order processing, and customer service [7].

How Facebook Communication Strategies Facilitate Ukay-Ukay Live Selling (2024)

The study How Facebook Communication Strategies Facilitate

Ukay-Ukay Live

The study titled How Facebook Communication Strategies Facilitate Ukay-Ukay Live Selling and Buying (2024), published in the European Proceedings of Social and Behavioural Sciences, investigates the communication strategies used by ukay-ukay sellers in the Philippines during Facebook Live selling sessions and their influence on buyers' behavior. The research addresses a gap by demonstrating how digital platforms have transformed traditional ukay-ukay selling—from physical stores and face-to-face bargaining—into interactive, real-time online commerce through Facebook Live. This platform has become vital for small online businesses, creating a novel interactive experience distinct from conventional retail [8].

Unlike typical online shopping, where customers view static images and read product descriptions, Facebook Live provides immediate, direct communication between sellers and buyers. The study used a qualitative phenomenological design, involving in-depth interviews with five sellers and five buyers, alongside observational analysis of live sessions to confirm the application of communication strategies. It focused on verbal and non-verbal techniques, their effectiveness, and how these impacted buyer engagement, purchase decisions, and satisfaction [8].

Among the prominent strategies is entertainment, involving humor, games, giveaways, and promotions to keep viewers interested and encourage longer viewing times. This approach fosters a social and relaxed shopping environment, turning transactions into interactive social interactions [8]. Sellers also leveraged price promotions including discounts and flash sales to create urgency, enticing buyers to make quick purchasing decisions, especially since affordability is a key attraction in the ukay-ukay market [8].

Informal, conversational communication was another tactic. Sellers engaged buyers through comments, jokes, greetings, and terms of endearment to build personal connections, making the experience more relatable and communal rather than a mere business transaction [8]. Product presentation was carefully managed through live demonstrations using mannequins or live models, describing materials, colors, sizes, and conditions to help viewers assess the items more accurately [8].

Transparency was crucial, as sellers openly disclosed defects such as stains or fabric damage, which fostered trust and credibility by reducing buyer uncertainty and disappointment associated with secondhand clothes [8]. Visual non-verbal cues like seller appearance and background setup enhanced professionalism and trustworthiness during broadcasts [8].

Facebook's sharing feature was integral for expanding session reach, allowing viewers to share broadcasts with their networks and generating a wider audience without high advertising costs [8].

The live interactions reduced typical online shopping uncertainties by allowing real-time questions and immediate answers. This interaction built social proof and community among buyers, making shopping a shared, engaging experience [8]. The entertainment elements extended viewing time, exposing buyers to more products and potentially encouraging impulse purchases [8].

Buyer-seller responsiveness was critical for transaction completion, requiring sellers to promptly acknowledge comments, manage reservations accurately, and clarify payment and delivery instructions. Challenges arose with unreliable buyers, referred to as "joy miners," who reserved but did not conclude purchases, leading sellers to institute clear rules for managing transactions [8].

Despite advantages, sellers faced technical difficulties such as unstable internet, poor lighting, and audio issues, which impacted viewer confidence and session quality. Additionally, handling high volumes of comments and varying viewer numbers demanded strong organizational skills and adaptability, prompting sellers to refine their approach continually [8].

This study contributes to local literature by highlighting that Facebook Live offers affordable yet interactive marketing opportunities for small ukay-ukay businesses. It shows the need for sellers to emphasize communication, entertainment, transparency, and relationship-building beyond just pricing. Recommendations include improving communication skills, honest product disclosure, effective presentation, prompt buyer responses, and clear transaction guidelines. For policymakers and educators, the findings stress promoting digital literacy, technical training, and better internet infrastructure to support the sustainable growth of online ukay-ukay commerce [8].

Level of Satisfaction Among Women on Online Ukay-Ukay (2023) The study Level of Satisfaction Among Women on Online Ukay- Ukay (2023), published in the International

The study titled Level of Satisfaction Among Women on Online Ukay-Ukay (2023), published in the Journal of Academic Research and Innovative Engineering Education (IJARIIE), examines the satisfaction levels of female consumers regarding their online ukay-ukay shopping experiences in the Philippines. This research is important because women constitute a significant segment of ukay-ukay buyers, yet their unique experiences, expectations, and satisfaction factors have been understudied. As online shopping continues to reshape secondhand clothing purchases, understanding what influences women's satisfaction with online ukay-ukay sellers and transactions becomes essential. Employing quantitative methods, the study measured satisfaction across several key dimensions: product quality, pricing, delivery service, seller communication, and perceived value for money [9].

Product quality was identified as the most critical factor impacting satisfaction. Respondents were particularly attentive to the condition of items, the accuracy of product descriptions, and consistency between online photos and actual delivered products. Because buyers cannot physically inspect or try on items before purchase, the reliability

of seller-provided information is crucial. Discrepancies between advertised and received items often lead to disappointment, distrust, and poor seller evaluations [9].

Price competitiveness also significantly affected customer satisfaction. While affordability primarily attracts buyers to online ukay-ukay—offering access to branded or unique pieces at lower costs—the research highlighted that customers were willing to pay higher prices if quality, description accuracy, dependable service, and transaction security were assured. This indicates that pricing alone does not determine satisfaction; buyers assess price fairness relative to product and service quality [9].

Efficient delivery was another essential factor. Negative experiences from delayed shipments, unclear schedules, high shipping fees, and poor packaging hurt satisfaction. Buyers expect timely processing and shipment, especially for limited or single-piece items. Well-packaged orders that arrive on time, along with accurate tracking, build buyer confidence and improve evaluations of sellers [9].

Seller communication played a substantial role in shaping satisfaction. Prompt and clear responses to inquiries, thorough product information, and transparent interaction throughout the transaction enhanced the customer experience. Communication quality was vital for details like measurements, fabric condition, flaws, shipping, and returns. Sellers who neglected inquiries or inadequately addressed problems faced negative ratings regardless of pricing [9].

Overall satisfaction among women was moderate to high, though challenges like inconsistent sizing, difficulty judging item condition from photos, incomplete descriptions, and limited return options remained concerns. Buyers preferred precise garment measurements rather than generic size labels, which helped reduce uncertainty [9].

Detailed product descriptions, multiple photos from different angles, sizing charts, and clear return policies were highly valued. These elements allowed consumers to make more informed decisions and mitigated risks associated with buying secondhand online. Trust-building mechanisms such as seller ratings, customer reviews, testimonials, and active social media engagement further reduced perceived risks and increased satisfaction. Buyers tended to trust sellers with positive feedback, consistent transactions, and visible customer interactions [9].

Demographically, younger women, especially those aged 18 to 30, reported higher satisfaction levels compared to older groups, possibly due to greater digital literacy, comfort with online shopping, and cultural acceptance of thrift fashion. Repeat customers also expressed more satisfaction than first-time buyers, suggesting that experience reduces uncertainty and fosters more confident purchasing strategies [9].

The study contributed important insights into female consumer behavior and online secondhand clothing consumption in the Philippines, showing that satisfaction depends on multiple factors beyond price, including quality, accurate information, reliable delivery, transparent communication, and trustworthiness. Sellers who prioritize these areas can differentiate themselves in a competitive market, encouraging repeat purchases and positive word-of-mouth [9].

For future research, the study suggests exploring gender-specific consumer patterns, satisfaction drivers across different ukay-ukay segments, and how satisfaction influences loyalty. It also recommends examining differences among social media sellers, e-commerce sites, and live-selling models, as well as considering influences of sustainability awareness, income, body size, and prior shopping experience on satisfaction [9].

Filipino Gen Z Consumer Behavior Towards Ukay-Ukay (2024)

The study titled Filipino Gen Z Consumer Behavior Towards Ukay-Ukay and Their Sustainable Fashion Practices (2024), published in the De La Salle University Senior High School Research Convention Proceedings, explores how Filipino Generation Z consumers engage with ukay-ukay and the factors motivating their sustainable fashion practices. This study is timely, given that Generation Z represents one of the fastest-growing demographics in the ukay-ukay market. Their participation has contributed to a cultural shift, transforming secondhand clothing from an

association with social stigma or financial necessity into a mainstream and accepted fashion choice. Using survey research, the study examined attitudes, purchasing habits, and values that shape Filipino Gen Z's relationship with ukay-ukay and sustainable fashion consumption [10].

The findings revealed three main motivations driving Gen Z's ukay-ukay participation: affordability, sustainability awareness, and unique fashion expression. Affordability remains a key reason, especially as this demographic faces financial constraints like student loans, entry-level incomes, and rising living costs. Ukay-ukay offers access to stylish, branded, and diverse clothing at lower costs. However, affordability is no longer the sole factor; sustainability consciousness is equally critical. Many respondents see their secondhand purchases as environmentally responsible, connecting them to values such as climate action, waste reduction, and mindful consumption. The third motivation is unique fashion expression, where Gen Z uses ukay-ukay to craft distinct personal styles that set them apart from the mass-produced fast-fashion trends [10].

The study also found that Filipino Gen Z consumers predominantly use digital platforms like Facebook Marketplace, Instagram, and TikTok to discover, promote, and purchase secondhand clothing. These platforms match Gen Z's digital-native profiles by offering easy access to sellers, product details, styling inspirations, and curated selections beyond traditional physical stores. Through these channels, consumers browse conveniently, compare prices, communicate directly with sellers, and participate in online thrift fashion communities. The research further showed that Gen Z buyers actively research sellers, reviewing ratings, feedback, and social media presence to verify credibility and reduce shopping risks. Many also follow ukay-ukay content creators who provide styling tips, haul videos, recommendations, and education about sustainable fashion, contributing to a digital culture that normalizes and celebrates secondhand clothing among younger users [10].

The study emphasized that Gen Z's ukay-ukay engagement aligns with broader circular economy and sustainable consumption values. Respondents expressed concern about textile waste, resource-intensive fast fashion, and ethical issues in the apparel industry. By purchasing secondhand, they help prolong garment lifespans, reduce demand for newly made clothes, and divert textiles from landfills. This value-driven consumption distinguishes Gen Z from earlier generations, who often viewed ukay-ukay as an economic necessity. For Gen Z, it is both a budget-friendly option and a form of environmental activism and personal expression. The alignment between values and behavior also fosters stronger consumer loyalty; Gen Z consumers tend to favor sellers who articulate sustainability messages, advocate responsible consumption, and demonstrate environmental care. This suggests that future growth of the ukay-ukay market depends partly on maintaining credibility and resonating with Gen Z sustainability values [10].

This study contributes significantly to understanding sustainable fashion, secondhand shopping, and Filipino consumer behavior. It documents a generational shift where Gen Z consumers play a pivotal role in advancing sustainable fashion through digital engagement, self-expression, and environmental awareness. Their participation shows that secondhand clothing can be a key component in sustainable consumption when linked to affordability, identity, and eco-friendly values.

The findings also hold implications for policymakers, educators, and businesses aiming to promote circular economy practices. Policymakers can leverage the results to support textile waste reduction initiatives and strengthen secondhand clothing systems. Educators may integrate ukay-ukay and sustainable fashion topics into environmental curricula to demonstrate the environmental impact of everyday buying decisions. Meanwhile, businesses can attract Gen Z by offering transparent product information, responsible selling methods, and authentic sustainability commitments. The study suggests that Gen Z's engagement with ukay-ukay could serve as a gateway for broader sustainability education and behavioral shifts. For future studies, it recommends longitudinal research to track how Gen Z's sustainable fashion values, shopping habits, and environmental concerns evolve as they mature and gain spending power [10]**.**

Experiences of Online Sellers of Pre-Loved Clothes (Zenodo, 2023)

The study Experiences of Online Sellers of Pre-Loved Clothes (2023), archived on Zenodo, investigated the lived experiences, challenges, and adaptive strategies of online sellers of pre-loved clothing in the Philippines. This qualitative research provided a seller-centered perspective that complemented existing consumer-focused studies, offering a more comprehensive understanding of the digital ukay ecosystem. Using in-depth interviews and observations, the study captured the daily realities of individuals who transitioned into online ukay entrepreneurship, documenting both opportunities and obstacles encountered within this informal yet dynamic market [11].

Several critical challenges faced by sellers were identified. Inventory sourcing emerged as a primary difficulty, with sellers reporting irregular availability of quality items, competition among sellers for desirable products, and the physical labor required to sort, clean, and prepare garments for sale. Pricing was challenging as sellers balanced market competitiveness with profitability, considering the variable condition of each item and unpredictable customer demand. Additionally, platform dependency created vulnerabilities; changes in algorithms on platforms like Facebook, Instagram, or Shopee could significantly impact visibility and sales. Fraudulent buyer behaviors, such as fake payment screenshots and abandoned purchases, also posed ongoing financial risks demanding constant vigilance from sellers [11].

Despite these challenges, successful sellers adopted adaptive strategies including multi-channel selling across Facebook Marketplace, Instagram, Shopee, and TikTok. This multi-platform presence mitigated platform dependence while maximizing sales opportunities. Key competitive advantages included content creation capabilities like product photography, video demonstrations, and engaging captions. Sellers also built customer loyalty through personalized communication, follow-ups, and loyalty programs, which contributed to income stability. Financial management skills such as inventory tracking, cash flow monitoring, and emergency fund maintenance were essential to managing income variability common in online selling [11].

The study further highlighted the entrepreneurial resilience of online ukay sellers, many of whom balanced selling activities with other forms of employment or education. For some, online ukay selling was their primary livelihood; for others, it served as supplemental income or a step toward full-time entrepreneurship. Sellers developed diverse skills spanning product sourcing, digital marketing, customer service, photography, and financial management. This adaptability reflected broader Filipino cultural values of resourcefulness, flexibility, and improvisation, enabling business continuity despite operating in an informal and lightly regulated environment with limited institutional support [11].

Contributing significantly to local literature, this study centered seller perspectives in understanding the digital ukay economy, revealing that online selling involves complex skills beyond mere reselling, including digital literacy, business management, marketing, and communication. For organizations supporting online entrepreneurs, the findings suggested capacity-building in digital marketing, financial management, fraud prevention, customer relations, and legal compliance. Policymakers were advised to recognize online ukay selling's economic importance and consider formalization and supportive policies rather than prohibitive measures to foster safer, sustainable, and organized digital commerce [11].

Consumer Behavior Towards Online Clothes Shopping in San Fernando City, La Union

Torres (2021) investigated the online clothing purchasing behavior of consumers in urban barangays of San Fernando City, La Union. The study aimed to profile online clothing shoppers and identify factors influencing their buying decisions, focusing on convenience, website features, time efficiency, and online transaction security. It employed a descriptive research design using a survey questionnaire distributed to individuals with prior experience purchasing clothes through online platforms. This method allowed the collection of actual consumer behavior data within the community during the study period [12].

The study surveyed 382 respondents who were active online clothing shoppers residing in selected urban barangays of San Fernando City. The participants were chosen based on their engagement in online clothes buying to ensure relevant insights. Data collected included demographics, access to computers and the internet, and self-assessed

internet skills. Descriptive analysis was used to summarize and present a clear overview of the local online shopping environment and shopper characteristics [12].

Findings revealed that most online shoppers were female employees aged 21 to 25 years with a monthly income of at least Php 5,000. Many respondents had internet access and computer use at home or work, enabling convenient participation in online shopping. They also rated their internet skills as good, indicating confidence in navigating online shopping sites and completing orders. These factors contributed to their active engagement and demonstrated the importance of technological accessibility in supporting the adoption of online shopping [12].

Several factors were found to strongly influence buying behavior, such as convenience—the ability to shop anytime and from any location without visiting physical stores. Website design, including attractiveness, organization, and usability, positively affected the shopping experience and encouraged purchases. The results highlighted that consumers evaluated both product offerings and the quality and accessibility of the online platform before making purchasing decisions [12].

Time-saving advantages were significant, as online shopping eliminated the need to physically visit stores and spend extended time browsing. Consumers utilized online platforms to conveniently explore products and decide on purchases from home or workplace. Additionally, the perceived safety and security of personal and financial data were crucial in fostering customer trust and encouraging continued online shopping. Thus, addressing transaction security became vital for online clothing sellers to retain consumers [12].

Based on these findings, Torres (2021) recommended entrepreneurs to adopt online selling as a viable strategy in the clothing industry. Enhancing website design, ensuring secure payment systems, and promoting convenience and efficiency were emphasized to meet consumer preferences. The study identified a promising market in San Fernando City among young, employed female consumers with adequate internet access and skills. Overall, the research underscored the critical roles of consumer demographics, technological accessibility, convenience, website features, time efficiency, and transaction security in shaping local online clothing purchasing behavior [12].

Emergence of Online Clothing Business and Its Effect on Local Businesses in Tuguegarao City

Banera, Binarao, De Castro, Furigay, and Mosquito (2022) examined the impact of the growth of online clothing businesses on local physical clothing enterprises in Tuguegarao City. The study aimed to describe the experiences of local entrepreneurs and assess how the rise of online competitors affected their business performance. Employing a descriptive research design, the study presented the current situation of traditional physical stores amid increasing online competition [13].

The study surveyed 30 local clothing business owners selected through purposive sampling, all of whom had direct experience with the effects of online competition. Data were collected using survey questionnaires that inquired about financial performance indicators and coping strategies. The descriptive approach summarized the perceptions and observed business changes over time [13].

Findings revealed that the emergence of online clothing businesses significantly negatively impacted the financial performance of local stores. Respondents reported considerable decreases in sales, profits, working capital, customer demand, and their ability to retain customers. The overall category mean score (2.7267) indicated a notable decline rather than a marginal one, suggesting a considerable shift of consumers from physical to online shopping, intensifying pressure on local stores [13].

The study identified coping strategies employed by entrepreneurs in response to these challenges. All respondents (100%) utilized sales talks to maintain customer interest, 96.67% offered discounts to attract and retain buyers, while 70% provided exclusive bargains to compete with online promotions. These findings demonstrated efforts by local businesses to adapt to evolving competitive conditions despite their primarily physical store operations [13].

Based on these results, the researchers recommended that local business owners consider embracing online clothing sales. They encouraged the use of various social media platforms to promote and sell products, increasing customer reach and responding to changing shopping preferences. The combination of traditional store operations with online selling was suggested as a way to enhance financial performance and competitiveness in the digital retail landscape. This study underscores the importance of adaptation and innovation among local enterprises to survive growing online competition [13]

Effectiveness of Online Services Offered by Pre-Loved Clothes Sellers in Batangas City

Group 1 (2021) conducted a study examining the effectiveness of online services provided by pre-loved clothes sellers in Batangas City. This research aimed to identify the key factors influencing consumer decisions to purchase pre-loved clothing online and how these factors contributed to sellers' success. The study focused on elements such as fashion appeal, brand recognition, price competitiveness, delivery efficiency, and overall customer satisfaction within the context of Batangas City [14].

The study analyzed how these factors affected consumers' purchasing decisions from online pre-loved clothes sellers. Fashion and brand were assessed based on the trendiness and recognition of clothing items. Price competitiveness was evaluated in terms of affordability compared to alternatives. Delivery efficiency focused on speed and reliability, while customer satisfaction measured buyers' experiences from browsing to product receipt [14].

Findings highlighted social media platforms—particularly Facebook Marketplace and Instagram—as the primary channels for selling pre-loved clothes. These platforms enabled sellers to easily upload product photos, interact with potential buyers, and manage order inquiries efficiently. The popularity and accessibility of these platforms made them convenient marketplaces for discovering and purchasing secondhand clothing, underscoring the importance of selecting suitable online platforms for success [14].

The research also revealed that the effectiveness of online services by pre-loved clothes sellers largely depended on maintaining a strong online presence. Active and consistent posting of items, prompt responses to messages, and timely updates about new arrivals and promotions significantly contributed to attracting and retaining customers. Flexible pricing strategies, such as periodic price adjustments or promotional deals, were effective in helping sellers stand out in a competitive market [14].

Lastly, the study emphasized the importance of cultivating positive customer relationships to encourage repeat purchases. Sellers who communicated clearly, responded courteously to inquiries, and addressed issues promptly earned greater trust and loyalty from customers. These factors were crucial for sustained customer satisfaction and long-term success in the online pre-loved clothing business [14]. Overall, the study concluded that a combination of strong online presence, strategic pricing, and quality customer relations enhanced the effectiveness of online services for pre-loved clothes sellers in Batangas City.

Promotional Strategies and Consumers' Purchase Intention on Garment Bazaar Retailers in the National Capital Region

Torres, Santos, and Lim (2023) investigated how promotional strategies employed by garment bazaar retailers influenced consumers' purchase intentions within the National Capital Region (NCR), Philippines. The study focused on assessing the effects of promotional activities on consumers' attitudes, subjective norms, perceived behavioral control, and purchasing intentions, especially amid the transition from traditional retail to online platforms during and after the COVID-19 pandemic. This research is relevant to the Margaux Collections Ordering System because it provides insights into how promotional tactics and platform choice may affect customers' willingness to purchase clothing products [15].

Using a survey research design, the study gathered data from 510 consumers across 16 cities and one municipality in the NCR. Respondents had prior experience purchasing garments during and after the pandemic, a period marked by increased reliance on digital selling channels. Data collection was done via online questionnaires, which measured perceptions of promotional effectiveness, consumer attitudes, social influences, perceived control, and purchase intentions. This methodology enabled pattern identification in consumer perceptions and examined the relationship between promotional strategies and buying behavior [15].

The study examined various selling platforms, including physical stores, virtual stores, company websites, and social media, acknowledging that consumers might respond differently to promotions depending on the platform. For instance, a promotion effective on social media might not have the same impact on a company website or physical store. This finding is significant for Margaux Collections, which could leverage multiple digital channels to promote both brand-new and preloved clothing. The ordering system could serve as a centralized platform where consumers access comprehensive product information after being engaged by promotional content elsewhere [15].

Findings indicated a low to moderate correlation between perceived promotional effectiveness and consumers' attitudes, subjective norms, behavioral control, and purchase intention. This suggests that while promotions can influence consumers' perceptions and foster purchase willingness, they alone do not determine buying behavior. Other factors such as product quality, price, convenience, trust, and personal preferences, as well as the platform characteristics, also play critical roles. For Margaux Collections, this implies that promotional efforts must complement a well-functioning ordering system [15].

The study further identified perceived promotional effectiveness as a precursor to favorable consumer attitudes and intentions. When promotions are perceived as relevant, appealing, and useful, they can enhance consumer interest and motivate purchase consideration. For Margaux Collections, attractive product announcements, discount offers, new arrivals, and other marketing content could spark initial customer interest, with the ordering system providing the detailed information and transaction tools to facilitate actual purchases [15].

Age emerged as a significant segmentation factor influencing purchase intention. Different age groups showed diverse preferences for product presentation, promotional styles, pricing, and preferred sales platforms. Margaux Collections could tailor promotional content accordingly, with younger consumers responding better to visually engaging digital content, while other groups may prioritize detailed descriptions or affordability [15].

The study also found that purchase intention varies by selling platform type. Physical and virtual stores, websites, and social media platforms offer distinct consumer experiences that affect purchasing behavior differently. Hence, a one-size-fits-all promotional approach is ineffective; marketing strategies must be adapted to platform specifics. For Margaux Collections, social media could be employed for product awareness, while the ordering system manages browsing and sales [15].

Recognizing retail shifts accelerated by the COVID-19 pandemic, the research emphasized the need for retailer adaptability as consumer behavior evolves across digital environments. Understanding these dynamics is essential for Margaux Collections to effectively serve its online customers [15].

Overall, Torres, Santos, and Lim (2023) concluded that garment retailers must customize promotional strategies per platform and customer demographics to maximize purchase intention. Coordinated promotional activities paired with comprehensive product information and efficient ordering systems can better convert customer interest into sales. These insights support Margaux Collections' development of an integrated online ordering and promotional platform aligned with current retail trends [15].

Web-Based Ordering System with Inventory Integration for a Small Beverage Business Catubag, Dela Cruz, Reyes, Santos, and Torres (2024)

Catubag, Dela Cruz, Reyes, Santos, and Torres (2024) developed a web-based ordering system integrated with inventory management tailored for a small lemonade beverage business. The study aimed to streamline ordering and transactions to improve order accuracy, inventory control, and overall operational efficiency. Employing a developmental research design, the system was created, tested, and evaluated as part of an undergraduate capstone project, demonstrating a practical and affordable solution for small businesses. This approach is particularly relevant to the Margaux Collections Ordering System, highlighting the benefits of consolidating ordering and inventory functions into a unified platform [16].

The respondents comprised the owners and staff who directly used and assessed the system, as they managed daily ordering, transactions, and inventory. Data were collected through system testing, observation of daily operations, and feature evaluations. Key system components included a carting function, payment gateway simulation, inventory management, sales data aggregation, product performance monitoring, and automated report generation. The developmental process enabled refinement of the system to suit actual operational needs and constraints [16].

Findings indicated that the web-based ordering system enhanced operational efficiency. The carting feature improved order accuracy by allowing customers to review selections before finalizing purchases, helping prevent incorrect orders. This function is directly applicable to Margaux Collections, where customers purchasing both brand-new and preloved clothing could verify items, quantities, and prices before checkout [16].

The study also demonstrated the value of integrating payment processes within the ordering workflow. Mock payment gateways simulated secure and flexible transaction methods, supporting various payment options. While Margaux Collections employs GCash payments, these findings emphasize the importance of a structured payment process to guide customers and facilitate systematic payment verification by the business owner [16].

Inventory integration significantly contributed to enhanced stock monitoring, reducing stockouts and overstock scenarios. Accurate real-time inventory management is vital for small businesses operating with limited stock, especially for unique preloved items, where single quantities are common. By maintaining updated inventory data, Margaux Collections can prevent the sale of unavailable products [16].

Moreover, integrating inventory management reduces errors caused by manual stock updates after receiving orders through multiple channels, preventing duplicate sales and delays. Sales data aggregation and product performance tracking enable identification of popular items and informed decision-making on inventory, pricing, and promotions. Automated report generation assists in summarizing sales trends and operational metrics, supporting strategic business management [16].

The research underscored that small enterprises successfully adopt digital tools when systems are user-friendly, functional, affordable, and aligned with existing processes. Expensive or overly complex platforms are unnecessary; instead, practical solutions tailored to business scale and needs suffice. This insight strongly supports the design philosophy behind the Margaux Collections Ordering System [16].

Additionally, digital systems enhance customer experience by simplifying ordering, reducing manual communication, and providing consistent product availability and transaction follow-up. For Margaux Collections, this would translate into a smoother online shopping experience for both brand-new and preloved clothing customers [16].

Lastly, the study recommended that small businesses adopt web-based ordering systems integrated with inventory management to optimize operations. It also advocated for using accessible technologies like PHP, MySQL, and Bootstrap, known for their affordability and flexibility—making them suitable for small-scale applications like Margaux Collections' platform. Real-time inventory updates within the ordering system are crucial for ensuring accurate product availability, enhancing customer satisfaction, and minimizing inventory-related issues [16].

Overall, Catubag, Dela Cruz, Reyes, Santos, and Torres (2024) offered compelling evidence supporting an integrated web-based ordering and inventory system as an effective approach to improving small-business operations. Their findings are directly applicable to Margaux Collections, underscoring how such a system can centralize product browsing, ordering, payment, and stock management to drive operational efficiency and customer convenience [16].

Factors Associated with the Patronage of Ukay-Ukay Products in Davao City — Villa et al. (2013)

Villa et al. (2013) conducted a study to investigate the factors influencing Filipino consumers' patronage of ukay-ukay, or secondhand clothing, in Davao City. The research involved 388 respondents and provided valuable insights into consumer behavior regarding preloved clothing. This local study is relevant to the Margaux Collections Ordering System because it offers a detailed understanding of why Filipino consumers continue to purchase secondhand fashion and what product characteristics affect their buying decisions [17].

The study found that affordability was the primary reason respondents chose ukay-ukay products. Many consumers preferred secondhand clothing due to its generally lower price compared to brand-new apparel. This finding demonstrates that price is a critical factor in consumer decision-making, especially for those seeking fashion within a limited budget. For Margaux Collections, this supports the inclusion of preloved products as a way to attract cost-conscious customers looking for practical options [17].

However, affordability was not the sole factor influencing patronage. The study also identified durability, variety, and availability of branded clothing at affordable prices as significant considerations. Customers appeared to evaluate the overall value of items, considering factors beyond price, such as style, longevity, and brand. These insights imply that Margaux Collections should highlight these qualities in product listings to better inform customer choices [17].

Variety was another important factor. The range of types, styles, brands, and designs available in ukay-ukay stores attracted consumers by increasing the likelihood of finding items matching personal preferences and budgets. This supports the need for organized product categories in the Margaux Collections platform, allowing customers to efficiently browse inventory based on relevant attributes such as size, price, condition, and product type [17].

The research showed that consumers viewed ukay-ukay clothing as a practical and affordable shopping alternative that met needs for style, durability, and functionality. This suggests that preloved items should be presented not merely as used goods, but as valuable fashion options, enhancing their appeal on Margaux Collections [17].

Social influence also played a role in purchasing decisions. Recommendations from family, friends, and coworkers boosted consumer confidence in buying preloved clothing. This underscores the importance of ensuring a reliable customer experience within the Margaux Collections system to foster positive word-of-mouth and repeat patronage [17].

Additionally, the study emphasized the need to provide comprehensive product information. Since preloved clothing varies widely in condition, size, and appearance, consumers require detailed data to make informed decisions. Margaux Collections can address this by offering organized product listings with clear photographs, pricing, size details, descriptions, and condition notes [17].

Transparency about product conditions is particularly critical, as preloved items often show signs of wear such as minor defects or fading. Clear disclosure sets realistic customer expectations, reducing misunderstandings and increasing satisfaction—an approach supported by the design of Margaux Collections [17].

Furthermore, the study's insights extend to comparing brand-new and preloved products. With consumers considering multiple factors in their purchase decisions, offering both categories on one platform enables customers to compare options based on price, condition, and other criteria, thereby supporting more informed and flexible shopping choices [17].

The local context of this study enhances its value for Margaux Collections, as it focuses specifically on Filipino consumers, revealing preferences for affordability, variety, durability, and accessibility in the preloved clothing market. This underscores the potential market for secondhand fashion within the Philippines, reinforcing the importance of integrating preloved items into the Margaux Collections platform [17].

Overall, Villa et al. (2013) provide strong rationale for including preloved clothing on the Margaux Collections site. Their findings about affordability, product variety, durability, brand availability, and social influence highlight the need for well-organized product information and trusted customer service, all of which can enhance accessibility and convenience for Filipino shoppers [17].

2.3 Related Literature

The two preceding sections reviewed foreign and local studies that tested similar systems in real business settings. This section turns to literature that explains the ideas behind those systems — the models and concepts that described why an ordering platform, a digital payment method, or a preloved product listing worked the way it did. While the studies in Sections 2.1 and 2.2 focused on what was built and how it performed, the literature discussed here focused on why these systems made sense in the first placed. Five areas were discussed: web-based ordering and inventory systems, digital payment and e-wallet adoption, online shopping cart and checkout behavior, and secondhand fashion consumption. Each of these connected directly toa specific part of the Margaux Collections Ordering System, from its checkout flow down to how preloved items were described on the product page.

Web-Based Ordering and Inventory Systems

Tanaman, Baylosis, Abiles, Catungal, and Encarnacion (2023) built a web-based inventory management system for a small business in Pagadian City that had previously relied on manual, paper-based stock recorded. Their system let the owner tracked incoming and outgoing products, monitored stock across branches, and generate reports without leaving a spreadsheet open all day. What stands out in their worked was not the technology itself — PHP and MySQL were common choices — but the motivation: a small business owner needed something that fit how she already worked, rather than a system that demanded she changed her entire process. That same logic applied to Margaux Collections, whose owner currently tracked stock through Facebook posted and manual counted.

The study by Tanaman et al. (2023) was particularly relevant because it showed that the success of a digital system did not depended on how advanced its features were, but on how well it matches the actual workflow of the business. In the case of the Pagadian City business, the owner did not needed a complex enterprise resource planning system. What she needed was a simple tool that could replace her paper recorded and reduced the time she spent on stock monitoring. The same principle guided the design of the Margaux Collections Ordering System. Since the business was operated by a single owner without staff, the system was intentionally kept simple and focused on the core tasks of product listing, order tracking, and inventory updated.

Otobo and Alegbe (2024) reached a similar conclusion while designing an inventory system for small and medium production companies. They found that SMEs were constrained by limited resources and staff, so a system built for them had to be simple enough that non-technical users could operate it without training. Their design prioritized real-time stock visibility and automated reporting over more advanced features like demand forecasting, arguing

that added complexity often goes unused in a small-business context. This was a reasonable trade-off for Margaux Collections as well, where the priority was accurate stock counted and clear order status rather than predictive analytics.

The insight from Otobo and Alegbe (2024) was especially useful because it highlighted a common mistake in system development: building features that looked impressive on paper but were never actually used in practice. For a small business like Margaux Collections, every feature should have a clear purpose and should make the owner's daily work easier. This was why the proposed system focused on practical functions such as product management, order processing, and inventory monitoring, rather than advanced analytics or automated recommendations. The goal was not to create the most sophisticated system possible, but to create a system that the owner could rely on every day.

Pathak et al. (2024) took a slightly different approach in developing a web-based inventory system for a retail stored, using a prototyping method that involved the actual end users at each stage — requirements gathering, interface design, and testing. Black-box testing confirmed that the system's functioned worked as intended, while a small usability tested with six respondents surfaced friction points before launch. This iterative, user-involved process mirrors the testing procedures described in Section 3.8 of this study, where the Margaux Collections system was likewise validated with the business owner and a small group of customers before full deployment.

The approach taken by Pathak et al. (2024) reinforces an important lesson: the best way to build a useful system was to involve the people who would actually used it.

When end users were consulted during requirements gathering and interface design, the resulting system was more likely to match their needed and expectations. This was why the Margaux Collections Ordering System was being developed with direct input from the business owner. Her feedback on product listing, checkout flow, and inventory management helped ensured that the system reflected how she actually ran her business, rather than how a developer imagines she should run it.

Taken together, these studies on web-based ordering and inventory systems provided a clear message: small businesses benefited most from systems that were simple, practical, and designed around their actual workflow. For Margaux Collections, this meant that the proposed system should focus on reducing manual work, improving stock accuracy, and making daily operations easier for the owner. The literature supported the decision to build a system that was functional and user-friendly rather than overly complex or feature-heavy.

Digital Payment and E-Wallet Adoption

Because the proposed system settles payment through a GCash QR code, literature on e-wallet adoption was directly relevant. Belmonte, Prasetyo, Cahigas, Nadlifatin, and Gumasing (2024) surveyed 500 Filipino Generation Z and Millennial respondents using an extended Technology Acceptance Model and found that perceived trust was actually a stronger predictor of e-wallet used than perceived value — in other words, people were more willing to pay a little more for a service if they trust it, than they were to use a cheaper option they were unsure of. Perceived security and social influenced also played meaningful roles. Applied to Margaux Collections, this suggests that visible payment confirmation and a clearly labeled order and payment status (as already built into the system's order_status and payment_status fields) were not just administrative details — they were what made customers comfortable enough to pay through GCash in the first placed.

The finding by Belmonte et al. (2024) that trust mattered more than value was especially important for a small business like Margaux Collections. Since the business did not had the brand recognition of large e-commerce platforms, customers might be more hesitant to complete payments through GCash unless they felt confident that their order would be processed correctly. This was why the system included clear payment instructions, visible order

status updates, and a straightforward checkout flow. These features were not just about convenience; they were about building trust with customers who might be unfamiliar with online ordering systems.

Another useful insight from the study was the role of social influenced in e-wallet adoption. Belmonte et al. (2024) found that people were more likely to use e-wallets when they see others in their social circle using them. For Margaux Collections, this suggests that word-of-mouth and customer recommendations could play a significant role in encouraging GCash payments. When customers had a positive experience with the ordering system and see that their friends or family members also used it, they were more likely to trust the platform and completed their purchases. This reinforces the importance of providing a smooth and reliable shopping experience from start to finish.

The literature on e-wallet adoption also highlighted the importance of perceived security. Even if a payment method was convenient, customers might hesitate to use it if they were concerned about fraud or unauthorized transactions. This was why the Margaux Collections Ordering System included clear payment confirmation steps and maintained accurate records of all transactions. By making the payment process transparent and easy to verify, the system helped reduce customer anxiety and encouraged more people to use GCash for their orders.

Overall, the literature on digital payment and e-wallet adoption supported the design choices made in the proposed system. the focus on trust, security, and simplicity aligns with the factors that influenced customer willingness to use e-wallets in the Philippines. For Margaux Collections, this meant that the GCash integration was not just a technical feature — it was a key part of building customer confidence and encouraging online transactions.

Online Shopping Cart and Checkout Behavior

A recurring theme in e-commerce literature was that most lost sales did not happen because a customer disliked a product, but because something about the checkout process made them hesitate. Kukar-Kinney and Close (2010), whose framework was still widely cited, identified registration requirements and lengthy forms as leading causes of cart abandonment. Rajamma, Paswan, and Hossain (2009) extended this idea with the concept of perceived transaction inconvenience — the frustration created by technical glitches, being logged out mid-purchase, or having to re-enter information. More recently, Wang, Ye, Ning, Cheah, and Lim (2022) modeled this hesitation using a stimulus-organism-response framework and found that even the anticipation of a better deal elsewhere was enough to make a shopper stall at checkout and, eventually, leave without buying.

The study by Kukar-Kinney and Close (2010) was particularly relevant because it showed that even small inconveniences in the checkout process could have a significant impact on customer behavior. When customers were required to create an account before making a purchase, many of them simply leave the site without completing their order. This was why the Margaux Collections Ordering System allowed customers to checkout without full account registration. The goal was to reduce friction and made it as easy as possible for customers to complete their purchases.

Rajamma, Paswan, and Hossain (2009) added another important dimension to this discussion by introducing the concept of perceived transaction inconvenience. Their research showed that customers were sensitive to any delays, errors, or complications that occur during the checkout process. Even minor issues such as slow page loading or confusing form fields could cause customers to abandon their carts. This insight supported the decision to keep the Margaux Collections checkout process simple and straightforward. By limiting the number of required fields and avoiding unnecessary steps, the system reduced the likelihood of customer frustration and cart abandonment.

The more recent study by Wang et al. (2022) provided a psychological perspective on checkout behavior. Their stimulus-organism-response framework explains how external factors such as website design, pricing, and checkout

flow influenced a customer's internal state, which in turn affected their purchasing decisions. One of their key findings were that the anticipation of a better deal elsewhere could cause customers to hesitate at checkout and eventually leave without buying. This suggests that the Margaux Collections Ordering System should not only be easy to use but also provided clear value to customers. Transparent pricing, accurate product descriptions, and reliable order processing all contributed to a positive checkout experience that encourages customers to complete their purchases.

None of these studies were conducted in a Philippine or small-business context, and their statistical models go far beyond what this study attempts. Still, the underlying lesson carries over directly: the fewer steps and the fewer surprises a checkout process had, the less likely a customer was to abandon it. This was the reasoning behind keeping the Margaux Collections checkout toa small number of fields — delivery details, order method, and payment method — rather than requiring full account registration before a purchase could be completed.

The literature on checkout behavior also supported the inclusion of a shopping cart module in the proposed system. By allowing customers to review their selected items before confirming their order, the system helped prevent mistakes and gave customers a sense of controlled over their purchases. This was especially important for a business that sold both brand-new and preloved items, as customers might wanted to compare different products before making a final decision. The shopping cart feature also provided a clear summary of the order, including item names, prices, and quantities, which helped reduced confusion and improved the overall checkout experience.

Secondhand Fashion and Sustainable Consumption

Since Margaux Collections sold both brand-new and preloved items, literature on secondhand fashion consumption helped explain how the platform should present these products differently. Gilal et al. (2024) reviewed 96 studies published between 2012 and 2024 on secondhand clothing consumption and found that buyer decisions in this category were shaped by a distinct set of factors — seller credibility, item condition, and perceived risk — that did not carry the same weight when a customer was buying something new. Sepe (2025) added a behavioral layer to this, showing through an integrated model that a customer's attitude toward secondhand clothing and their personal sense of environmental responsibility together predict whether they would actually buy it, not just express interest in doing so.

The systematic review by Gilal et al. (2024) was particularly useful because it identified the specific concerns that influenced secondhand clothing purchases. Unlike brand-new items, preloved clothes carry additional uncertainties related to condition, authenticity, and seller trustworthiness. Customers wanted to know that the item they were buying matches the description and that the seller was reliable. This was why the Margaux Collections Ordering System included separated product types for brandnew and preloved items, along with detailed condition descriptions and care notes for preloved products. By addressing these concerns directly in the product listing, the system helped reduced perceived risk and encourages more customers to consider secondhand options.

Sepe (2025) added another important dimension to this discussion by examining the role of sustainability in secondhand fashion consumption. The study found that customers who were environmentally conscious were more likely to buy preloved clothing, not just because it was affordable but because it aligns with their values. This suggests that the Margaux Collections Ordering System could appeal toa broader audience by highlighting the sustainability benefited of buying preloved items. For example, product descriptions could include information about how purchasing secondhand clothing helped reduced waste and supported responsible consumption. This not only informs customers but also reinforces the positive impact of their purchasing decisions.

The literature on secondhand fashion also highlighted the importance of seller credibility. When customers buy preloved items, they were often more concerned about the seller's reputation than when buying brand-new products. This was because preloved items could vary significantly in condition, and customers wanted to feel confident that

they were making a safe purchase. The Margaux Collections Ordering System addressed this concern by providing clear contact information, order tracking, and a buyer-seller communication module. These features helped built trust between the seller and customers, making it easier for customers to ask questions and resolved any concerns before completing their purchase.

Taken together, these findings supported a design decision already presented in the proposed system: preloved and brand-new items were tagged by product_type and could carry separated care notes or condition descriptions (Section 3.4.5). Treating preloved listings as their own category, rather than folding them into the general catalog, gave sellers a way to address the credibility and condition concerns that this literature identified as central to a secondhand buyer's decision. This approach not only improved the shopping experience for customers but also helped the seller manage inventory more effectively by clearly distinguishing between different types of products.

The literature on secondhand fashion consumption also provided insight into the growing market for preloved clothing in the Philippines. As more consumers became aware of sustainability issues and sought affordable fashion options, the demand for secondhand items were expected to continue rising. This trend supported the decision to include preloved clothing in the Margaux Collections Ordering System, as it positions the business to take advantage of this growing market. By providing a well organized and trustworthy platform for buying preloved items, the system could attract both budget-conscious customers and those who value sustainable fashion.

Usability and System Acceptance in Small Business Platforms

Davis, Venkatesh, and Morris (2003) explained how usefulness and ease of use determined whether people actually accepted and used a new system. Their Technology Acceptance Model showed that even a capable system was of little value if it felt difficult or confusing to operate. They found that users consistently avoided complicated tools and preferred platforms that matched what they already knew how to do. These findings reinforced the design approach chosen for Margaux Collections. Because the owner managed the business alone without technical training, the system was kept straightforward and intuitive. Every feature was arranged so that daily tasks like listing items, checking orders, and updating stock required only simple, familiar steps. The literature confirmed that simplicity was not just a matter of convenience it was the most important factor in whether the system would actually be used every day.

Product Information Quality and Online Purchase Confidence

Park, Lennon, and Stoel (2005) examined how the quality of product information affected buying decisions in online clothing retail. Their study showed that when shoppers could not examine items personally, they depended almost entirely on written descriptions and images to judge quality, fit, and value. Incomplete or vague information made buyers uncertain, and that uncertainty caused many to abandon their purchase entirely. They concluded that detailed, accurate product details served as a substitute for physical inspection. This finding directly guided how listings were structured in the Margaux Collections system. Every product page included clear descriptions, size details, condition notes, and photographs. By providing complete and honest information, the system helped reduce hesitation and gave buyers the confidence they needed to choose preloved items without seeing them in person.

Trust and Reliability in Online Seller–Buyer Relationships

Ba and Pavlou (2002) studied how trust formed between buyers and sellers who had never met in person. Their research found that reliability, consistency, and clear communication helped bridge the gap created by physical distance. When sellers provided accurate information, kept their promises, and responded promptly, buyers became more willing to transact and even paid reasonable prices with confidence. On the other hand, uncertainty grew when details were missing or inconsistent. This literature supported the transparency features built into the Margaux Collections system — visible order status, clear payment confirmation, and straightforward communication options. These features did more than improve efficiency; they helped establish the kind of reliability that turned first-time buyers into returning customers. The system became a tool that demonstrated consistency and built trust with every transaction.

Inventory Accuracy and Operational Efficiency in Retail

Kull, Boyaci, and Krishnan (2013) explored how inventory accuracy influenced overall business performance. Their study showed that even small errors in stock records caused noticeable problems — missed sales, overbooking, customer disappointment, and wasted effort. They found that simple, real-time updates reduced these errors far more effectively than complex systems that were difficult to maintain. Accurate inventory data also helped the business owner decide which items to offer and when to promote them. This insight explained why the Margaux Collections system prioritized reliable stock tracking above advanced forecasting or analytics. Every time an order was placed or marked sold, inventory levels updated immediately. Keeping stock records accurate and visible helped prevent mistakes, saved time, and ensured that customers only saw items that were truly available.

Consumer Expectations and Transparency in Resale Markets

Chen, Zhang, and Wang (2022) reviewed buyer expectations specifically in second-hand and resale marketplaces. Their findings showed that buyers in this category placed especially high value on honesty, openness, and accurate condition reporting. They understood that preloved items carried signs of use, but they expected sellers to describe those signs clearly and fairly. When condition was disclosed openly, buyers felt respected and trusted the seller more. When important details were omitted, confidence dropped sharply, even if the price was low. This literature strongly supported the decision to include separate condition fields and transparent descriptions for every preloved listing in Margaux Collections. By being upfront about each item's actual condition, the system helped build trust before the purchase was made. It also reduced misunderstandings and created a smoother, more reliable experience for both buyers and the seller.

Summary of Related Literature

Across all four areas, the related literature reinforces a single point: a small business ordering system was judged less by the sophistication of its code and more by whether it removes friction — in stock tracking, in payment, at checkout, and in how products were described. This framing directly supported the feature choices detailed in

Chapter 3 of this study. The literature showed that successful systems were those that prioritized simplicity, trust, and user experience over advanced features that might not be used in practice. For Margaux Collections, this meant that the proposed system should focus on making daily operations easier for the owner and providing a smooth, reliable shopping experience for customers.

The reviewed literature also highlighted the importance of designing systems around the actual needs of the business. Whether it was inventory management, payment processing, checkout behavior, or secondhand fashion consumption, the studies consistently showed that the most effective systems were those that addressed realworld challenges faced by small businesses. This was why the Margaux Collections Ordering System was being developed with direct input from the business owner and was tailored to the specific workflow of her business. By aligning the system's features with the insights from the literature, the study aimed to create a practical and effective solution that could improve both business operations and customer satisfaction.

2.4 Synthesis of the Reviewed Literature

The foreign studies, local studies, and related literature reviewed in this chapter were drawn from different countries, different methods, and in most cases, different scales of business — yet they consistently arrive at the same handful of conclusions. Order and inventory systems succeed when they were simple enough for a non-technical owner to run day today. Digital payments were adopted when customers trust them, not simply when they were convenient. Shoppers abandon a purchase not because of the product but because of friction at checkout. And secondhand items sold when sellers actively addressed the concerns — condition, credibility, authenticity — that made buying something used different from buying something new.

One of the most important patterns that emerges from the reviewed literature was the emphasis on simplicity and practicality. Whether the study focused on inventory management, e-wallet adoption, checkout behavior, or secondhand fashion consumption, the findings consistently showed that small businesses benefited most from systems that were easy to use and directly addressed their operational challenges. This was particularly relevant for Margaux Collections, as the business was operated by a single owner without staff. The proposed system was designed to be straightforward and focused on the core tasks of product listing, order processing, and inventory management, rather than including advanced features that might not be used in practice.

Another common theme across the literature was the importance of trust in digital transactions. Studies on e-wallet adoption and secondhand fashion consumption both highlight that customers were more likely to engage with a platform when they felt confident that their transactions were secure and that the seller was reliable. For Margaux Collections, this meant that features such as clear payment instructions, visible order status updates, and a buyer-seller communication module were not just administrative details — they were essential for building customer confidence and encouraging online transactions. The literature supported the decision to prioritize trust-building features in the proposed system, as these were key factors in customer willingness to use e-wallets and purchase preloved items.

The literature on checkout behavior provided another valuable insight: the checkout process was often the point where customers decide whether to complete a purchase or abandon their cart. Studies consistently showed that even minor inconveniences such as lengthy forms, registration requirements, or technical glitches could cause customers to leave without buying. This was why the Margaux Collections Ordering System was designed to minimize friction at checkout by limiting the number of required fields and avoiding unnecessary steps. The goal was to make the checkout process as smooth and straightforward as possible, reducing the likelihood of cart abandonment and improving the overall shopping experience for customers.

When it comes to secondhand fashion consumption, the literature emphasizes that preloved items carry unique concerns that did not applied to brand-new products. Customers were more sensitive to issues related to item condition, seller credibility, and perceived risk when buying secondhand clothing. This was why the proposed system included separated product types for brand-new and preloved items, along with detailed condition descriptions and care notes for preloved products. By addressing these concerns directly in the product listing, the system helped reduce perceived risk and encouraged more customers to consider secondhand options. The literature also highlights the growing market for preloved clothing, particularly among environmentally conscious consumers, which supported the decision to include preloved items in the Margaux Collections Ordering System.

What the reviewed literature did not offer, however, was a system built around all four of these concerns at once. The foreign and local studies examined in Sections 2.1 and 2.2 largely described ordering or inventory systems built for a single type of business — either a conventional retailer or a straightforward inventory operation — without addressing the specific combination that Margaux Collections required: a catalog that separated new and preloved items, a checkout process that stayed short enough to avoid hesitation, and a payment flow built specifically around GCash rather than a generic payment gateway. The literature reviewed in Section 2.3 explains why each of these pieces mattered individually, but no single existing study, local or foreign, brings them together for a small Philippine fashion resale business operating primarily through Facebook.

This was the gap the Margaux Collections Ordering System was meant to fill. It was not a novel piece of technology — the PHP-and-MySQL stack, the shopping cart module, and the GCash integration all built on patterns already established in the literature discussed above. Its contribution was in how these established, individually tested ideas were combined and adapted for one specific business: a small, Facebook-based fashion retailer selling both new and preloved clothing, where the owner needed a system she could actually operated without a technical background. In that sense, the literature reviewed across this chapter did not just supported the idea of building a system — it points fairly specifically to the features this particular system needed to have in order to worked.

The synthesis of the reviewed literature also highlighted the importance of aligning system design with the actual workflow of the business. Studies on inventory management and system development consistently showed that the most effective systems were those that were designed around the need and constraints of the end users. For Margaux Collections, this meant that the proposed system should reflect how the owner currently managed her business, rather than forcing her to change her entire process. By involving the business owner in requirements gathering, interface design, and testing, the system was more likely to match her expectations and provided a practical solution to her operational challenges.

Finally, the reviewed literature underscores the value of creating a system that was both functional and user-friendly. Whether it was inventory management, payment processing, checkout behavior, or secondhand fashion consumption, the studies consistently showed that small businesses benefited most from systems that were easy to use and directly addressed their operational challenges. This was why the Margaux Collections Ordering System was designed to be straightforward and focused on the core tasks of product listing, order processing, and inventory management. By prioritizing simplicity, trust, and user experience, the system aimed to improve both business operations and customer satisfaction, providing a practical and effective solution for a small fashion resale business in the Philippines.

Filipino faculty members' online shopping

The concerns raised about fake buyers and unclear communication were part of the reason the Margaux Collections system tied every order to a verified GCash reference number rather than relying on a buyer's word alone. This gave the owner a concrete basis for confirming a transaction before releasing an order for pickup or delivery.

Villa et al. (2013), Ukay-Ukay in Davao City
This local evidence reinforced why Margaux Collections chose to sell brand-new and preloved items side by side rather than treating preloved stock as an afterthought. If affordability and variety were what drew buyers to ukay-ukay products in Davao City, a system that let customers browse both categories in one place, rather than through separate Messenger threads, was more likely to keep that same buyer engaged long enough to complete a purchase.

Rufano and Gervacio, E-Commerce Satisfaction
This gap between expectation and reality was something the ordering system tried to narrow by keeping each stage of a transaction visible to the customer. Once an order was placed, the buyer could see its current status rather than having to ask the seller directly, which reduced the kind of uncertainty the study identified as a common source of dissatisfaction.

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CHAPTER 3

METHODOLOGY

3.1 Research Design

This study followed a developmental research design, since the main goal was not to describe or measure existing conditions but to design, built, and tested a working software system. The output of the study was the Margaux Collections Ordering System, a web-based platform intended to help a small fashion business managed orders and inventory for two distinct product lines: brand-new items such as bikinis, coordinated sets, and accessories, and preloved items such as pants, skirts, and shorts. Because the project centers on solving an actual operational problem rather than simply observing or explaining one, a developmental approach fit the nature of the worked.

A developmental design was chosen for the following reasons:

It was appropriate when the primary goal was to build and refine a working system rather than to describe or correlate variables.

It allowed the design process to be iterative, with continuous testing and adjustment based on feedback rather than a single fixed plan.

It supported a structured evaluation of the finished system's performance, usability, reliability, and security once it was used in an actual business setting.

Guided by this design, the study evaluated the Margaux Collections Ordering System against four practical questions: whether it accurately recorded and processes orders, whether it managed both new and preloved inventory without confusion, whether the interface was usable for both the shop owner and her customers, and whether it fit

naturally into the Facebook-based marketing and selling process the business already relied on. Answering these questions through actual development and testing, rather than through description alone, was what made the developmental design the appropriate choice for this study.

3.2 System Development Methodology

The Margaux Collections Ordering System was developed using the Agile methodology, applied through short, iterative development cycles rather than a single long build-and-release phase. Agile was selected over a strictly sequential model such as Waterfall because the requirements of a small business system tend to shift as the owner sees working features and reacts to them. Specifically, Agile was chosen because it:

allowed worked to be broken into short, focused iterations instead of one long development phase; kept the business owner involved throughout development rather than only at the start and the end; made it easier to adjust features as requirements become clearer; and built in regular testing and review, so problems were caught while they were still small.

The methodology was carried out through five phases, described below.

3.2.1 REQUIREMENTS ANALYSIS

The first phase focused on understanding how the business actually operated before any code was written. This involved:

conducting interviews with the business owner to learn about her current process for receiving orders, tracking stock, and dealing with the mix of new and preloved items; observing the existing manual workflow — mostly conducted through Facebook Messenger and handwritten or spreadsheet-based recorded — to identify recurring pain points such as missed orders or unclear stock counted; reviewing existing product lists, price sheets, and order records to understand how items and transactions were already being organized; and translating these observations into user stories (for example, "As a customer, I wanted to view preloved items separately from new items so that I know what I am buying") and, from there, into concrete technical requirements, such as supporting two distinct product categories, allowing full product and order management, and providing a dashboard for daily sales and stock status.

This phase produced a working requirements specification that guided the design and development work that followed.

3.2.2 SYSTEM DESIGN

With the requirements defined, the next phase focused on planning the system's structure. This included laying out the overall architecture across the front end, back end, and database; sketching wireframes for both the customer-facing shop and the admin panel; designing the database schema and the relationships between tables; and preparing a Data

Flow Diagram and Entity Relationship Diagram to map out how information would move through the system before any of it was built.

3.2.3 DEVELOPMENT

Development work followed the plans set in the design phase. The front end was built using HTML, CSS, and JavaScript to handle product browsing, the shopping cart, checkout, and order history. The back end was implemented in PHP, handling authentication, order processing, inventory updated, and the analytics features. The MySQL database was built and connected securely to the PHP scripts, and periodic self-review of the code was carried out to catch errors early rather than after the fact.

3.2.4 TESTING

Testing was carried out at multiple levels throughout development rather than only at the end. Individual PHP functions and database queries were checked in isolation through unit testing; the front end, back end, and database were then tested together through integration testing to confirm that the layers worked correctly as a whole; the system was checked for common security weaknesses such as injection attacks and unauthorized access; and, once a working version was available, the business owner and a few customers tried the system directly to evaluate whether it was accurate and easy to use. A fuller account of the testing process was given in Section 3.6.

3.2.5 DEPLOYMENT (PROTOTYPE)

Once testing showed the system was stable enough for real use, a working prototype was deployed toa local test environment and presented to the business owner for hands-on evaluation. She was able to place tested orders, updated inventory, and review the analytics dashboard herself. Her feedback from this stage informed a round of minor refinements before the system was finalized, which kept the final product closely aligned with how the business actually operated rather than with assumptions made early in the project.

3.3 System Architecture

The Margaux Collections Ordering System followed a three-layer architecture — a presentation layer, an application layer, and a data layer — so that the interface, the business logic, and the stored data remained separated from one another. Keeping these concerns apart made the system easier to build, tested, and maintained, since a changed in one layer, such as a redesign of the product page, did not required touching how orders were processed or how data was stored.

3.3.1 PRESENTATION LAYER

The presentation layer was what customers and the shop owner actually see and interact with. It was built using HTML5 for page structure, CSS3 for styling and layout, and JavaScript for dynamic behavior such as live cart updated and form validation. On the customer side, this layer provided a product browsing page that separated new items (bikinis, coordinated sets, accessories) from preloved items (pants, skirts, shorts), a shopping cart, a checkout page for entering delivery details and choosing a payment method, and an order history page. On the admin side, it provided a dashboard summarizing sales and inventory, along with pages for managing products, orders, and customer accounts. The interface used responsive CSS so it displayed properly on desktops, tablets, and phones, which mattered given that most customers reach the shop through links shared on Facebook and browse primarily on mobile.

3.3.2 APPLICATION LAYER

The application layer was the back end, built in PHP, and sat between the interface and the database. It handled user authentication, order processing (including order numbering, total computation, and inventory updated), status changed such as moving an order from "pending" to "for delivery," and a set of security measures: prepared statements toprevented SQL injection, secure session handling with timeouts, and validation of all user-submitted input. This layer also enforced rolebased access — customers could browse, added items to their cart, checked out, and view their own order history, while the owner, as administrator, could manage products, orders, users, and the analytics dashboard — so that sensitive operations remained restricted to the account that should be able to perform them.

3.3.3 DATA LAYER

The data layer stored user accounts, the product catalog, order transactions, and the individual line items within each order. The database followed a normalized structure to avoid duplicating information, such as repeating product details inside every order records, and indexes were applied to fields that were queried often, such as order and product IDs, to keep lookups fast as the number of orders grew.

3.3.4 CORE SYSTEM MODULES

Within these three layers, the system's functionality was organized into modules, each responsible for one part of the ordering workflow:

Authentication Module — handled account registration, login, and session management, and, at minimum, secures admin access to the owner's dashboard even if customer accounts were kept simple.

Product Management Module — let the owner added, edit, and deactivated products, tagged items by category and condition (for example, "preloved, good condition"), and kept stock counted current as items sold.

Shopping Cart Module — managed a per-session cart, checking stock before items were added, letting quantities be adjusted, and calculating running totals.

Order Processing Module — assigned order numbers, recorded customer and product details, computed totals, supported cash and GCash payment, and updated stock once an order was confirmed.

Preloved-Specific Management Module — since preloved fashion was central to the business, this module let the owner tagged condition and added notes specific to secondhand items, distinguishing them from new stock.

User Management Module — allowed the owner to view, search, and managed customer accounts.

Analytics and Sales Reporting Module — gave the owner a dashboard of revenue, best-selling items, order volume, and low-stock alerts to support restocking and pricing decisions.

Together, these modules covered the full cycle of listing products, taking orders, and monitoring how the shop was performing.

3.3.5 DATABASE DESIGN

The database was built in MySQL and organized around five core tables:

Users — user_id (primary key), name, email, hashed password, contact number, addressed, role, and date created.

Products — product_id (primary key), product name, description, price, product type (new or preloved), condition, image path, stock quantity, and date added.

Orders — order_id (primary key), user_id (foreign key), customer details, fulfillment method (meet-up, drop-off, shipping, or pick-up via rider), payment method and status, order status, total amount, pinned map location where relevant, and order date.

Order_Items — item_id (primary key), order_id and product_id (foreign keys), quantity, and unit price at the time of purchase.

User_Payment_Accounts — account_id (primary key), user_id (foreign key), account type (GCash), account name and number, and addressed.

Primary and foreign keys enforced the relationships between these tables, the schema was normalized to avoid redundant data, and indexes were placed on frequently queried fields such as order and product IDs, which together kept the data consistent and queries efficient as the shop's order volume grew.

3.4 Tools and Technologies Used

The development of the Margaux Collections Ordering System utilized various programming languages, frameworks, development tools, and technologies that supported the implementation, testing, and maintenance of the system. These tools were selected based on their suitability for developing a web-based ordering system, managing customer and product information, supporting interactive features, and maintaining an organized development workflow. The following tools and technologies were used throughout the development process:

PHP — was used as the primary server-side programming language for implementing the system's business logic and backend functions. It handled processes such as user authentication, product management, order processing, inventory updates, payment verification, and communication between the application and database.

MySQL — was used as the relational database management system for storing and retrieving system information. It managed records related to users, products, categories, inventory, orders, order items, payments, messages, notifications, and other system data. The relational structure helped maintain consistency and establish connections among related records.

HTML5 and CSS3— were used to structure and design the system's web pages and user interfaces. HTML5 provided the structure for the storefront and administrative pages, while CSS3 was used to implement the dark noir visual theme, layouts, typography, colors, spacing, and responsive design. These technologies allowed the system to provide a consistent interface across desktop and mobile screen sizes.

JavaScript — was used to provide interactive and dynamic behavior within the system. It supported functions such as cart updates, form validation, interface interactions, and other client-side processes. Its use helped make the ordering process more responsive and convenient for users.

Chart.js — was used to create graphical representations of sales and performance data within the administrative analytics dashboard. The charts helped present business information in a more understandable visual format, allowing the administrator to review sales-related data and identify relevant trends.

XAMPP — was used as the local development and testing environment. It provided the Apache web server, PHP runtime, and MySQL database services required to run and test the web application locally. It also allowed the developers to verify system functions before deployment.

Visual Studio Code — served as the primary source-code editor throughout the development process. It was used to create, edit, organize, and review the PHP, HTML, CSS, JavaScript, and other project files.

Git and GitHub — were used for version control and collaborative development. Git allowed the developers to track changes, manage different versions of the project files, and revert changes when necessary, while GitHub provided a centralized repository that helped keep the codebase synchronized across the development machines.

3.5 Data Gathering Techniques

During the requirements analysis phase, several data gathering techniques were used to obtain relevant information about the operations, existing processes, and system needs of Margaux Collections before the development of the proposed ordering system. Interviews were conducted with the business owner through structured and semi-structured questions to understand how orders were received and processed, particularly through Facebook Messenger and in-person transactions. The interviews also covered existing inventory management practices, common challenges encountered during order processing, payment procedures, and the features that the business owner considered important for the proposed system. Observation was also conducted by examining the actual ordering, product preparation, and packing processes. This allowed the researchers to identify operational difficulties such as disorganized product listings, manual recording errors, difficulties in monitoring inventory, and challenges in distinguishing preloved products from brand-new stock.

Document analysis was conducted by reviewing existing product lists, price sheets, and previous order records to understand how products were categorized, how prices were determined, and how customer orders and transactions had been recorded. These documents provided additional information that helped the researchers identify the data that needed to be included in the proposed system. **Surveys** were also used, where feasible, to gather feedback from existing customers regarding their interest in online ordering and their preferred system features. The survey responses helped identify customer expectations, including the need for clearer product photographs, organized

product information, and a separate section for preloved items. The information gathered through these techniques served as a basis for defining the system requirements, designing the database structure, and developing features that addressed the actual operational needs of Margaux Collections and its customers.

3.6 Testing Procedures

The Margaux Collections Ordering System underwent four types of testing to determine whether the developed system functioned correctly, performed reliably under normal conditions, and adequately protected customer and business information. The testing procedures were conducted throughout the development process to identify functional errors, usability issues, integration problems, and potential security concerns. Each testing type focused on specific aspects of the system, and the results were used to identify necessary corrections and refinements before the system was considered ready for deployment and evaluation.

3.6.1 UNIT TESTING

Individual PHP functions, such as order creation and stock updates, were tested with a range of inputs to confirm they returned the expected results. Database queries were checked to make sure records were inserted, updated, and retrieved correctly, and basic front-end validation, such as required-field checks, was verified as well. This confirmed that each component worked correctly on its own before being combined with the rest of the system.

3.6.2 INTEGRATION TESTING

Once individual components passed unit testing, they were tested together to confirm that the front end, back end, and database worked correctly as a whole. This included confirming that adding an item to the cart correctly stored it, that placing an order created the right order and order-item recorded while updating stock accordingly, and that changed made in the admin panel, such as price edits, were reflected properly on the customer-facing side.

3.6.3 SECURITY TESTING

Because the system stored personal and transactional data, it was checked against common vulnerabilities, including SQL injection (mitigated through prepared statements), cross-site scripting (mitigated by escaping user input), cross-site request forgery (mitigated with tokens where applicable), and session hijacking (mitigated through proper session handling). Access controlled were also tested to confirm that customers cannot reach the admin panel and that authentication cannot be bypassed.

3.6.4 USER ACCEPTANCE TESTING (UAT)

After the prototype was deployed, the business owner and a small group of customers used the system through real or simulated tasks — placing orders, updating products, and reviewing the dashboard — and their feedback on ease of used, accuracy, and overall appearance was collected. This step confirmed that the system holds up not just in testing conditions, but in the kind of everyday used it was actually built for.

3.7 Evaluation Criteria

The finished system was assessed against five criteria, described below.

3.7.1 FUNCTIONALITY

The system had to correctly perform its required functioned, including accurately recording orders, properly updating inventory, correctly computing totals, and clearly displaying order history and status.

3.7.2 USABILITY

The interface had to be intuitive for both customers and the owner, keeping the learning curve low, providing clear navigation and feedback, and remaining fully usable on mobile devices.

3.7.3 SECURITY

The system had to protect user data and restrict access appropriately, with information properly hashed, transmitted securely, and accessible only toauthorize users.

3.7.4 PERFORMANCE

The system had to respond quickly under normal used, loading pages within a reasonable time and handling multiple concurrent users without noticeable slowdown.

3.7.5 RELIABILITY

The system had to process orders correctly and maintained data integrity, with orders and inventory changed recorded consistently and reversible only through a proper process, such as a formal cancellation.

Supporting Diagrams

The following diagrams illustrated the structure and expected behavior of the system in more detail.

Data Flow Diagram (DFD — Level 0)

The Level 0 Data Flow Diagram showed how customers and the administrator interacted with the system and how their data ultimately reached the MySQL database. Customers submitted registration details, login credentials, order information, and checkout confirmations; the system validated product availability, created order records, and updated inventory before storing everything in the database and returning confirmations, product listings, or order status updates. The administrator, on the other hand, submitted product updates, inventory adjustments, and order status changes, which the system processed and reflected in the database in real time. Read together, the diagram laid out the system's overall workflow in a way that was easier to follow than a written description alone.

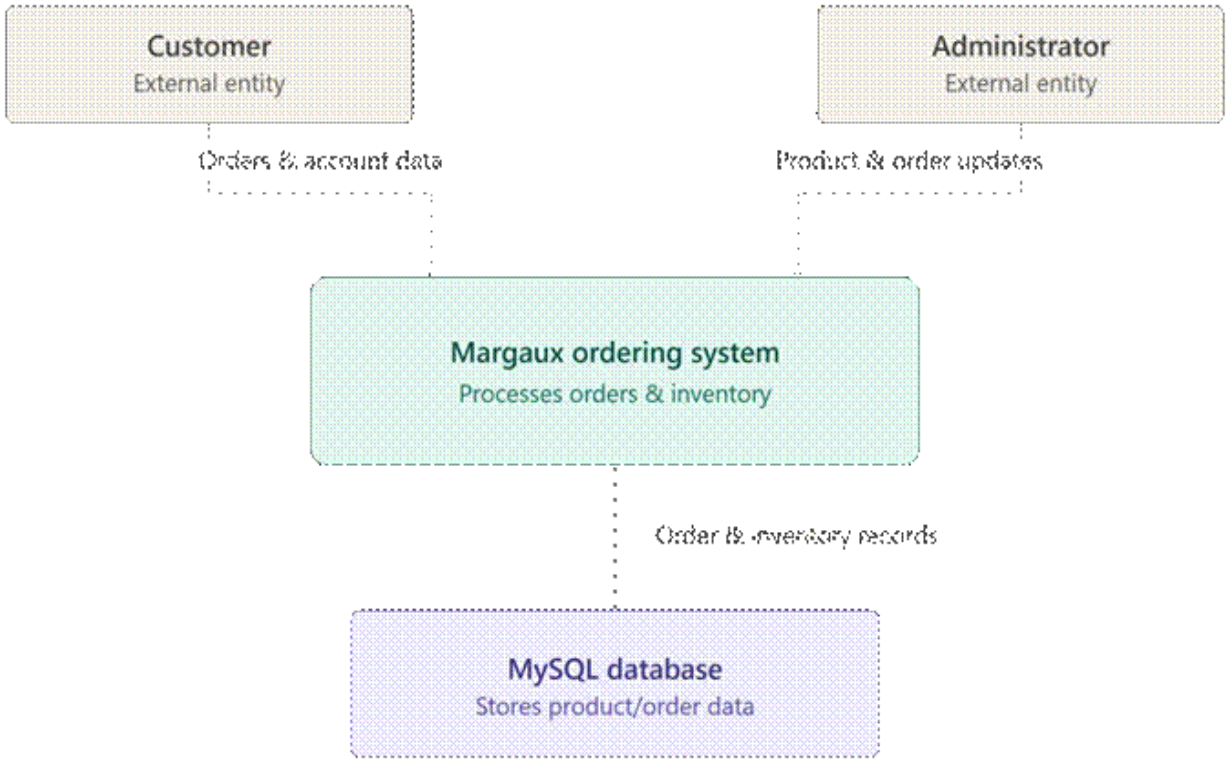


Figure 3.1. Level 0 Data Flow Diagram of the Margaux Collections Ordering System

Entity Relationship Diagram (ERD)

The Entity Relationship Diagram defined the core entities in the database User, Product, Order, OrderItem, and UserPaymentAccount — along with how they related to one another. A user could place multiple orders, an order could contain multiple order items, a product could appear across multiple order items, and a user could save multiple payment accounts. Laying this out clearly helped confirm that the tables connected correctly and that data could be retrieved accurately, which in turn supported a database design that was easier to maintain as the system grew.

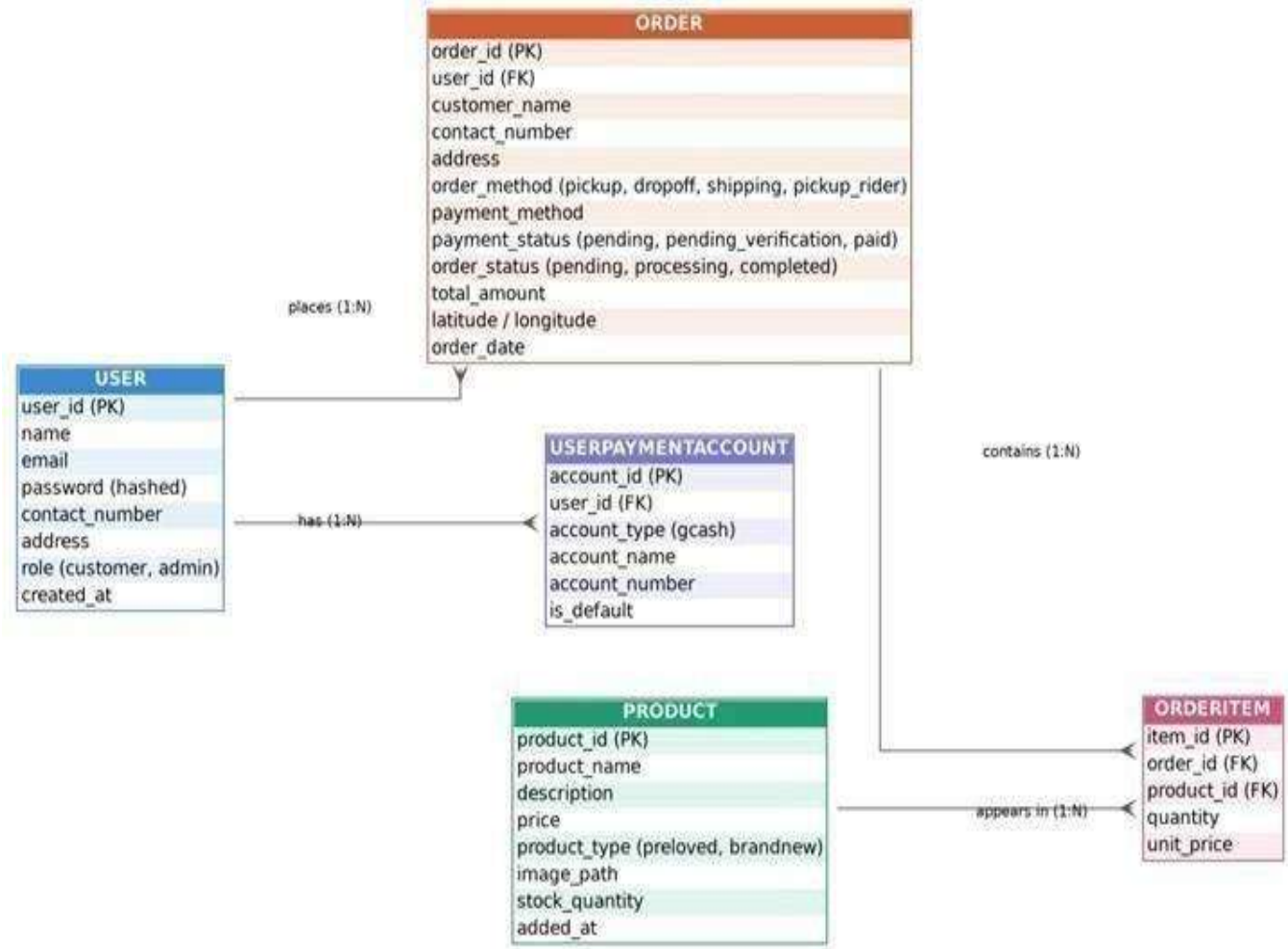


Figure 3.2. Entity Relationship Diagram of the Margaux Collections Ordering System

Use Case Diagram

The Use Case Diagram presented the system's two main actors, the customer and the administrator, and the actions available to each. Customers could register, log in, browse products, managed their cart, checked out, placed orders, viewed order history, updated their profile, and save payment methods. Administrators could log in to the admin panel, manage products, manage orders, manage users, and view analytics. Together these use cases outlined the functional scope of the system and made clear what each type of user was able to do within it.

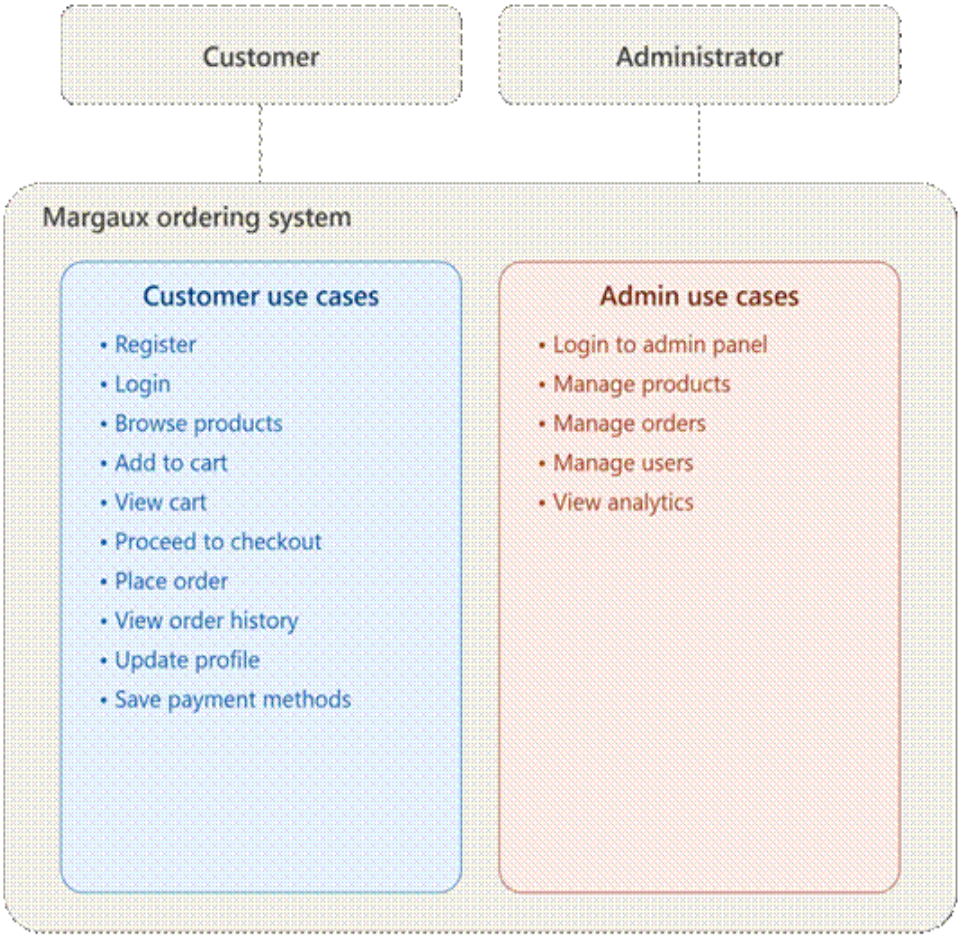


Figure 3.3. Use Case Diagram of the Margaux Collections Ordering System

CHAPTER 4 SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

4.2 OVERVIEW OF THE DEVELOPED SYSTEM

Margaux Collections was a web-based e-commerce and boutique management system developed for a fashion retail business to modernize its product presentation, sales transactions, and order handling. The system replaced manual and social-media-based

selling practices with a structured online platform that allowed customers to browse the boutique's product catalog, place orders, and complete payment through a GCash QR code checkout process.

Unlike generic e-commerce templates, Margaux Collections was designed with a distinct visual identity tailored to the boutique's brand. A dark noir theme was implemented throughout the storefront and administrative interfaces using custom CSS variables for color theming and the Playfair Display and Jost typefaces for headings and body text, respectively, giving the platform a fashion-forward and elegant presentation consistent with the boutique's market positioning.

The system was built entirely from scratch, encompassing the database schema, authentication pages, customer-facing storefront, and a full administrative panel. The application processed product listings, customer orders, and homepage promotional content, and presented this information through role-based interfaces for customers and administrators. The major objectives of the developed system were as followed:

Provided an online storefront where customers could browse and purchase boutique products.

Automate the recording, storage, and tracking of customer orders.

Support a secure, QR-code-based GCash payment process, alongside Cash on Pickup and Cash on Delivery options depending on the customer's chosen order fulfillment method.

Enabled administrators to manage products, orders, customer accounts, and site content through a dedicated panel.

Allowed administrators to control homepage promotional slots to feature selected products or campaigns.

Provide a messaging feature for direct communication between customers and administrators.

Maintain a consistent, brand-aligned visual design across all customer and administrative pages.

The development of Margaux Collections followed modern web development practices, incorporating responsive interface design, secure database management, user authentication, and version-controlled collaborative development. The completed system consisted of several interconnected modules that worked together to manage products, process orders, and maintained the boutique's online presence.

Figure 4.1 illustrated the overall workflow of the developed system.

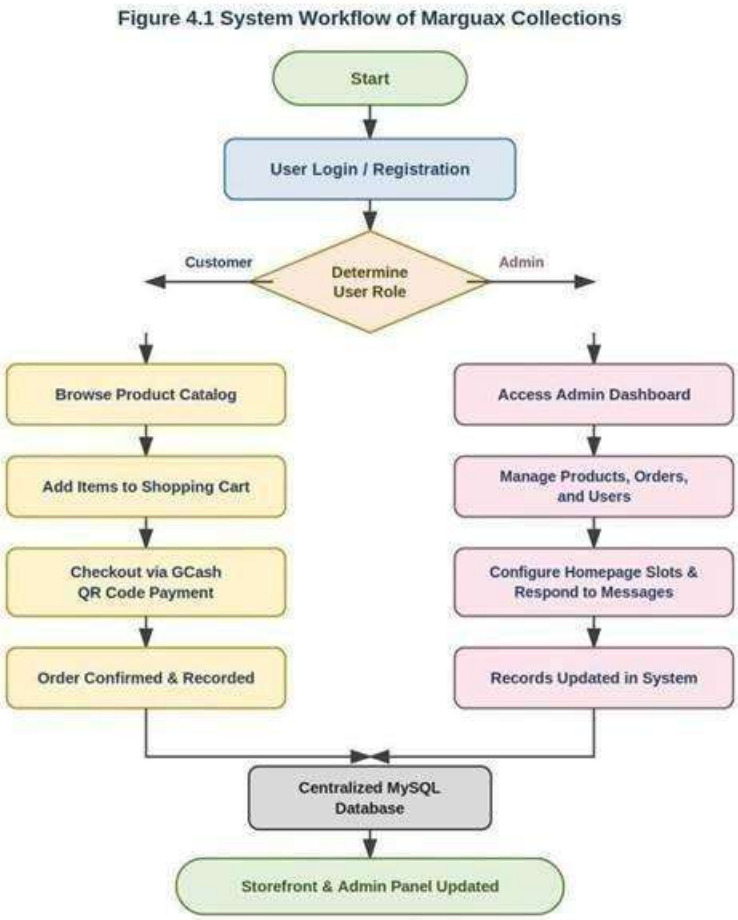


Figure 4.1. System Workflow of Margaux Collections

The workflow began with user authentication, after which customers could browse products, add items to their cart, and proceed to checkout using the GCash QR payment method. Administrators, upon logging in, could manage products, monitor and update order status, configured homepage promotional slots, and respond to customer messages through the administrative dashboard.

4.3 SYSTEM DEVELOPMENT METHODOLOGY
4.1 DEVELOPMENT METHODOLOGY

The development of Margaux Collections adopted the Agile Software Development Methodology, an iterative approach that emphasized incremental development, continuous testing, evaluation, and refinement throughout the

software development life cycle. Agile was selected because the system requirements, particularly its functional features, user interface, and visual design, evolved as the development progressed. This approach provided the developers with the flexibility to add, modify, improve, or remove features based on the identified needs of the business without significantly disrupting the overall development process. It also allowed the team to respond to issues discovered during implementation and testing, ensuring that the final system remained aligned with the operational requirements of Margaux Collections.

The Agile methodology allowed the development process to be organized into a series of iterative cycles in which individual modules were planned, developed, tested, evaluated, and refined before being integrated into the completed system. Each iteration provided an opportunity to identify technical issues, usability concerns, and unnecessary features at an early stage. This iterative process proved particularly useful when previously implemented features required significant modification. One example was the eventual removal of the tiered membership access feature, which had initially been included in the system design but was later determined to be unnecessary for the actual business requirements. Removing the feature simplified the customer experience, database structure, and administrative controls while allowing development efforts to focus on the system's core functions.

Requirements Gathering

The requirements gathering phase involved identifying and documenting the operational needs of the Margaux Collections boutique. The development team determined the required functions by considering the business's existing selling and ordering processes, the needs of customers, and the administrative tasks performed by the business owner. These requirements included the desired look and feel of the online storefront, product catalog management, inventory monitoring, customer account management, order processing, payment verification, and administrative controls. The preferred GCash payment method was also considered in designing the checkout and transaction verification process. The information gathered during this phase served as the basis for defining the system's core modules, database structure, user roles, and overall functionality. In addition, the requirements guided the development of the system's dark noir visual theme, which was intended to maintain a consistent and recognizable brand identity for Margaux Collections.

System Planning

Following the requirements gathering phase, the project scope, feature set, development priorities, and schedule were established. The planning phase identified the major components that needed to be developed, including the customer-facing storefront, product catalog, shopping cart, checkout and payment verification, user accounts, inventory management, order management, messaging, and administrative dashboard. The technical resources required for development were also identified, including the local XAMPP development environment, MySQL database platform, PHP programming language, and Git and GitHub for version control and collaborative development. This planning process provided a clear direction for the development team and helped ensure that the system was developed according to the identified business requirements.

System Design

During the system design phase, the database schema, system structure, page layouts, navigation flow, and administrative panel were designed based on the requirements identified during the previous phase. The database structure was planned to organize information related to users, products, inventory, orders, payments, messages, notifications, and other supporting functions while maintaining appropriate relationships among records. The customer-facing pages and administrative interfaces were also designed to provide clear navigation and organized access to system functions. In addition, the color palette, typography, spacing, and other visual elements of the dark noir theme were defined through reusable CSS variables and design rules. This approach helped maintain a consistent visual identity across the storefront and administrative panel while allowing the interface to be modified more efficiently during later development and refinement.

Software Development

The software development phase involved implementing the planned system components using PHP and MySQL within a XAMPP local server environment. Development proceeded incrementally, beginning with the creation of the database schema and authentication functions, followed by the development of the customer storefront, product catalog, shopping cart, and checkout process with GCash QR-based payment. The administrative panel was then developed to include the dashboard, analytics, order management, product management, user management, inventory-related functions, promotional content, and messaging modules. Each module was developed according to the established requirements and was gradually integrated with the other components of the system. During development, several features and interface elements were refined based on testing results and the actual needs of the boutique. This included modifying existing functions and removing the tiered membership feature when it was determined to be unnecessary for the intended operation of the system.

Testing

The testing phase involved evaluating the functionality, usability, integration, and consistency of the developed system. Each module was tested individually after it was completed to identify errors and confirm that its intended functions operated correctly. Additional testing was conducted after the modules were integrated to determine whether the different components worked properly together. Particular attention was given to changes that affected multiple parts of the system. For example, when the tiered membership feature was removed, the developers checked the database structure, authentication processes, administrative panel, and customer-facing pages to ensure that the removal did not create broken links, unnecessary records, or inconsistencies in system behavior. Interface and stylesheet issues were also identified and corrected during testing

to maintain the consistency of the dark noir theme. The testing process allowed the developers to refine the system and address issues before deployment.

Deployment

Following the testing and refinement phases, the Margaux Collections Ordering System was configured and deployed within a local XAMPP server environment. The deployment process involved setting up the application files, MySQL database, server configuration, and required system components to ensure that the application could operate properly in its intended environment. The completed system was made accessible to its intended users for demonstration, evaluation, and operational testing. The deployment phase also included verifying that the database connection, authentication functions, product listings, ordering process, GCash payment verification, and administrative features continued to operate correctly in the configured environment. This ensured that the final system was properly prepared for use by the Margaux Collections business and its intended customers.

Deployment Process

The deployment process began with the configuration of the XAMPP environment, where the Apache web server and MySQL database service were prepared to host the Margaux Collections Ordering System locally. The completed database schema, which included the Users, Products, Categories, Orders, Order Items, Homepage Slots, and Messages tables, was exported and imported through phpMyAdmin to establish a consistent database state across the development machines used by the project. This process helped ensure that the application operated using the same database structure and records during development, testing, and deployment.

The application files, including the customer-facing storefront, administrative panel, shared stylesheet files, configuration files, and other required resources, were placed inside the XAMPP `htdocs` directory so that they could be accessed through the local Apache server. The database connection settings were configured to connect the application to the local MySQL service. The folder structures and file permissions for product images and other uploaded content were also verified to ensure that both the storefront and administrative panel could properly retrieve and display stored media. These configurations helped ensure that the system's product listings, images, administrative functions, and other resources operated correctly within the deployed environment.

Since development was conducted across two separate laptops, Git and GitHub were used to synchronize the project's source code and maintain consistency between the development environments. During this process, several practical technical issues were encountered and resolved. One issue involved multiple `mysqld.exe` processes running simultaneously on one machine, which prevented the XAMPP MySQL module from starting properly. The issue was resolved by terminating the duplicate MySQL process before restarting the MySQL service through XAMPP. Occasional merge conflicts were also encountered in the stylesheet files when changes from different development environments were combined. These conflicts were resolved through manual review and reconciliation

of the affected CSS rules to ensure that necessary changes were preserved while maintaining the intended dark noir visual theme of the system.

After the application files and database were successfully configured and deployed on the local server, end-to-end testing was conducted to verify the functionality of the completed system. Both customer and administrator workflows were simulated to ensure that the different components operated correctly after deployment. Customer-side functions, including product browsing, product viewing, shopping cart management, and GCash QR-based checkout, were tested, while administrative functions such as order management, product management, and homepage slot configuration were also verified. The testing process confirmed that the deployed system maintained the expected behavior observed during development and that the database, application files, user interfaces, and system functions were properly integrated within the local XAMPP environment

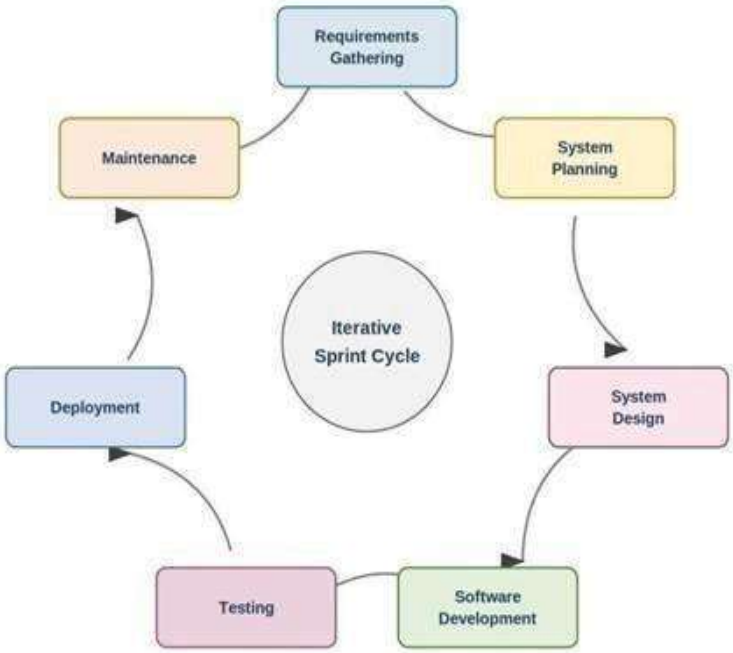
Maintenance

The maintenance phase involved monitoring the performance, functionality, and consistency of the Margaux Collections Ordering System after its initial development and deployment. This phase focused on identifying and resolving issues that were discovered through continued testing and actual system use. Maintenance activities included correcting styling inconsistencies, addressing data-related issues, refining existing features, and ensuring that the customer-facing storefront and administrative panel continued to operate as intended. These activities helped maintain the system's reliability and usability while ensuring that its functions remained aligned with the operational requirements of the boutique.

Particular attention was given to resolving stylesheet conflicts between the customer-facing storefront and administrative interfaces to preserve the consistency of the dark noir visual theme. Database synchronization issues encountered during collaborative development were also addressed to ensure that information remained consistent across the development environments. In addition, existing system features were reviewed and refined whenever testing identified areas that could be improved. Through continuous maintenance and refinement, the system remained functional, organized, and suitable for supporting the online ordering and management needs of Margaux Collections.

Figure 4.2 presented the Agile Software Development Life Cycle adopted in the study.

Figure 4.2 Agile Software Development Life Cycle



4.4 SYSTEM ARCHITECTURE

Margaux Collections utilized a Three-Tier Client-Server Architecture, which divided the system into three interconnected layers: the presentation layer, application layer, and data layer. The presentation layer handled the user interface and interactions of customers and administrators with the system, including product browsing, ordering,

checkout, and administrative functions. The application layer manages the system's business logic and core processes, such as authentication, product and order management, payment verification, inventory management, messaging, and other application functions. Meanwhile, the data layer was responsible for storing, retrieving, and managing system information through the MySQL database. This architectural approach improved the organization, maintainability, security, and scalability of the system by assigning specific responsibilities to each layer. It also allowed individual components to be modified, tested, or improved with minimal impact on the other layers, making the system easier to manage and maintain throughout its development and future enhancement.

Figure 4.3 illustrated the architecture of the developed system.

Figure 4.3 Three-Tier System Architecture of Marguax Collections

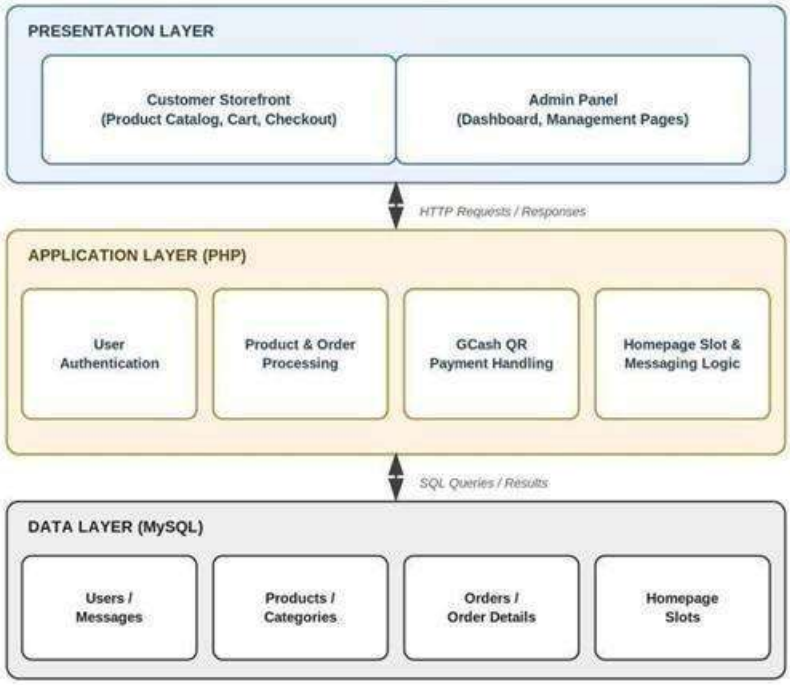


Figure 4.3. Three-Tier System Architecture of Margaux Collections

Presentation Layer

The presentation layer served as the primary interface between users and the Margaux Collections Ordering System. It was responsible for presenting the product catalog, product details, shopping cart, checkout process, order information, and administrative dashboards in an organized and user-friendly manner. It also handled user interactions, allowing customers to browse available brand-new and preloved clothing, review product information, place orders, and access relevant transaction details. For administrators, the layer provided interfaces for managing products, users, orders, inventory, and promotional content. The presentation layer also implemented the system's dark noir visual theme consistently throughout the storefront and administrative pages while using

a responsive layout to provide a suitable viewing experience across different screen sizes and devices.

Major interface components included:
Login and Registration Pages

Customer Storefront and Product Catalog

Shopping Cart and Checkout (GCash QR Payment)

Admin Dashboard

Product Management

Order Management

User Management

Analytics

Messages

Conversations

Notifications

User Payment Accounts

OTP Tokens Password

Resets

Application Layer
The application layer contained the business logic responsible for processing user requests, validating input data, and managing the interactions between the storefront, the administrative panel, and the database. Its primary responsibilities included:

User Authentication

Product and Category Management • Order Processing

Payment Verification (GCash QR)

Homepage Promotional Slot Management
Customer–Admin Messaging

Analytics and Reporting

The application server, built using PHP, acted as the intermediary between the presentation layer and the MySQL database server. *Data Layer*
The data layer stored all business information in a centralized relational database managed through MySQL. It was responsible for maintaining data integrity and supporting efficient retrieval of the information required by the storefront and administrative panel. Stored information included:

User Accounts

Products and Categories

Orders and Order Items

Homepage Promotional Slots

Customer–Admin Messages

The architecture ensured that communication between the application server and the database remained organized and consistent, restricting unauthorized access to administrative functioned and customer data.

4.5 HARDWARE AND SOFTWARE REQUIREMENTS

The implementation of Margaux Collections required appropriate hardware and software resources to ensure efficient performance during product browsing, order processing, and administrative operations.

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i3	Intel Core i5 or higher

Memory (RAM)	4 GB	8 GB or higher
Storage	128 GB SSD	256 GB SSD or higher
Monitor	1366 x 768 Resolution	Full HD Display
Internet Conne	10 Mbps	25 Mbps or higher
Mobile Device	Android 8	Android 11 or higher

Software	Purpose
Windows 10 / 11	Operating System

Visual Studio Code	Source Code Editor
MySQL (via XAMPP)	Database Management System

PHP	Backend Development
HTML, CSS, JavaScript	Frontend Development
Google Chrome	Web Browser
Git and GitHub	Version Control and Collaboration
XAMPP	Local Development Server

4.5.1 HARDWARE REQUIREMENTS

Table 4.1 presented the recommended hardware specifications.

4.5.2 SOFTWARE REQUIREMENTS

Table 4.2 presented the software requirements.

The selection of these software tools was based on their stability, accessibility, and suitability for developing and locally hosting a PHP and MySQL web application.

4.6 DATABASE DESIGN

The database served as the foundation of the Margaux Collections Ordering System by organizing, storing, and managing essential information required for the operation of the online platform. It contained records related to product listings, customer accounts, orders, order details, payments, inventory, promotional content, messages, notifications, and other supporting system functions. A relational database model was implemented using MySQL, allowing related information to be stored in separate but interconnected tables. This structure helped reduce data redundancy, improve data organization, and maintain consistency among related records.

The database design was developed based on the functional requirements of the system and the actual processes involved in the operations of Margaux Collections. Primary entities were established for important business information, while supporting tables were created for authentication, communication, payment, and security-related functions. Relationships between tables were established using primary and foreign keys to ensure that records remained properly connected. For example, customer accounts could be associated with their orders, while order details could be linked to the corresponding products. This relational structure allowed the system to retrieve and update information efficiently while minimizing the possibility of duplicate or inconsistent data.

The use of MySQL also provided a structured environment for managing the increasing amount of information generated by the system. Product records could be updated as items became available or were sold, while order and payment records could be maintained for transaction monitoring and administrative reference. Customer and administrative information could also be managed through appropriate database relationships and security measures. Overall, the database design provided a reliable and organized foundation for the Margaux Collections Ordering System and supported its major functions, including product management, customer management, inventory monitoring, order processing, payment verification, communication, and administrative operations.

Figure 4.4 presented the Entity Relationship Diagram (ERD) of the system.

Figure 4.4. Entity Relationship Diagram of Margaux Collections The primary entities of the database included:

Users

Products

Categories

Orders

Order Items
Homepage Slots

Messages

Conversations

Notifications

User Payment Accounts

OTP Tokens Password

Resets

4.6.1 Database Tables ***Users Table***

Field	Description
user_id	Primary Key

name	Complete Name
email	Login Email Address
password	Encrypted Password
role	User Access Level (Customer / Admin)

This table stored account credentials and role information for both customers and administrators.

Products Table

This table stored the boutique's product listings.

Field	Description
product_id	Primary Key
product_name	Product Name
category_id	Foreign Key — Product Category

price	Selling Price
Field	Description
stock	Available Inventory
image	Product Image

description	Product Description
product_type	Product Type (Pre-loved / Brandnew)
sold_out_at	Timestamp When Product Went Out of Stock

Categories Table

This table stored the product classifications used to organize the boutique's catalog.

Field	Description
category_id	Primary Key
name	Category Name

Orders Table

This table recorded customer purchase transactions.

Field	Description

order_id	Primary Key
Field	Description
user_id	Foreign Key

order_date	Date of Purchase

order_method	Order Fulfillment Method (Meet up / Drop off / Shipping Delivery / Pick-up Via Rider)
payment_method	Selected Payment Method (Cash on Pickup / Cash on Delivery / GCash)

total_amount	Total Order Amount
latitude	Pinned Delivery Latitude (Shipping / Pick-up Via Rider only)
longitude	Pinned Delivery Longitude (Shipping / Pick-up Via Rider only)

payment_status	Payment Status (Pending / Pending Verification / Paid)
order_status	Order Fulfillment Status

Order Items Table

This table recorded the individual items included in each order.

Field	Description

item_id	Primary Key
order_id	Foreign Key
product_id	Foreign Key
quantity	Quantity Ordered
price	Unit Price at Time of Purchase

Homepage Slots Table

This table stored administrator-configured promotional content displayed on the storefront's homepage.

Field	Description
slot_id	Primary Key

label	Slot Label (e.g. Featured, Dresses, Tops)
image_path	Uploaded Promotional Image Path
price_label	Optional Display Price Text (e.g. "Starting at ₱299")
updated_at	Date Last Updated

Messages Table
This table stored messages exchanged between customers and administrators.

Field	Description
message_id	Primary Key
conversation_id	Foreign Key — Linked Conversation
sender_type	Sender Role (Customer / Admin)

sender_id	Foreign Key — Sending User
message	Message Text
created_at	Date and Time Sent

In addition to these primary entities, the database included several supporting tables that handled authentication, communication, notifications, and account security. These included the Conversations table, which grouped customer–admin messages into threads referenced by the Messages table; the Notifications table, which stored order-related alerts and updates sent to customers; the OTP Tokens and Password Resets tables, which supported one-time password verification for login, registration, and password recovery; and the User Payment Accounts table, which stored customers' saved GCash payment details for checkout. These supporting tables were excluded from Figure 4.4 for clarity and to maintain a more readable representation of the primary database structure. However, they remained connected to the appropriate user, order, and transaction records through established database relationships.

The relationships among the database tables allowed the application to associate customers with their orders, connect order details with specific products, maintain transaction histories, and manage the promotional content displayed on the storefront. Foreign key constraints were implemented to maintain referential integrity among related records and to ensure that data remained properly associated with its corresponding users, products, orders, and messages. These relationships also helped prevent inconsistent or incomplete records while supporting the accurate retrieval and management of information throughout the system. As a result, the database structure provided a reliable foundation for the major functions of the Margaux Collections Ordering System, including product management, customer communication, ordering, payment processing, notifications, and account security.

CHAPTER 5 CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presented the overall conclusion of the development of the Margaux Collections Ordering System. It synthesized the results obtained from the development, testing, and evaluation of the system and assessed the extent to which the identified research objectives were achieved. The chapter also summarized how the developed system addressed the operational needs of the boutique, particularly in terms of online product presentation, ordering, inventory management, payment verification, and administrative monitoring. The findings provided an overall assessment of the system's functionality, usability, and relevance to the business operations of Margaux Collections.

Furthermore, this chapter discussed the contributions and impact of the developed system on the boutique's transition from traditional and manual selling processes to a more organized web-based ordering environment. It highlighted how the system could have improved the accessibility of products, streamlined transaction procedures, supported inventory monitoring, and provided customers with a more convenient and reliable shopping experience. The chapter also presented recommendations for further system improvements, future research directions, and implementation strategies that could have strengthened the system's effectiveness, security, reliability, and long-term sustainability for Margaux Collections.

5.2 Summary of Findings

The development and evaluation of the Margaux Collections Ordering System yielded findings that were generally aligned with the objectives of the project. The developed system successfully provided an online storefront where customers could browse the boutique's available product catalog, view relevant product information, and place orders through a GCash QR-based checkout process. The system provided a more organized alternative to manual ordering and allowed customers to access product information and complete their purchases through a centralized online platform. This demonstrated that the system was capable of supporting the basic online selling requirements of Margaux Collections.

The administrative panel also effectively supported the management of important business operations, including product, order, and user management. The administrator was able to manage product information, monitor customer orders, and maintain user-related records through a centralized interface. The system also included configurable promotional content on the homepage, which allowed the boutique to highlight selected products, announcements, or promotional materials. These functions provided the business owner with greater control over the content and operations of the online store and helped reduce the dependence on separate manual processes for managing products and orders.

The system's dark noir visual theme was also implemented consistently across both the customer-facing storefront and administrative interfaces. During development, stylesheet conflicts and inconsistencies affected certain aspects of the interface, but these issues were identified and resolved through testing and refinement. The correction of these styling problems helped maintain a more consistent visual identity throughout the system and ensured that the design remained aligned with the branding of Margaux Collections. This demonstrated the importance of continuous testing and interface refinement in achieving both functional and visual consistency in a web-based system.

The removal of the tiered membership feature also contributed to simplifying the overall customer purchasing experience and reducing unnecessary system complexity. The feature was removed after considering its relevance to the actual operations and requirements of the boutique. Its removal reduced the number of membership-related processes that needed to be managed by customers and administrators and also simplified the associated database structures and administrative controls. As a result, the system could have focused more directly on its primary functions, such as product browsing, ordering, payment verification, and inventory and user management.

Lastly, collaborative development through Git and GitHub supported the maintenance and organization of the project across multiple development machines. The use of version control allowed the developers to manage changes, maintain project files, and coordinate development activities more systematically. During the development process, synchronization issues were encountered, particularly when changes were made across different machines, but these issues were identified and resolved through proper version-control procedures. This experience demonstrated the importance of collaborative development tools in maintaining consistency, tracking changes, and supporting efficient teamwork throughout the development of the Margaux Collections Ordering System.

5.3 Conclusion

Margaux Collections proved to be a functional, user-friendly, and brand-consistent solution for transitioning the boutique's traditional sales process into an organized online platform. The developed system provided a structured digital storefront that allowed customers to browse available brand-new and preloved clothing products in a more convenient and organized manner. It also incorporated a secure GCash-based checkout and payment verification process, which helped provide greater transparency and reliability during online transactions. In addition, the comprehensive administrative panel enabled the business owner to manage product listings, monitor inventory, process orders, verify payments, and maintain important transaction records more efficiently.

The system successfully addressed several operational challenges experienced in the boutique's previous selling process. By centralizing product information and ordering activities, it reduced the need to rely solely on manual records and scattered conversations through messaging platforms. Customers were able to access important product

information, including photographs, prices, sizes, descriptions, and condition details, before placing their orders. This was particularly beneficial for preloved clothing because customers needed accurate information regarding the actual condition of each item before making a purchase. The system therefore provided not only a platform for completing transactions but also a structured way of presenting products and communicating essential information to customers.

The GCash-based checkout process further strengthened the system's ability to support reliable online transactions. Customers were able to submit their payment information through the established ordering procedure, while the business owner could verify transactions using the provided payment reference number. This process helped reduce uncertainty regarding advance payments and provided a more organized method of matching payments with customer orders. As a result, the system could have contributed to greater accountability and reduced the possibility of errors associated with manually confirming payments through screenshots and separate messaging conversations.

The administrative panel also played an important role in improving the business's internal operations. Through centralized management of products, inventory, orders, and payment information, the owner could have monitored business activities more efficiently. The system helped ensure that product information remained organized and that available items could be managed more systematically. This was especially important for preloved clothing because many items were unique or available only in limited quantities. Proper inventory management could have reduced the possibility of accepting orders for products that were already sold or unavailable.

Furthermore, the developed system maintained the visual identity of Margaux Collections while incorporating the necessary functions of an online ordering platform. The design allowed the boutique to maintain a professional and consistent presentation of its products and brand. This was important because the online platform served not only as a transaction tool but also as a digital representation of the business. By combining organized product listings, attractive product presentation, and functional ordering features, the system could have helped present preloved clothing as valuable and desirable fashion products rather than simply as used items.

The project also demonstrated that a small-scale fashion retail business could have benefited from a purpose-built web application designed around its actual operational needs. Instead of relying entirely on generic or template-based e-commerce solutions, Margaux Collections was able to use a system that was specifically developed to address its product categories, ordering procedures, payment method, inventory requirements, and administrative processes. This demonstrated the potential of customized information systems to support small businesses that required functions tailored to their particular operations.

Overall, the Margaux Collections Ordering System successfully demonstrated how a purpose-built web application could have supported the digital transformation of a small fashion retail business. By integrating an organized storefront, product and inventory management, GCash-based payment verification, order processing, and administrative functions, the system provided a more structured and convenient approach to online selling. It addressed both customer-facing and business-management needs while maintaining the boutique's brand identity.

The project therefore demonstrated that an appropriately designed and affordable web-based ordering system could have improved the efficiency, organization, accessibility, and reliability of Margaux Collections' sales operations without requiring dependence on generic e-commerce platforms.

5.4 Recommendations System

The Margaux Collections Ordering System could have been further improved by implementing additional features that strengthened inventory management, account security, and reporting capabilities. Automated low-stock notifications could have been integrated to alert the administrator when product quantities reached a specified level, allowing timely inventory updates and reducing the possibility of stock shortages. Account security could also have been enhanced through additional authentication measures, such as one-time password (OTP) verification, to provide better protection for administrator and customer accounts. In addition, reporting functions could have been improved by allowing order and sales information to be exported in formats such as PDF or CSV, which could have supported recordkeeping, monitoring, and business analysis.

For future research and system development, automated payment verification could have been explored to reduce the need for manual confirmation of GCash transactions. This feature could have allowed the system to verify payment information more efficiently and could have reduced delays in order processing. The feasibility of developing a mobile-native application could also have been investigated to make the system more accessible to customers who had limited access to desktop computers. Furthermore, a follow-up study could have been conducted to evaluate the long-term impact of the system on the boutique's sales performance, customer satisfaction, operational efficiency, and customer retention. The results of such a study could have provided additional information for identifying areas that required further improvement.

For successful implementation and continued operation, boutique staff could have been provided with proper training on the consistent use of the administrative panel. Training could have helped ensure that product information, inventory records, orders, and payment details were entered and updated accurately. Regular database backups could also have been performed to protect important business information from accidental loss, system errors, or other technical problems. Maintaining updated backups and conducting periodic system maintenance could have supported the reliability, security, and long-term sustainability of the Margaux Collections Ordering System.

5.5 Impact of the Study

The Margaux Collections Ordering System contributed to the practical application of web development principles in supporting the operations of a small-scale fashion retail business. The study demonstrated how a purpose-built e-commerce platform could have addressed the specific needs of the boutique, particularly in product presentation, online ordering, inventory management, payment verification, and administrative monitoring. By providing a brand-consistent and organized digital storefront, the system could have helped improve the accessibility of the boutique's products while providing customers with a more convenient way to browse and purchase both brand-new and preloved clothing. The project also demonstrated that small businesses could have benefited from customized web-based solutions that were designed according to their actual operational requirements rather than relying entirely on generic e-commerce platforms.

The study also contributed to improving the efficiency and organization of the boutique's business processes. The integration of product listings, inventory records, ordering functions, payment verification, and administrative tools could have reduced dependence on manual processes and separate communication channels. This could have helped minimize errors, improve the accuracy of product and order information, and provide the business owner with a more systematic method of managing daily transactions. For customers, the system could have provided clearer product information, organized categories, convenient ordering procedures, and more transparent transaction updates. These improvements could have contributed to a more reliable and satisfactory online shopping experience.

Furthermore, the development of the system demonstrated the importance of iterative refinement in software development. Throughout the development process, adjustments such as removing unnecessary features, correcting stylesheet and interface issues, improving system functions, and refining the overall user experience could have been necessary to ensure that the final system remained functional and aligned with the needs of Margaux Collections. These refinements demonstrated that developing an effective information system required continuous evaluation, testing, and improvement rather than relying solely on the initial design. Overall, the study showed that a customized and continuously refined web-based ordering system could have supported the digital transformation of a small fashion business while maintaining its unique brand identity and addressing its specific operational challenges.

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